



Typescript

Complete Course



React js 19 tutorial Hi

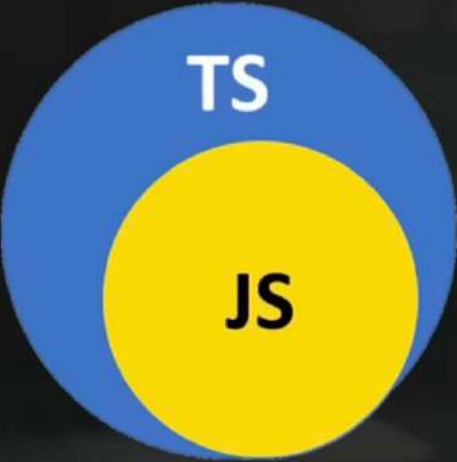


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What is TypeScript



- Better version of JavaScript
- With more features
- TypeScript is superset of JS
- All Features of JS
- And extra features and functionality

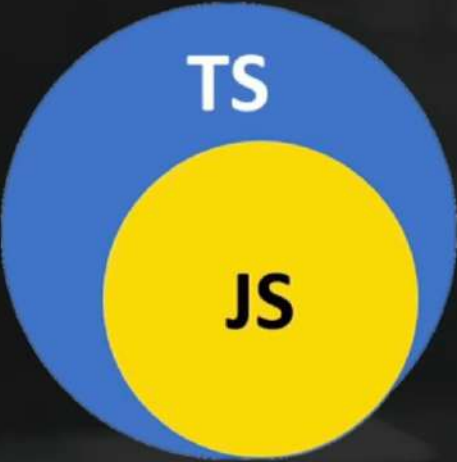
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Why TypeScript



A Venn diagram consisting of two overlapping circles. The larger circle on the left is blue and contains the text 'TS' in white. The smaller circle on the right is yellow and contains the text 'JS' in black. The two circles overlap in the center.

- Static Typing or Define Data type.
- Improved Code Maintainability.
- Advance Features.
- Code Quality

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typescript-tut

EXPLORER

OPEN EDITORS

TS app.ts

TYPESCRIPT-TUT

node_modules

TS app.ts

package-lock.json

package.json

OUTLINE

TIMELINE

TS app.ts

TS app.ts > [@]userName

1 var userName:string="Anil sidhu"

PROBLEMS

OUTPUT

DEBUG CONSOLE

TERMINAL

PORTS

SPELL CHECKER

zsh

anilsidhu@anils-MacBook-Air typescript-tut % npx tsc app.ts

Ln 1, Col 33

Spaces: 4

UTF-8

LF

TypeScript

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+

← → ↻

https://www.typescriptlang.org/download/

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via npm

TypeScript is available as a [package on the npm registry](#), available as "typescript".

You will need a copy of [Node.js](#) as an environment to run the package. Then you use a dependency manager like [npm](#), [yarn](#) or [pnpm](#) to download TypeScript into your project.

`npm install typescript --save-dev`

npm

yarn

pnpm

All of these dependency managers support lockfiles, ensuring that everyone on your team is using the same version of the language. You can then run the TypeScript compiler using one of the following commands:

`npx tsc`

npm

yarn

pnpm

with Visual Studio

For most project types, you can get TypeScript as a package in Nuget for your MSBuild projects, for example an ASP.NET Core app.

When using Nuget, you can [install TypeScript through Visual Studio](#) using:

- The Manage NuGet Packages window (which you can get to by right-clicking on a project node)
- The Nuget Package Manager Console (found in Tools > NuGet Package Manager > Package Manager Console) and then running:
`Install-Package Microsoft.TypeScript.MSBuild`

For project types which don't support Nuget, you can use the [TypeScript Visual Studio extension](#). You can [install the extension](#) using Extensions > Manage Extension in Visual Studio.



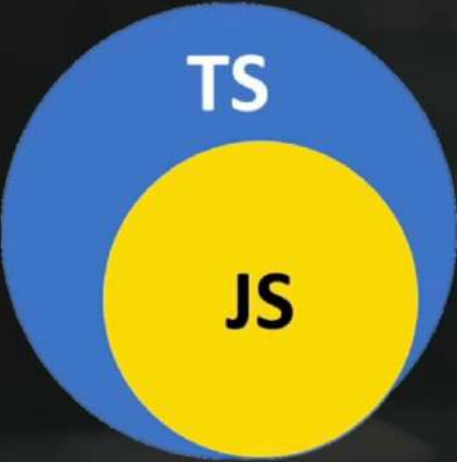
The examples below are for more advanced use cases.

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Important TS features



A Venn diagram consisting of two overlapping circles. The larger circle on the left is blue and labeled 'TS'. The smaller circle on the right is yellow and labeled 'JS'. The two circles overlap in the center.

- Define Data type.
- Interfaces
- Decorators
- Generics
- Namespaces
- Type Inference
- Advance Features.
- Code Quality
- Etc

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EXPLORED

OPEN EDITORS

TS app.ts

JS app.js

TYPESCRIPT-TUT

node_modules

JS app.js

TS app.ts

package-lock.json

package.json

OUTLINE

TIMELINE

typescript-tut

TS app.ts

JS app.js

TS app.ts > ...

1 var userName:string="Anil sidhu"

2

3 console.log(userName)

PROBLEMS

OUTPUT

DEBUG CONSOLE

TERMINAL

PORTS

SPELL CHECKER

zsh

anilsidhu@anils-MacBook-Air typescript-tut % npx tsc app.ts

anilsidhu@anils-MacBook-Air typescript-tut % npx tsc app.ts

anilsidhu@anils-MacBook-Air typescript-tut % npx tsc app.ts --watch

Ln 2, Col 1

Spaces: 4

UTF-8

LF

TypeScript

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1

EXPLORER

...

OPEN EDITORS 1 unsaved

TS app.ts

JS app.js

user.html

TYPESCRIPT-TUT

node_modules

JS app.js

TS app.ts

package-lock.json

package.json

user.html

OUTLINE

TIMELINE

Launchpad

typescript-tut

TS app.ts JS app.js user.html

JS app.js > ...

var age = 30;

var college = "IET alwar";

console.log(userName);

console.log(age);

console.log(college);

npm, tsc

watch

PROBLEMS 2 OUTPUT DEBUG CONSOLE TERMINAL PORTS SPELL CHECKER 2

node + -

[12:38:34 AM] File change detected. Starting incremental compilation...

[12:38:35 AM] Found 0 errors. Watching for file changes.

Ln 8, Col 1 (19 selected) Spaces: 4 UTF-8 LF {} JavaScript Port : 5500



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EXTENSIONS: MAR...
live
Live Server 4ms
Launch a development lo...
Ritwick Dey
Live Share 20M
Real-time collaborative d...
Microsoft
Live Preview 8.2M
Hosts a local server in yo...
Microsoft
{Deprecated} L... 5.3M
Adds audio calling capab...
Microsoft
Live Sass Com... 2.5M
Compile Sass or Scss to ...
Ritwick Dey
Live Sass Com... 2.2M
Compile Sass or Scss to ...
Glenn Marks
Live Server (Fiv... 1.7M

Launchpad

typescript-tut

TS app.ts X <> user.html X

```
TS app.ts > ...
1  var username:string="Anil Sionu"
2  var age:number=30;
3  var college:string="IET alwar"
4
5
6  console.log(username)
7  console.log(age)
8  console.log(college)
9
10
```

PROBLEMS 1 OUTPUT DEBUG CONSOLE TERMINAL PORTS SPELL CHECKER 1

node + - [X] [Y] ... ^ X

```
[12:38:34 AM] File change detected. Starting incremental compilation...
[12:38:35 AM] Found 0 errors. Watching for file changes.
```



Ln 8, Col 20 Spaces: 4 UTF-8 LF {} TypeScript Port : 5500

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typescript-tut

EXTENSIONS: MAR...

TS app.ts

<> user.html X JS app.js

<> user.html > html

2 <html lang="en">

3 <head>

4 <meta charset="utf-8">

5 <meta name="viewport" content="width=device-width, initial-scale=1.0">

6 <title>Document</title>

7 <script src="./app.js" ></script>

8 </head>

9 <body>

10 <h1>User file</h1>

11 </body>

12 </html>

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS SPELL CHECKER

node + - [] [] ... ^ X

[12:35:49 AM] File change detected. Starting incremental compilation...

[12:35:49 AM] Found 0 errors. Watching for file changes.

Ln 12, Col 8 Spaces: 4 UTF-8 LF () HTML Go Live

Launchpad 0 0

Ln 12, Col 8 Spaces: 4 UTF-8 LF () HTML Go Live



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✎ 🗨



Data Types in TypeScript



▶ ▶

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Document

← → ↻ 🔍 https://www.typescriptlang.org/download/ 📄 ☆ 🔍 🌐 📄 📄 📄 📄

Globally Installing TypeScript

It can be handy to have TypeScript available across all projects, often to test one-off ideas. Long-term, codebases should prefer a project-wide installation over a global install so that they can benefit from reproducible builds across different machines.

via npm

You can use npm to install TypeScript globally, this means that you can use the `tsc` command anywhere in your terminal.

To do this, run `npm install -g typescript` This will install the latest version (currently 5.8).

via Visual Studio Marketplace

You can install TypeScript as a Visual Studio extension, which will allow you to use TypeScript across many MSBuild projects in Visual Studio.

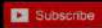
The latest version is available [in the Visual Studio Marketplace](#).

Working with TypeScript-compatible transpilers

There are other tools which convert TypeScript files to JavaScript files. You might use these tools for speed or consistency with existing build tooling.

Each of these projects handle the file conversion, but do not handle the type-checking aspects of the TypeScript compiler. So it's likely





typescript-tut

EXPLORER

OPEN EDITORS

- TS app.ts 1
- JS app.js
- user.html

TYPESCRIPT-TUT

- node_modules
- JS app.js
- TS app.ts 1
- package-lock.json
- package.json
- user.html

PROBLEMS 3 OUTPUT DEBUG CONSOLE TERMINAL PORTS SPELL CHECKER 2

node_modules/typescript/lib/lib.dom.d.ts:28611:15 - error TS2451: Cannot redeclare block-scoped variable 'name'.

28611 declare const name: void;

app.ts:4:5

4 var name="sam"

'name' was also declared here.

[12:39:53 AM] Found 2 errors. Watching for file changes.

Ln 4, Col 14 Spaces: 4 UTF-8 LF {} TypeScript Port : 5500



5 Edit: Code x New Tab x +

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✎ 🗨

Primitive Types



- number
- string
- boolean
- null
- undefined
- bigint
- symbol



➤

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Address bar: <https://slides.com/anilsidhu/code-f82053/edit>

Categories



- Primitive Types
- Object Types
- Special Types
- Advanced Types
- Function Types



Navigation arrows: back, forward

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What is Data Types



```
1 var userName:string="Anil sidhu"
2 var age:number=30;
3 var male:boolea=true;
```



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Special Types



- any
- unknown
- void
- never



Navigation icons: back, forward, search, share, etc.

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Object Type



- Array
- Tuple
- Object



Navigation arrows: < >

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New Tab

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Symbol

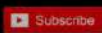
```
1 const sym1 = Symbol();
2 const sym2 = Symbol('description');
3 const sym3 = Symbol('description');
4
5 console.log(sym1 === sym2); // false
6 console.log(sym2 === sym3); // false
```











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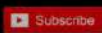




Number Data Type

- Apply number data type
- Redeclare issue
- How to Add number with data type
- How to use binary and hexadecimal numbers
- Convert string to number
- Type Inference with number
- Decimal with number data type





Edit: Code

Typescript Data Types

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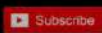


Function Types

Defines the type of a function.







Edit: Code x Typescript Data Types x +

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TS

Advanced Types

- Union
- Intersection
- Type Alias
- Enum
- Literal Types



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TS number.ts U ×

TS number.ts > ...

```
1 //var num1:number=10;
2 //var num2:number=30;
4 var num3:number=30
5
6 var total:number=num1+num3;
7 console.log(total);
8
9 var oct:number=0o100001;
10 var hexa:number=0b00001;
11 var binary:number=0x00001;
12
13 console.log(oct+10);
14 |
15
16
17
```

JS number.js U ×

JS number.js > ...

```
1 //var var_name : data_type = value
2 var num1 = 10;
3 var num2 = 30;
4 var num3 = 30;
5 var total = num1 + num3;
6 console.log(total);
7 var oct = 32769;
8 var hexa = 1;
9 var binary = 0x00001;
10 console.log(oct + 10);
11
```

PROBLEMS 2 OUTPUT DEBUG CONSOLE TERMINAL PORTS GITLENS SPELL CHECKER 2

/dev/fd/13:18: command not found: compdef

anilsidhu@anils-MacBook-Air typescript-tut % node number.js

40

anilsidhu@anils-MacBook-Air typescript-tut % node number.js

40

32779

anilsidhu@anils-MacBook-Air typescript-tut %

Google Chrome

Ln 14, Col 1 Spaces: 4 UTF-8 LF {} TypeScript 5500



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typescript-tut

TS number.ts 3, U X

```
TS number.ts > [e] oct
1 //var var_name : data_type = value
2 var num1:number=10;
3 var num2:number=30
4 var num3:number=30
5
6 var total:numero
7 console.log(tot
8
9 var oct:number=0001;
10 var hexa:number=00001;
11 var binary:number=00001;
12
13
```

Octal literals are not allowed. Use the syntax '0o1'. ts(1121)

View Problem (\F8) No quick fixes available

JS number.js U X

```
JS number.js > ...
1 //var var_name : data_type = value
2 var num1 = 10;
3 var num2 = 30;
4 var num3 = 30;
5 var total = num1 + num3;
6 console.log(total);
7 var oct = 1;
8 var hexa = 1;
9 var binary = 1;
10
```

PROBLEMS 5 OUTPUT DEBUG CONSOLE TERMINAL PORTS GITLENS SPELL CHECKER 2

number.ts:11:19 - error TS1121: Octal literals are not allowed. Use the syntax '0o1'.

11 var binary:number=00001;

[1:04:20 PM] Found 3 errors. Watching for file changes.

first-program* Launchpad 3 0 2

Ln 9, Col 17 Spaces: 4 UTF-8 LF {} TypeScript 5500



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typescript-tut

TS number.ts U X

TS number.ts > ...
1 //var var_name : data_type = value
2 var num1:number=10;
3 var num2:number=30;
4
5 var total=num1+num2;
6 console.log(total);
7

JS number.js U X

JS number.js > ...
1 //var var_name : data_type = value
2 var num1 = 10;
3 var num2 = 30;
4 var total = num1 + num2;
5 console.log(total);
6

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS GITLENS SPELL CHECKER

/dev/fd/13:18: command not found: compdef
anilsidhu@anils-MacBook-Air typescript-tut % node number.js
40
anilsidhu@anils-MacBook-Air typescript-tut %

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Ln 6, Col 18 Spaces: 4 UTF-8 LF {} TypeScript 5500



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String and Boolean



- Apply string data type
- Ways to define string
- Convert in string data type
- Apply boolean data type
- Possible boolean values
- Inference and declaration issues
- Run with html file



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typescript-tut

TS number.ts U X

TS number.ts > ...

```
17 // var item2Converted=Number(item2)
18 var item2Converted=+item2
19 console.log(item+ +item2Converted);
20
21
22 //
23 // var data =30;
24 // data=40
25 // data="anil sidhu"
26
27 var data: number | string =30
28 data="anil"
29
30
31
32
33
```

JS number.js U X

JS number.js > ...

```
1 //var var_name : data_type = value
2 var num1 = 10;
3 var num2 = 30;
4 var num3 = 30;
5 var total = num1 + num3;
6 console.log(total);
7 var oct = 32769;
8 var hexa = 1;
9 var binary = 0x00001;
10 console.log(oct + 10);
11 var item = 100;
12 var item2 = "50";
13 // var item2Converted=Number(item2)
14 var item2Converted = +item2;
15 console.log(item + +item2Converted);
16 //
17 // var data =30;
```

PROBLEMS 2 OUTPUT DEBUG CONSOLE TERMINAL PORTS GITLENS SPELL CHECKER 2

150

anilsidhu@anils-MacBook-Air typescript-tut % node number.js

40

32779

10050

anilsidhu@anils-MacBook-Air typescript-tut % node number.js

40

32779

150

anilsidhu@anils-MacBook-Air typescript-tut %

first-program* Launchpad 0 0 1 2

Ln 28, Col 12 Spaces: 4 UTF-8 LF {} TypeScript 5500

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typescript-tut

TS number.ts U X

TS number.ts > ...

```
9 var oct:number=0o100001;
10 var hexa:number=0b00001;
11 var binary:number=0x00001;
12
13 console.log(oct+10);
14
15 var item:number=100;
16 var item2="50";
17 // var item2Converted=Number(item2)
18 var item2Converted=+item2
19 console.log(item+ +item2);
20
21
22
23
24
```

JS number.js U X

JS number.js > ...

```
1 //var var_name : data_type = value
2 var num1 = 10;
3 var num2 = 30;
4 var num3 = 30;
5 var total = num1 + num3;
6 console.log(total);
7 var oct = 32769;
8 var hexa = 1;
9 var binary = 0x00001;
10 console.log(oct + 10);
11 var item = 100;
12 var item2 = "50";
13 // var item2Converted=Number(item2)
14 var item2Converted = +item2;
15 console.log(item + +item2);
16
```

PROBLEMS 2 OUTPUT DEBUG CONSOLE TERMINAL PORTS GITLENS SPELL CHECKER 2

150

anilsidhu@anils-MacBook-Air typescript-tut % node number.js

40

32779

10050

anilsidhu@anils-MacBook-Air typescript-tut % node number.js

40

32779

150

anilsidhu@anils-MacBook-Air typescript-tut %

first-program* Launchpad 0 1 2

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typescript-tut

TS string.ts U X

JS string.js U X

TS string.ts > ...

JS string.js > ...

1 var str:string="Hello How are you";

2 var str2:string='Hello How are you';

3 var str3:string=`Hello How are you`;

4

5 var age :number=30;

6 var userName:string="Anil sidhu";

7

8 var info:string=`my name is \${userName} and age is

9 \${age} years`

10 console.log(info);

11

1 var str = "Hello How are you";

2 var str2 = 'Hello How are you';

3 var str3 = "Hello How are you";

4 var age = 30;

5 var userName = "Anil sidhu";

6 var info = "my name is ".concat(userName, " and age

7 console.log(info);

8

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS GITLENS SPELL CHECKER

● anilsidhu@anils-MacBook-Air typescript-tut % node string.js

my name is Anil sidhu and age is

30 years

○ anilsidhu@anils-MacBook-Air typescript-tut %

Ln 11, Col 1 Spaces: 4 UTF-8 LF () TypeScript 5500



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typescript-tut

TS string.ts U X

JS string.js U X

TS string.ts > [0] info

JS string.js > ...

1 var str:string="Hello How are you";

2 var str2:string='Hello How are you';

3 var str3:string=`Hello How are you`;

4

5 var age :number=30;

6 var userName:string="Anil sidhu";

7

8 var info:string=`my name is \${userName} and age is

9 \${age} years`

1 var str = "Hello How are you";

2 var str2 = 'Hello How are you';

3 var str3 = "Hello How are you";

4 var age = 30;

5 var userName = "Anil sidhu";

6 var info = "my name is ".concat(userName, " and age

7

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS GITLENS SPELL CHECKER

[2:24:55 PM] File change detected. Starting incremental compilation...

[2:24:55 PM] Found 0 errors. Watching for file changes.

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typescript-tut

EXPLORER

OPEN EDITORS

- TS string.ts U

TYPESCRIPT-TUT

- node_modules
- JS app.js
- TS app.ts
- JS first.js
- TS first.ts
- JS number.js
- TS number.ts
- package-lock.json
- package.json
- TS string.ts U
- user.html

OUTLINE

TIMELINE

PROBLEMS OUTPUT DEBUG CONSOLE **TERMINAL** PORTS GITLENS SPELL CHECKER

```
anilsidhu@anils-MacBook-Air typescript-tut % tsc string.ts --watch
```

Ln 1, Col 1 Spaces: 4 UTF-8 LF () TypeScript Port: 5500



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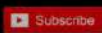


Null And Undefined

- What is Null Data Type
- How to use it
- What is Undefined Data Type
- How to use Undefined
- Possible Values
- Run with html file
- Interview Question







typescript-tut

TS string.ts U X

TS string.ts > ...

```
10
11 var num:number=100;
12 // var data:string = num.toString();
13 // var data:string = " " + num;
14 var boolData=true;
15 // var data:string = boolData.toString();
16 var data:string = "" +boolData;
17
18
19 var item:boolean=true
20
```

JS string.js U X

JS string.js > ...

```
1 var str = "Hello How are you";
2 var str2 = 'Hello How are you';
3 var str3 = "Hello How are you";
4 var age = 30;
5 var userName = "Anil sidhu";
6 var info = "my name is ".concat(userName, " and age
7 console.log(info);
8 var num = 100;
9 // var data:string = num.toString();
10 // var data:string = " " + num;
11 var boolData = true;
12 // var data:string = boolData.toString();
13 var data = "" + boolData;
14 var item = true;
15
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS GITLENS SPELL CHECKER

[2:28:59 PM] File change detected. Starting incremental compilation...

[2:28:59 PM] Found 0 errors. Watching for file changes.

first-program* Launchpad 0 0

Ln 19, Col 22 Spaces: 4 UTF-8 LF () TypeScript Port: 5500



Subscribe

typescript-tut

TS string.ts U X

JS string.js U X

TS string.ts > ...

JS string.js > ...

var str:string= 'hello how are you' ;

1 var str = "Hello How are you";

4

2 var str2 = 'Hello How are you';

5 var age :number=30;

3 var str3 = "Hello How are you";

6 var userName:string="Anil sidhu";

4 var age = 30;

7

5 var userName = "Anil sidhu";

8 var info:string=`my name is \${userName} and age is \$

6 var info = "my name is ".concat(userName, " and age

9 console.log(info);

7 console.log(info);

10

8 var num = 100;

11 var num:number=100;

9 // var data:string = num.toString();

12 // var data:string=num.toString();

10 var data = " " + num;

13 var data:string = " " + num

11

14

15

16

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS GITLENS SPELL CHECKER

anilsidhu@anils-MacBook-Air typescript-tut % node string.js

my name is Anil sidhu and age is

30 years

anilsidhu@anils-MacBook-Air typescript-tut % node string.js

my name is Anil sidhu and age is 30 years

anilsidhu@anils-MacBook-Air typescript-tut %

first-program* Launchpad 0 0

Ln 14, Col 1 Spaces: 4 UTF-8 LF () TypeScript 5500

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EXPLORER

OPEN EDITORS

GROUP 1

TS null.ts U

GROUP 2

JS null.js U

TYPESCR...

node_modules

undefine.

JS app.js

TS app.ts

JS first.js

TS first.ts

JS null.js U

TS null.ts U

JS number.js

TS number.ts

package-lock.json

package.json

JS string.js

TS string.ts

user.html

OUTLINE

TIMELINE

typescript-tut

TS null.ts U X

TS null.ts > ...

1 var data : null | string =null;

2 var userName:null|string=null

3 var login=true

4 if(login){

5 | userName="Anil sidhu"

6 }

7

8 console.log(typeof data);

9

JS null.js U X

JS null.js > ...

1 var data = null;

2 var userName = null;

3 var login = true;

4 if (login) {

5 | userName = "Anil sidhu";

6 }

7 console.log(typeof data);

8

PROBLEMS

OUTPUT

DEBUG CONSOLE

TERMINAL

PORTS

GITLENS

SPELL CHECKER

/dev/fd/13:18: command not found: compdef

● anilsidhu@anils-MacBook-Air typescript-tut % node null.js

object

● anilsidhu@anils-MacBook-Air typescript-tut % node null.js

object

● anilsidhu@anils-MacBook-Air typescript-tut % node null.js

string

○ anilsidhu@anils-MacBook-Air typescript-tut %

Ln 9, Col 1

Spaces: 4

UTF-8

LF

{ } TypeScript

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EXPLORER

OPEN EDITORS

GROUP 1

TS null.ts U

GROUP 2

JS null.js U

TYPESCRIPT-TUT

node_modules

JS app.js

TS app.ts

JS first.js

TS first.ts

JS null.js U

TS null.ts U

JS number.js

TS number.ts

package-lock.json

package.json

JS string.js

TS string.ts

user.html

OUTLINE

TIMELINE

TS null.ts U X

TS null.ts > ...

1 var data : null| string =null;

2 var userName:null|string=null

3 var login=true

4 if(login){

5 userName="Anil sidhu"

6 }

JS null.js U X

JS null.js > ...

1 var data = null;

2 var userName = null;

3 var login = true;

4 if (login) {

5 userName = "Anil sidhu";

6 }

7

PROBLEMS

OUTPUT

DEBUG CONSOLE

TERMINAL

PORTS

GITLENS

...

[3:05:06 PM] File change detected. Starting incremental compilation...

[3:05:06 PM] Found 0 errors. Watching for file changes.

Ln 6, Col 2

Spaces: 4

UTF-8

LF

{ } TypeScript

Go Live



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typescript-tut

EXPLORER

OPEN EDITORS

TS null.ts

TYPESCRIPT-TUT

- node_modules
- app.js
- app.ts
- first.js
- first.ts
- null.ts
- number.js
- number.ts
- package-lock.json
- package.json
- string.js
- string.ts
- user.html

OUTLINE

TIMELINE

first-program*

Launchpad

0 0

Ln 1, Col 1 (19 selected) Spaces: 4 UTF-8 LF {} TypeScript Go Live

```
1 var data:null=null;
2 var val:undefined=undefined
```



Subscribe

typescript-tut

TS undefine.ts U X

JS undefine.js U X

TS undefine.ts > ...

JS undefine.js > ...

1 var userName1:undefined|string=undefined

2 var val=null

3

4 userName1="Anil sidhu"

5

6 console.log(val==userName1);

7

1 var userName1 = undefined;

2 var val = null;

3 userName1 = "Anil sidhu";

4 console.log(val == userName1);

5

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS GITLENS SPELL CHECKER

undefined

anilsidhu@anils-MacBook-Air typescript-tut % node undefine.js

Anil sidhu

anilsidhu@anils-MacBook-Air typescript-tut % node undefine.js

string

anilsidhu@anils-MacBook-Air typescript-tut % node undefine.js

undefined

anilsidhu@anils-MacBook-Air typescript-tut % node undefine.js

false

anilsidhu@anils-MacBook-Air typescript-tut %

first-program* Launchpad 0 0

Ln 6, Col 27 Spaces: 4 UTF-8 LF {} TypeScript Go Live

Subscribe

typescript-tut

TS undefine.ts U X

TS undefine.ts > ...
1 var userName1:undefined|string=undefined
2
3 // userName1="Anil sidhu"
4
5 console.log(typeof userName1);
6

JS undefine.js U X

JS undefine.js > ...
1 var userName1 = undefined;
2 // userName1="Anil sidhu"
3 console.log(typeof userName1);
4

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS GITLENS SPELL CHECKER

anilsidhu@anils-MacBook-Air typescript-tut % node undefine.js
anilsidhu@anils-MacBook-Air typescript-tut % node undefine.js
undefined
anilsidhu@anils-MacBook-Air typescript-tut % node undefine.js
Anil sidhu
anilsidhu@anils-MacBook-Air typescript-tut % node undefine.js
string
anilsidhu@anils-MacBook-Air typescript-tut % node undefine.js
undefined
anilsidhu@anils-MacBook-Air typescript-tut %

first-program* Launchpad 0 0

Ln 3, Col 26 Spaces: 4 UTF-8 LF {} TypeScript Go Live





EXPLORED

OPEN EDITORS

GROUP 1

TS undefine.ts 1, U

GROUP 2

JS undefine.js U

TYPESCRIPT-TUT

TS app.ts

JS first.js

TS first.ts

JS null.js U

TS null.ts U

JS number.js

TS number.ts

{ package-lock.json

{ package.json

JS string.js

TS string.ts

JS undefine.js U

TS undefine.ts 1, U

<> user.html

OUTLINE

TIMELINE

TS undefine.ts 1, U

TS undefine.ts

var userName1: undefined

1 var userName1:undefined=undefined

2

3 console.log(item);

4

JS undefine.js U

JS undefine.js > ...

1 var userName1 = undefined;

2 console.log(item);

3

PROBLEMS 1

OUTPUT

DEBUG CONSOLE

TERMINAL

PORTS

GIT LENS

SPELL CHECKER

anilsidhu@anils-MacBook-Air typescript-tut % node null.js

object

anilsidhu@anils-MacBook-Air typescript-tut % node null.js

object

anilsidhu@anils-MacBook-Air typescript-tut % node null.js

string

anilsidhu@anils-MacBook-Air typescript-tut % node undefine.js

anilsidhu@anils-MacBook-Air typescript-tut % node undefine.js

undefined

anilsidhu@anils-MacBook-Air typescript-tut %

Ln 1, Col 14

Spaces: 4

UTF-8

LF

{ } TypeScript

Go Live





typescript-tut

EXPLORER

OPEN EDITORS

tsconfig.json

TYPESCRIPT-TUT

- node_modules
- app.js
- app.ts
- null.js
- null.ts
- number.ts
- package-lock.json
- package.json
- string.ts
- tsconfig.json
- undefine.js
- undefine.ts
- user.html

tsconfig.json

```
{
  "compilerOptions": {
    // "disableSourceOfProjectReferenceRedirect": true, /* Disable preferring source files
    // "disableSolutionSearching": true, /* Opt a project out of multi-proje
    // "disableReferencedProjectLoad": true, /* Reduce the number of projects lo

    /* Language and Environment */
    "target": "ES2024", /* Set the JavaScript language vers
    // "lib": [], /* Specify a set of bundled library
    // "jsx": "preserve", /* Specify what JSX code is generat
    // "libReplacement": true, /* Enable lib replacement. */
    // "experimentalDecorators": true, /* Enable experimental support for
    // "emitDecoratorMetadata": true, /* Emit design-type metadata for de
    // "jsxFactory": "", /* Specify the JSX factory function
    // "jsxFragmentFactory": "", /* Specify the JSX Fragment referer
    // "jsxImportSource": "", /* Specify module specifier used to
    // "reactNamespace": "", /* Specify the object invoked for '
  }
}
```

PROBLEMS 1 OUTPUT DEBUG CONSOLE TERMINAL PORTS GITLENS SPELL CHECKER 1

anilsidhu@anils-MacBook-Air typescript-tut % tsc string.ts

anilsidhu@anils-MacBook-Air typescript-tut % tsc

Ln 27, Col 1 Spaces: 2 UTF-8 LF {} JSON with Comments Go Live



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typescript-tut

EXPLORER

OPEN EDITORS

TYPESCRIPT-TUT

- node_modules
- JS app.js
- TS app.ts
- JS first.js
- TS first.ts
- JS null.js
- TS null.ts
- JS number.js
- TS number.ts
- { } package-lock.json
- { } package.json
- JS string.js
- TS string.ts
- JS undefine.js
- TS undefine.ts
- <> user.html

OUTLINE

TIMELINE

PROBLEMS

OUTPUT

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GITLENS

SPELL CHECKER

zsh

anilsidhu@anils-MacBook-Air typescript-tut % tsc --init

first-program

Launchpad

0 0

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TypeScript Config File



- How to generate config file
- Use of config file
- Convert All TS file to js together
- Fix errors after config file
- Common use of config
- Interview Questions



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typescript-tut

tsconfig.json U X TS test.ts U TS null.ts M TS string.ts M TS number.ts M

EXPLORED

OPEN EDITORS

- tsconfig.json U
- TS test.ts U
- TS null.ts M
- TS string.ts M
- TS number.ts M

TYPESCRIPT-TUT

- JS null.js M
- TS null.ts M
- JS number.js M
- TS number.ts M
- { } package-lock.json
- { } package.json
- JS string.js M
- TS string.ts M
- TS test.ts U
- tsconfig.json U
- JS undefine.js M
- TS undefine.ts
- <> user.html

OUTLINE

TIMELINE

tsconfig.json > { } compilerOptions

```
2  "compilerOptions": {
50  // "allowJs": true,
51  // "checkJs": true,
52  // "maxNodeModuleJsDepth": 1,
53
54  /* Emit */
55  "declaration": true,
56  // "declarationMap": true,
57  // "emitDeclarationOnly": true,
58  // "sourceMap": true,
59  // "inlineSourceMap": true,
60  // "noEmit": true,
61  // "outFile": "./",
62  "outDir": "./output",
63  // "removeComments": true,
64  // "importHelpers": true,
65  // "downlevelIteration": true,
66  // "sourceRoot": "",
67  // "mapRoot": "",
```

PROBLEMS 2 OUTPUT DEBUG CONSOLE TERMINAL PORTS GITLENS SPELL CHECKER 2

anilsidhu@anils-MacBook-Air typescript-tut % tsc
anilsidhu@anils-MacBook-Air typescript-tut % tsc
anilsidhu@anils-MacBook-Air typescript-tut % tsc
anilsidhu@anils-MacBook-Air typescript-tut % tsc
anilsidhu@anils-MacBook-Air typescript-tut %

first-program* Launchpad 0 0 2 Ln 55, Col 27 Spaces: 2 UTF-8 LF { } JSON with Comments Go Live



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EXPLORER

OPEN EDITORS

- tsconfig.json U
- TS test.ts U
- TS null.ts M
- TS string.ts M
- TS number.ts M

TYPESCRIPT

- node_modules
- output
 - app.js U
 - null.js U
 - number.js U
 - string.js U
 - test.js U
 - underline.js U
- app.js M
- app.ts
- null.js M
- null.ts M
- number.js M
- number.ts M

OUTLINE

TIMELINE

tsconfig.json U

TS test.ts U X

TS null.ts M

TS string.ts M

TS number.ts M

TS test.ts > [0] test10

```
1 var test10:string="abc"
```

> outd Aa ab .* No results ↑ ↓ ≡ ×

PROBLEMS 2 OUTPUT DEBUG CONSOLE TERMINAL PORTS GITLENS SPELL CHECKER 2

```
anilsidhu@anils-MacBook-Air typescript-tut % tsc
anilsidhu@anils-MacBook-Air typescript-tut % tsc
anilsidhu@anils-MacBook-Air typescript-tut % tsc
anilsidhu@anils-MacBook-Air typescript-tut % tsc
anilsidhu@anils-MacBook-Air typescript-tut %
```

first-program* Launchpad 0 0 0 2

Ln 1, Col 23 Spaces: 4 UTF-8 LF {} TypeScript Go Live



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EXPLORER

OPEN EDITORS

- tsconfig.json U
- TS null.ts M
- TS string.ts M
- TS number.ts M

TYPESCRIPT-TUT

- > node_modules
- > output
- JS app.js M
- TS app.ts
- JS null.js M
- TS null.ts M
- JS number.js M
- TS number.ts M
- { } package-lock.json
- { } package.json
- JS string.js M
- TS string.ts M
- tsconfig.json U
- JS undefine.js M
- TS undefine.ts
- > OUTLINE
- > TIMELINE

tsconfig.json > {} compilerOptions

```
2  "compilerOptions": {
54  /* Emit */
55  // "declaration": true,
56  // "declarationMap": true,
57  // "emitDeclarationOnly": true,
58  // "sourceMap": true,
59  // "inlineSourceMap": true,
60  // "noEmit": true,
61  // "outFile": "./",
62  "outDir": "./output",
63  // "removeComments": true,
64  // "importHelpers": true,
65  // "downlevelIteration": true,
66  // "sourceRoot": "",
67  // "mapRoot": "",
68  // "inlineSources": true,
69  // "emitBOM": true,
70  // "newline": "crlf",
```

PROBLEMS 2 OUTPUT DEBUG CONSOLE TERMINAL PORTS GITLENS SPELL CHECKER 2

anilsidhu@anils-MacBook-Air typescript-tut % tsc

anilsidhu@anils-MacBook-Air typescript-tut %

Ln 62, Col 30 Spaces: 2 UTF-8 LF { } JSON with Comments Go Live



Edit: Code

https://slides.com/anilsidhu/code-f82053/edit

Why BigInt?



- In JavaScript and TypeScript, Number has a maximum safe integer
- `Number.MAX_SAFE_INTEGER = 253 - 1`
- 9007199254740991
- Beyond this limit, precision issues arise.

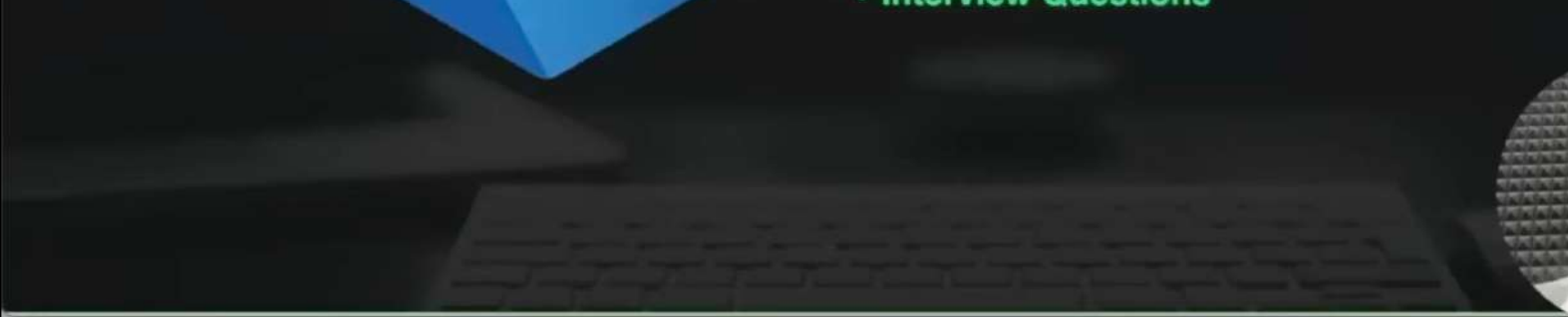


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https://slides.com/anilsidhu/code-f82053/edit



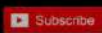




BigInt Data Type

- What is bigInt
- When we should use it
- Example
- Mixing issue
- Interview Questions





typescript-tut

EXPLORER

OPEN EDITORS

- tsconfig.json U
- TS string.d.ts out... U
- TS test.ts U
- TS null.ts M
- TS string.ts M
- TS number.ts M

TYPESCR...

- node_modules
- output
 - TS app.d.ts U
 - JS app.js U
 - TS null.d.ts U
 - JS null.js U
 - TS number.d.ts U
 - JS number.js U
 - TS string.d.ts U
 - JS string.js U
 - TS test.d.ts U
 - JS test.js U
 - TS undefine.d.ts U

OUTLINE

TIMELINE

tsconfig.json U

TS string.d.ts U X

TS test.ts U

TS null.ts M

TS string.ts M

TS number.ts M

output > TS string.d.ts > ...

```
1 declare var str: string;
2 declare var str2: string;
3 declare var str3: string;
4 declare var age: number;
5 declare var userName: string | null;
6 declare var info: string;
7 declare var num: number;
8 declare var boolData: boolean;
9 declare var str10: string;
10
```

> outd Aa ab .* No results ↑ ↓ ≡ X

PROBLEMS 2 OUTPUT DEBUG CONSOLE TERMINAL PORTS GITLENS SPELL CHECKER 2

```
anilsidhu@anils-MacBook-Air typescript-tut % tsc
anilsidhu@anils-MacBook-Air typescript-tut % tsc
anilsidhu@anils-MacBook-Air typescript-tut % tsc
anilsidhu@anils-MacBook-Air typescript-tut % tsc
anilsidhu@anils-MacBook-Air typescript-tut % tsc
anilsidhu@anils-MacBook-Air typescript-tut % tsc
```

first-program* Launchpad 0 0 0 2

Ln 1, Col 1 Spaces: 4 UTF-8 LF {} TypeScript Go Live



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← → typescript-tut

TS bigint.ts 2, U X

TS bigint.ts > [0] y

1 var bignumber:bigint=9007199254740991n

2 var x=1;

3 var y=2

4

5 console.log(bignumber+x);

6 console.log(bignumber+y);

7

JS bigint.js U X

output > JS bigint.js > ...

1 "use strict";

2 var bignumber = 9007199254740991n;

3 var x = 1;

4 var y = 2;

5 console.log(bignumber + x);

6 console.log(bignumber + y);

7

PROBLEMS 2 OUTPUT DEBUG CONSOLE TERMINAL PORTS GITLENS SPELL CHECKER

anilsidhu@anils-MacBook-Air typescript-tut % node output/bigint.js

9007199254740992

anilsidhu@anils-MacBook-Air typescript-tut % node output/bigint.js

9007199254740992

9007199254740992

anilsidhu@anils-MacBook-Air typescript-tut %

node

zsh



first-program* Launchpad 2 0

Ln 3, Col 8 Spaces: 4 UTF-8 LF () TypeScript Live

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EXPLORER

OPEN EDITORS

GROUP 1

TS bigint.ts 2, U

GROUP 2

JS bigint.js output U

TYPESCRIPT-TUT

output

TS test.d.ts

JS test.js

TS undefine.d.ts

JS undefine.js

JS app.js

TS app.ts

TS bigint.ts 2, U

JS null.js

TS null.ts

JS number.js

TS number.ts

{ } package-lock.json

{ } package.json

JS string.js

OUTLINE

TIMELINE

TS bigint.ts 2, U X

```
1 var bignumber=9007199254740991n
2 var x=1;
3 var y=2
4
5 console.log(bignumber+x);
6 console.log(bignumber+y);
7
```

JS bigint.js U X

```
1 "use strict";
2 var bignumber = 9007199254740991n;
3 var x = 1;
4 var y = 2;
5 console.log(bignumber + x);
6 console.log(bignumber + y);
7
```

PROBLEMS 2 OUTPUT DEBUG CONSOLE TERMINAL PORTS GITLENS SPELL CHECKER

anilsidhu@anils-MacBook-Air typescript-tut % node output/bigint.js
9007199254740992

anilsidhu@anils-MacBook-Air typescript-tut % node output/bigint.js
9007199254740992
9007199254740992

anilsidhu@anils-MacBook-Air typescript-tut %

node

zsh

first-program* Launchpad 2 0

Ln 1, Col 14 Spaces: 4 UTF-8 LF { } TypeScript Live



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EXPLORER

OPEN EDITORS

GROUP 1

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GROUP 2

- JS bigint.js output U

TYPESCRIPT-TUT

- output
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 - TS undefine.d.ts
 - JS undefine.js
 - JS app.js
 - TS app.ts
 - TS bigint.ts U
 - JS null.js
 - TS null.ts
 - JS number.js
 - TS number.ts
 - { } package-lock.json
 - { } package.json
 - JS string.js
- OUTLINE
- TIMELINE

TS bigint.ts

```
1 console.log("bigint example");
2
```

JS bigint.js

```
1 "use strict";
2 console.log("bigint example");
3
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS GITLENS SPELL CHECKER

Error: Cannot find module '/Users/anilsidhu/Desktop/tutorials/typescript-tut/bigint.js'

```
at Function._resolveFilename (node:internal/modules/cjs/loader:1225:15)
at Function._load (node:internal/modules/cjs/loader:1055:27)
at TracingChannel.traceSync (node:diagnostics_channel:322:14)
at wrapModuleLoad (node:internal/modules/cjs/loader:220:24)
at Function.executeUserEntryPoint [as runMain] (node:internal/modules/run_main:170:5)
at node:internal/main/run_main_module:36:49 {
  code: 'MODULE_NOT_FOUND',
  requireStack: []
}
```

Node.js v22.14.0

```
anilsidhu@anils-MacBook-Air typescript-tut % node output/bigint.js
bigint example
anilsidhu@anils-MacBook-Air typescript-tut %
```

node

zsh

Ln 2, Col 1 Spaces: 4 UTF-8 LF {} TypeScript



typescript-tut

EXPLORER

OPEN EDITORS

GROUP 1

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GROUP 2

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TYPESCRIPT-TUT

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- TS null.ts
- JS number.js
- TS number.ts
- { } package-lock.json
- { } package.json
- JS string.js
- TS string.ts
- TS symbol.ts U
- TS test.ts
- tsconfig.json
- JS undefine.js
- TS undefine.ts
- <> user.html

OUTLINE

TIMELINE

TS symbol.ts U X

```
1 var sym= Symbol();
2 var sym2= Symbol();
3
4 console.log(sym==sym2);
5 console.log(sym);
6 console.log(sym2);
7
8
9
10
```

JS symbol.js U X

```
output > JS symbol.js > ...
1 "use strict";
2 var sym = Symbol();
3 var sym2 = Symbol();
4 console.log(sym == sym2);
5 console.log(sym);
6 console.log(sym2);
7
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS GITLENS SPELL CHECKER

node

bash

anils-MacBook-Air:typescript-tut anilsidhu\$ node output/symbol.js
false
anils-MacBook-Air:typescript-tut anilsidhu\$ node output/symbol.js
false
Symbol()
Symbol()
anils-MacBook-Air:typescript-tut anilsidhu\$

Ln 6, Col 17 Spaces: 4 UTF-8 LF { } TypeScript Go Live



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Symbol Data Type



- What is Symbol Data Type
- How to use it
- Example
- Where we can use it
- Interview Questions



first-program Launchpad 0 0 Go Live

Subscribe

typescript-tut

TS bigint.ts U X

TS bigint.ts > [⌘] y
1 var bignumber:bigint=9007199254740991n
2 var x=1n;
3 var y=2n
4
5 console.log(bignumber+x);
6 console.log(bignumber+y);
7

JS bigint.js U X

output > JS bigint.js > ...
1 "use strict";
2 var bignumber = 9007199254740991n;
3 var x = 1n;
4 var y = 2n;
5 console.log(bignumber + x);
6 console.log(bignumber + y);
7

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS GITLENS SPELL CHECKER

anilsidhu@anils-MacBook-Air typescript-tut % node output/bigint.js
9007199254740992
anilsidhu@anils-MacBook-Air typescript-tut % node output/bigint.js
9007199254740992
9007199254740992
anilsidhu@anils-MacBook-Air typescript-tut % node output/bigint.js
9007199254740991n
9007199254740993n
anilsidhu@anils-MacBook-Air typescript-tut %

node
zsh



first-program*

Launchpad

0 0

Ln 3, Col 9 Spaces: 4 UTF-8 LF {} TypeScript

Live

typescript-tut

EXPLORER

OPEN EDITORS

GROUP 1

- TS symbol.ts U
- ✕ tsconfig.json

GROUP 2

- JS symbol.js out... U

TYPESCRIPT-TUT

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- { package.json
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- TS string.ts
- TS symbol.ts U
- TS test.ts
- tsconfig.json
- JS undefine.js
- TS undefine.ts
- ↔ user.html

OUTLINE

TIMELINE

symbol.ts U

tsconfig.json

```
<{} compilerOptions > target
2  "compilerOptions": {
9    // "disableSourceOfProjectReferenceR
10   // "disableSolutionSearching": true,
11   // "disableReferencedProjectLoad": t
12
13   /* Language and Environment */
14   "target": "ES5",
15   // "lib": [],
16   // "jsx": "preserve",
17   // "libReplacement": true,
18   // "experimentalDecorators": true,
19   // "emitDecoratorMetadata": true,
```

JS symbol.js U

```
output > JS symbol.js > ...
1  "use strict";
2  var sym = Symbol();
3  var sym2 = Symbol();
4  var sym3 = Symbol('abc');
5  var sym4 = Symbol('abc');
6  // console.log(sym==sym2);
7  console.log(sym3);
8  console.log(sym4);
9  console.log(sym3 == sym4);
10
```

PROBLEMS 1 OUTPUT DEBUG CONSOLE TERMINAL PORTS GITLENS SPELL CHECKER 1

node

bash

anils-MacBook-Air:typescript-tut anilsidhu\$ node output/symbol.js
false
anils-MacBook-Air:typescript-tut anilsidhu\$ node output/symbol.js
false
Symbol()
Symbol()
anils-MacBook-Air:typescript-tut anilsidhu\$ node output/symbol.js
false
anils-MacBook-Air:typescript-tut anilsidhu\$ node output/symbol.js
Symbol(abc)
Symbol(abc)
false
anils-MacBook-Air:typescript-tut anilsidhu\$

first-program* Launchpad 0 0 1 Ln 14, Col 20 Spaces: 2 UTF-8 LF {} JSON with Comments Go Live



typescript-tut

EXPLORER

OPEN EDITORS

GROUP 1

- TS symbol.ts U

GROUP 2

- JS symbol.js out... U

TYPESCRIPT-TUT

- JS null.js
- TS null.ts
- JS number.js
- TS number.ts
- { } package-lock.json
- { } package.json
- JS string.js
- TS string.ts
- TS symbol.ts U
- TS test.ts
- tsconfig.json
- JS undefine.js
- TS undefine.ts
- <> user.html

OUTLINE

TIMELINE

TS symbol.ts U X

```
6
7
8 // console.log(sym==sym2);
9 // console.log(sym);
10 // console.log(sym2);
11
12 console.log(sym3==sym4)
13
14
15
16
```

JS symbol.js U X

```
output > JS symbol.js > ...
1 "use strict";
2 var sym = Symbol();
3 var sym2 = Symbol();
4 var sym3 = Symbol('abc');
5 var sym4 = Symbol('abc');
6 // console.log(sym==sym2);
7 // console.log(sym);
8 // console.log(sym2);
9 console.log(sym3 == sym4);
10
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS GITLENS SPELL CHECKER

node

bash

anils-MacBook-Air:typescript-tut anilsidhu\$ node output/symbol.js
false
anils-MacBook-Air:typescript-tut anilsidhu\$ node output/symbol.js
false
Symbol()
Symbol()
anils-MacBook-Air:typescript-tut anilsidhu\$ node output/symbol.js
false
anils-MacBook-Air:typescript-tut anilsidhu\$

first-program* Launchpad 0 0

Ln 12, Col 23 Spaces: 4 UTF-8 LF {} TypeScript Go Live



EXPLORED

OPEN EDITORS

GROUP 1

TS symbol.ts U

GROUP 2

JS symbol.js out... U

TYPESCRIPT-TUT

JS null.js

TS null.ts

JS number.js

TS number.ts

{ } package-lock.json

{ } package.json

JS string.js

TS string.ts

TS symbol.ts U

TS test.ts

tsconfig.json

JS undefine.js

TS undefine.ts

<> user.html

OUTLINE

TIMELINE

typescript-tut

TS symbol.ts U X

TS symbol.ts > ...

1 var sym= Symbol();

2 var sym2= Symbol();

3

4 var sym3=Symbol('abc');

5 var sym4=Symbol('abc');

6

7 console.log(sym==sym2);

8 console.log(sym);

9 console.log(sym2);

10

11

12

...

JS symbol.js U X

output > JS symbol.js > ...

1 "use strict";

2 var sym = Symbol();

3 var sym2 = Symbol();

4 console.log(sym == sym2);

5 console.log(sym);

6 console.log(sym2);

7

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS GITLENS SPELL CHECKER

anils-MacBook-Air:typescript-tut anilsidhu\$ node output/symbol.js

false

anils-MacBook-Air:typescript-tut anilsidhu\$ node output/symbol.js

false

Symbol()

Symbol()

anils-MacBook-Air:typescript-tut anilsidhu\$

node

bash



first-program*

Launchpad

0 0

Ln 5, Col 24

Spaces: 4

UTF-8

LF

{ } TypeScript

Go Live

Edit: Code

https://slides.com/anilsidhu/code-f82053/edit

TS

How to use TS for Input fields

first-program Launchpad 0 0

Port : 5500

Subscribe

typescript-tut

EXPLORER

OPEN EDITORS

GROUP 1

- TS symbol.ts U
- tsconfig.json

GROUP 2

- JS symbol.js out... U

TYPESCRIPT-TUT

- JS null.js
- TS null.ts
- JS number.js
- TS number.ts
- { } package-lock.json
- { } package.json
- JS string.js
- TS string.ts
- TS symbol.ts U
- TS test.ts
- tsconfig.json
- JS undefine.js
- TS undefine.ts
- <> user.html

OUTLINE

TIMELINE

TS symbol.ts > ...

```
12 // console.log(sym3==sym4)
13
14 const id= Symbol('id');
15
16 const obj= {
17     id:100,
18     name:'anil sidhu'
19 }
20 console.log(obj.id);
21
22
23
```

JS symbol.js > ...

```
1 "use strict";
2 var sym = Symbol();
3 var sym2 = Symbol();
4 var sym3 = Symbol('abc');
5 var sym4 = Symbol('abc');
6 // console.log(sym==sym2);
7 // console.log(sym3);
8 // console.log(sym4);
9 // console.log(sym3==sym4)
10 const id = Symbol('id');
11 const obj = {
12     id: 100,
```

PROBLEMS 1 OUTPUT DEBUG CONSOLE TERMINAL PORTS GITLENS SPELL CHECKER 1

node

bash

anils-MacBook-Air:typescript-tut anilsidhu\$ node output/symbol.js

false

Symbol()

Symbol()

anils-MacBook-Air:typescript-tut anilsidhu\$ node output/symbol.js

false

Symbol(abc)

Symbol(abc)

anils-MacBook-Air:typescript-tut anilsidhu\$ node output/symbol.js

false

Symbol(abc)

Symbol(abc)

anils-MacBook-Air:typescript-tut anilsidhu\$ node output/symbol.js

100

anils-MacBook-Air:typescript-tut anilsidhu\$ node output/symbol.js

first-program* Launchpad 0 0 1 Ln 9, Col 4 (68 selected) Spaces: 4 UTF-8 LF {} TypeScript Go Live



typescript-tut

EXPLORER

OPEN EDITORS

GROUP 1

- TS symbol.ts 4, U

GROUP 2

- tsconfig.json M

TYPESCRIPT-TUT

- JS null.js
- TS null.ts
- JS number.js
- TS number.ts
- package-lock.json
- package.json
- JS string.js
- TS string.ts
- TS symbol.ts 4, U
- TS test.ts
- tsconfig.json M
- JS undefine.js
- TS undefine.ts
- user.html

OUTLINE

TIMELINE

TS symbol.ts 4, U

```
1 var sym= Symbol();
2 var sym2= Symbol();
3
4 var sym3=Symbol('abc');
5 var sym4=
6
7
8 // console
9 console.l
10 console.l
11
12 console.log(sym3==sym4)
```

output > JS symbol.js > ...

```
1 "use strict";
2 var sym = Symbol();
3 var sym2 = Symbol();
4 var sym3 = Symbol('abc');
5 var sym4 = Symbol('abc');
6 console.log(sym==sym2);
7 console.log(sym3);
8 console.log(sym4);
9 console.log(sym3 == sym4);
```

'Symbol' only refers to a type, but is being used as a value here. Do you need to change your target library? Try changing the 'lib' compiler option to es2015 or later. ts(2585)

View Problem (⌘F8) No quick fixes available

PROBLEMS 5 OUTPUT DEBUG CONSOLE TERMINAL PORTS GITLENS SPELL CHECKER 1

node

bash

anils-MacBook-Air:typescript-tut anilsidhu\$ node output/symbol.js
false
anils-MacBook-Air:typescript-tut anilsidhu\$ node output/symbol.js
false
Symbol()
Symbol()
anils-MacBook-Air:typescript-tut anilsidhu\$ node output/symbol.js
false
anils-MacBook-Air:typescript-tut anilsidhu\$ node output/symbol.js
Symbol(abc)
Symbol(abc)
false
anils-MacBook-Air:typescript-tut anilsidhu\$

first-program* Launchpad 4 0 1 Ln 10, Col 19 Spaces: 4 UTF-8 LF {} TypeScript Go Live

Subscribe

typescript-tut

inputs.html U X TS inputs.ts U

JS inputs.js U X

inputs.html > html > head > script

```
1 <!DOCTYPE html>
2 <html lang="en">
3 <head>
4   <meta charset="UTF-8">
5   <meta name="viewport" content="width=device-width, initial-scale=1.0">
6   <title>Html Fields with TypeScript</title>
7   <script src="output/inputs.js"></script>
8 </head>
9 <body>
10  <input type="text" id="username" placeholder="enter user name">
11  <br>
12  <br>
13  <input type="text" id="email" placeholder="enter user email">
14  <br>
15  <br>
16  <input type="text" id="age" placeholder="enter user age">
17  <br>
```

output > JS inputs.js

```
1 "use strict";
2
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS GITLENS SPELL CHECKER

[4:51:50 PM] Starting compilation in watch mode...

[4:51:50 PM] Found 0 errors. Watching for file changes.

first-program*

Launchpad

0 0

Ln 7, Col 34 Spaces: 4 UTF-8 LF {} HTML Port : 5500

Subscribe

typescript-tut

inputs.html U TS inputs.ts U

inputs.html > html > head

```
1 <!DOCTYPE html>
2 <html lang="en">
3 <head>
4   <meta charset="UTF-8">
5   <meta name="viewport" content="width=device-width, initial-scale=1.0">
6   <title>Html Fields with TypeScript</title>
7 </head>
8 <body>
9   <input type="text" id="username" placeholder="enter user name">
10  <br>
11  <br>
12  <input type="text" id="email" placeholder="enter user email">
13  <br>
14  <br>
15  <input type="text" id="age" placeholder="enter user age">
16  <br>
17  <br>
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS GITLENS SPELL CHECKER

anilsidhu@anils-MacBook-Air typescript-tut % tsc --watch

Ln 7, Col 8 Spaces: 4 UTF-8 LF {} HTML Port : 5500

EXPLORER

OPEN EDITORS

inputs.html U TS inputs.ts U

TYPESCRIPT-TUT

node_modules

output

JS app.js

TS app.ts

TS bigint.ts

inputs.html U

TS inputs.ts U

JS null.js

TS null.ts

JS number.js

TS number.ts

package-lock.json

package.json

JS string.js

TS string.ts

TS symbol.ts

TS test.ts

OUTLINE

TIMELINE



first-program*

Launchpad

0 0

Ln 7, Col 8 Spaces: 4 UTF-8 LF {} HTML Port : 5500

Subscribe

typescript-tut

EXPLORER

OPEN EDITORS

- inputs.html U
- TS inputs.ts U

TYPESCRIPT-TUT

- node_modules
- output
- JS app.js
- TS app.ts
- TS bigint.ts
- inputs.html U
- TS inputs.ts U
- JS null.js
- TS null.ts
- JS number.js
- TS number.ts
- package-lock.json
- package.json
- JS string.js
- TS string.ts
- TS symbol.ts
- TS test.ts

OUTLINE

TIMELINE

inputs.html

```
1 <!DOCTYPE html>
2 <html lang="en">
3 <head>
4   <meta charset="UTF-8">
5   <meta name="viewport" content="width=device-width, initial-scale=1.0">
6   <title>Html Fields with TypeScript</title>
7 </head>
8 <body>
9   <input type="text" id="username" placeholder="enter user name">
10  <br>
11  <br>
12  <input type="text" id="email" placeholder="enter user email">
13  <br>
14  <br>
15  <input type="text" id="age" placeholder="enter user age">
16  <br>
17  <br>
18  <button onclick="getInfo()" >Get Info</button>
19
20 </body>
21 </html>
```

Ln 20, Col 8 Spaces: 4 UTF-8 LF {} HTML Port : 5500



Subscribe

typescript-tut

inputs.html U

TS inputs.ts U X

JS inputs.js U X

TS inputs.ts > getInfo > age

```
1 function getInfo() {
2   const nameInput= document.getElementById('username') as HTMLInput
3
4   const name:string=nameInput.value
5
6   const emailInput= document.getElementById('email') as HTMLInput
7
8   const email:string=emailInput.value
9
10  const ageInput= document.getElementById('age') as HTMLInputElement
11
12  const age:string=ageInput.value
13
14  console.log(name,email,age);
15
16
17 }
```

output > JS inputs.js > ...

```
1 "use strict";
2 function getInfo() {
3   const nameInput = document.getEle
4   const name = nameInput.value;
5   const emailInput = document.getEl
6   const email = emailInput.value;
7   const ageInput = document.getElem
8   const age = nameInput.value;
9   console.log(name, email, age);
10 }
11
```

PROBLEMS

OUTPUT

DEBUG CONSOLE

TERMINAL

PORTS

GITLENS

SPELL CHECKER

```
[4:57:04 PM] File change detected. Starting incremental compilation...
[4:57:04 PM] Found 0 errors. Watching for file changes.
```

first-program*

Launchpad

0 0

Ln 12, Col 30

Spaces: 4

UTF-8

LF

{ } TypeScript

Port : 5500



Subscribe

typescript-tut

inputs.html U

TS inputs.ts U X

JS inputs.js U X

TS inputs.ts > getInfo > ageInput

1 function getInfo() {
2 const nameInput= document.getElementById('username') as HTMLInput
3
4 const name:string=nameInput.value
5
6 const emailInput= document.getElementById('email') as HTMLInput
7
8 const email:string=nameInput.value
9
10 const ageInput= document.getElementById('age') as HTMLInputElem
11
12 const age:string=nameInput.value
13
14
15 }

output > JS inputs.js > ...
1 "use strict";
2 function getInfo() {
3 const nameInput = document.getEle
4 const name = nameInput.value;
5 const emailInput = document.getEl
6 const email = nameInput.value;
7 const ageInput = document.getElem
8 const age = nameInput.value;
9 }
10

PROBLEMS

OUTPUT

DEBUG CONSOLE

TERMINAL

PORTS

GITLENS

SPELL CHECKER

[4:56:33 PM] File change detected. Starting incremental compilation...
[4:56:33 PM] Found 0 errors. Watching for file changes.
[]

first-program* Launchpad 0 0 Ln 13, Col 5 Spaces: 4 UTF-8 LF {} TypeScript Port : 5500



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typescript-tut

inputs.html U

TS inputs.ts U X

JS inputs.js U X

TS inputs.ts > getInfo

1 function getInfo(){
2 console.log("getInfo function called");
3
4 }

output > JS inputs.js > ...

1 "use strict";
2 function getInfo() {
3 console.log("getInfo fun
4 }
5

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS GITLENS SPELL CHECKER

[4:52:46 PM] File change detected. Starting incremental compilation...

[4:52:46 PM] Found 0 errors. Watching for file changes.

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Launchpad

0 0

Ln 2, Col 41 Spaces: 4 UTF-8 LF {} TypeScript Port : 5500



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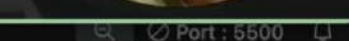




Array Data Type

- Collection of same Data type.
- Pre-defined data type.

```
1 let numbers: number[] = [1, 2, 3, 4, 5];
2 let names: Array<string> = ["Alice", "Bob", "Charlie"];
```



first-program

Launchpad

0 0 0

Port : 5500

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Array Data Type



- What is Array Data Type.
- How to use it.
- Example.
- Where we can use it.
- Interview Questions.



first-program Launchpad 0 0 Port : 5500

Subscribe

typescript-tut

inputs.html U

TS inputs.ts U X

JS inputs.js U X

TS inputs.ts > getInfo > age

```
1 function getInfo() {
2   const nameInput= document.getElementById('username') as HTMLInput
3
4   const name:string=nameInput.value
5
6   const emailInput= document.getElementById('email') as HTMLInput
7
8   const email:string=emailInput.value
9
10  const ageInput= document.getElementById('age') as HTMLInputElem
11
12  const age:string=ageInput.value
13
14  console.log(name,email,age);
15
16
17 }
```

PROBLEMS

OUTPUT

DEBUG CONSOLE

TERMINAL

PORTS

GITLENS

SPELL CHECKER

[4:57:07 PM] File change detected. Starting incremental compilation...

[4:57:07 PM] Found 0 errors. Watching for file changes.

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Launchpad

0 0

Ln 12, Col 30

Spaces: 4

UTF-8

LF

{ } TypeScript

Port : 5500

output > JS inputs.js > ...

```
1 "use strict";
2 function getInfo() {
3   const nameInput = document.getEle
4   const name = nameInput.value;
5   const emailInput = document.getEl
6   const email = emailInput.value;
7   const ageInput = document.getElem
8   const age = ageInput.value;
9   console.log(name, email, age);
10 }
11
```



Subscribe

typescript-tut

TS array.ts 1, U

JS array.js U x

TS array.ts > ...

output > JS array.js > ...

```
1 var users:string=['anil','sam','peter'];
2 var marks:number=[50,60,92,36];
3 var students:Array<string>=['bruce','arya']
4 students.push('sidhu')
5 marks.push(100)
6
7 var collegeName:ReadonlyArray<string>=['IET alwar','iit delhi']
8 collegeName.push()
9 console.log(students);
10
```

```
1 "use strict";
2 var users = ['anil', 'sam', 'peter'];
3 var marks = [50, 60, 92, 36];
4 var students = ['bruce', 'arya'];
5 students.push('sidhu');
6 marks.push(100);
7 var collegeName = ['IET alwar', 'iit delhi'];
8 console.log(students);
9
```

PROBLEMS 5 OUTPUT DEBUG CONSOLE TERMINAL PORTS GITLENS SPELL CHECKER 4

anils-MacBook-Air:typescript-tut anilsidhu\$ node output/array.js

['bruce', 'arya', 'sidhu']

anils-MacBook-Air:typescript-tut anilsidhu\$

first-program* Launchpad 1 0 4

Ln 8, Col 1 (18 selected) Spaces: 4 UTF-8 LF {} TypeScript Port : 5500



Subscribe

typescript-tut

TS array.ts 1, U

JS array.js U x

TS array.ts > ...

output > JS array.js > ...

```
1 var users:string=['anil','sam','peter'];
2 var marks:number=[50,60,92,36];
3 var students:Array<string>=['bruce','arya']
4 students.push('sidhu');
5 marks.push(100);
6
7 var collegeName=['IET alwar','iit delhi'];
8 collegeName.push('iit delhi');
9 console.log(students);
10
```

```
1 "use strict";
2 var users = ['anil', 'sam', 'peter'];
3 var marks = [50, 60, 92, 36];
4 var students = ['bruce', 'arya'];
5 students.push('sidhu');
6 marks.push(100);
7 var collegeName = ['IET alwar', 'iit delhi'];
8 console.log(students);
9
```

PROBLEMS 5 OUTPUT DEBUG CONSOLE TERMINAL PORTS GITLENS SPELL CHECKER 4

```
• anils-MacBook-Air:typescript-tut anilsidhu$ node output/array.js
[ 'bruce', 'arya', 'sidhu' ]
○ anils-MacBook-Air:typescript-tut anilsidhu$
```

first-program*

Launchpad

1 0 4

Ln 8, Col 18 Spaces: 4 UTF-8 LF {} TypeScript Port : 5500



Subscribe

typescript-tut

TS array.ts U X

JS array.js U X

TS array.ts > ...

output > JS array.js > ...

```
1 var users:string=['anil','sam','peter'];
2 var marks:number=[50,60,92,36];
3 var students:Array<string>=['bruce','arya']
4 students.push('sidhu')
5 marks.push(100)
6 console.log(students);
7
```

```
1 "use strict";
2 var users = ['anil', 'sam', 'peter'];
3 var marks = [50, 60, 92, 36];
4 var students = ['bruce', 'arya'];
5 students.push('sidhu');
6 marks.push(100);
7 console.log(students);
8
```

PROBLEMS 2 OUTPUT DEBUG CONSOLE TERMINAL PORTS GITLENS SPELL CHECKER 2

anils-MacBook-Air:typescript-tut anilsidhu\$ node output/array.js

['bruce', 'arya', 'sidhu']

anils-MacBook-Air:typescript-tut anilsidhu\$

first-program*

Launchpad

0 0 2

Ln 5, Col 16 Spaces: 4 UTF-8 LF {} TypeScript Port : 5500



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https://slides.com/anilsidhu/code-f82053/edit

- A **tuple** is a fixed-length,
- Ordered collection of **different types**.
- Each element in a tuple has a specific type.

```
1 let user: [number, string, boolean] = [1, "Anil Sidhu", true];
```



first-program Launchpad 0 25 0 Go Live

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5 Edit: Code x TypeScript Tutorial in Hindi 2/ x +

https://slides.com/anilsidhu/code-f82053/edit

- A **tuple** is a fixed-length,
- Ordered collection of **different types**.
- Each element in a tuple has a specific type.

```
1 let user: [number, string, boolean] = [1, "Anil Sidhu", true];
```

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Edit: Code

TypeScript Tutorial in Hindi 2

https://slides.com/anilsidhu/code-f82053/edit





Tuple Data Type

- What is Tuple Data Type.
- Tuple VS Array
- Example.
- Where we can use it.
- Interview Questions.



first-program

Launchpad

0 0 0

Go Live

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typescript-tut

TS tuple.ts U x

JS tuple.js U x

TS tuple.ts > ...

output > JS tuple.js > ...

```
1 var emp:string=['anil','sam','peter'];
2 var empData:[string,number,boolean,number]=['anil',30,true,10
3
4 empData.push('noida')
5
```

```
1 "use strict";
2 var emp = ['anil', 'sam', 'peter'];
3 var empData = ['anil', 30, true, 100000,];
4 empData.push('noida');
5
```

PROBLEMS 2 OUTPUT DEBUG CONSOLE TERMINAL PORTS GITLENS SPELL CHECKER 2

[12:11:28 PM] File change detected. Starting incremental compilation...

[12:11:28 PM] Found 0 errors. Watching for file changes.

first-program* Launchpad 0 0 2

Ln 4, Col 22 Spaces: 4 UTF-8 LF TypeScript Go Live

Subscribe

typescript-tut

TS tuple.ts U X

JS tuple.js U X

TS tuple.ts > [⌘] empData

output > JS tuple.js > [⌘] empData

```
1  i:string[]=['anil','sam','peter'];
2  iData:[string,number,boolean,number]=['anil',30,true,100000];
3
4
5
```

```
1  "use strict";
2  var emp = ['anil', 'sam', 'peter'];
3  var empData = ['anil', 30, true, 100000];
4  
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS GITLENS SPELL CHECKER

[12:09:34 PM] File change detected. Starting incremental compilation...

[12:09:34 PM] Found 0 errors. Watching for file changes.

first-program*

Launchpad

0 0

Ln 2, Col 42 Spaces: 4 UTF-8 LF () TypeScript Go Live

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TypeScript Tutorial In Hindi 2

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Feature	Array	Tuple
Type	Homogeneous (same type elements)	Heterogeneous (different type elements)
Length	Dynamic (can grow/shrink)	Fixed (predefined number of elements)
Order	Not enforced	Order is enforced (specific type at each index)

first-program

Launchpad

0 0 0

Go Live

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typescript-tut

TS object.ts U X

TS object.ts > [?] userData

```
1 var userData:{
2   name:string,
3   age:number,
4   company:string
5 }
6   name:'Anil sidhu',
7   age:30,
8   company:'Oracle'
9 }
```

JS object.js U X

output > JS object.js > ...

```
1 "use strict";
2 var userData = {
3   name: 'Anil sidhu',
4   age: 30,
5   company: 'Oracle'
6 };
7 
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS GITLENS SPELL CHECKER

[1:37:33 PM] File change detected. Starting incremental compilation...

[1:37:33 PM] Found 0 errors. Watching for file changes.

first-program* Launchpad 0 0

Ln 9, Col 2 Spaces: 4 UTF-8 LF {} TypeScript Go Live



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Object Data Type



- What is Object.
- Define data type in object
- Nested object
- Example.
- Interview Questions.



first-program Launchpad 0 0 0 Go Live

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typescript-tut

TS tuple.ts 1, U X

JS tuple.js U X

TS tuple.ts > empData

output > JS tuple.js > ...

1 var emp:string[]=['anil','sam','peter'];

2 var empData: readonly[string,number,boolean,number,string]=['

3

4 empData.push('noida');

5

6 console.log(empData);

7

8

1 "use strict";

2 var emp = ['anil', 'sam', 'peter'];

3 var empData = ['anil', 30, true, 100000, ''];

4 empData.push('noida');

5 console.log(empData);

6

PROBLEMS 3 OUTPUT DEBUG CONSOLE TERMINAL PORTS GITLENS SPELL CHECKER 2

● anils-MacBook-Air:typescript-tut anilsidhu\$ node output/tuple.js

['anil', 30, true, 100000, 'noida']

○ anils-MacBook-Air:typescript-tut anilsidhu\$

first-program* Launchpad 1 0 2

Ln 2, Col 22 Spaces: 4 UTF-8 LF TypeScript Go Live

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typescript-tut

TS object.ts 1, U

JS object.js U

TS object.ts > [?] userData

output > JS object.js > ...

```
26 var userData:{
27   name:string,
28   age:number,
29   company:string,
30   address:{
31     houseNo:string
32     sector:string
33     city:string
34   }
35 }={
36   name:'Anil sidhu',
37   age:30,
38   company:'Oracle',
39   address:{
40     houseNo:'house no 60',
41     sector:'70-a',
42     city:'gurgaon'
43   }
44 }
```

```
1 "use strict";
2 // var userData:{
3 //   name:string,
4 //   age:number,
5 //   company:string
6 // }={
7 //   name:'Anil sidhu',
8 //   age:30,
9 //   company:'Oracle'
10 // }
11 // userData.name="Sam";
12 // userData.age=25;
13 // console.log(userData);
14 // var userData:{
15 //   [key:string]:string|number| undefined
16 // }={
17 //   name:'Anil sidhu',
```

PROBLEMS 4

OUTPUT

DEBUG CONSOLE

TERMINAL

PORTS

GITLENS

SPELL CHECKER 4

- anils-MacBook-Air:typescript-tut anilsidhu\$ node output/object.js
- anils-MacBook-Air:typescript-tut anilsidhu\$ node output/object.js
- anils-MacBook-Air:typescript-tut anilsidhu\$ node output/object.js
- anils-MacBook-Air:typescript-tut anilsidhu\$ node output/object.js
- anils-MacBook-Air:typescript-tut anilsidhu\$

Ln 33, Col 20 Spaces: 4 UTF-8 LF {} TypeScript Go Live

code

sh

Subscribe

typescript-tut

TS object.ts U X

TS object.ts > ...

```
12
13 // console.log(userData);
14
15 var userData:{
16     [key:string]:string|number| undefined
17 }={
18     name:'Anil sidhu',
19     age:30,
20     company:undefined
21 }
22 userData.company="HCL";
23 userData.city="Noida"
24 console.log(userData);
```

JS object.js U X

output > JS object.js > ...

```
1 "use strict";
2 // var userData:{
3 //     name:string,
4 //     age:number,
5 //     company:string
6 // }={
7 //     name:'Anil sidhu',
8 //     age:30,
9 //     company:'Oracle'
10 // }
11 // userData.name="Sam";
12 // userData.age=25;
13 // console.log(userData);
14 var userData = {
15     name: 'Anil sidhu',
16     age: 30,
17     company: undefined
```

PROBLEMS 2 OUTPUT DEBUG CONSOLE TERMINAL PORTS GITLENS SPELL CHECKER 2

```
● anils-MacBook-Air:typescript-tut anilsidhu$ node output/object.js
{ name: 'Sam', age: 30, company: 'Oracle' }
● anils-MacBook-Air:typescript-tut anilsidhu$ node output/object.js
{ name: 'Anil sidhu', age: 30, company: 'Oracle' }
● anils-MacBook-Air:typescript-tut anilsidhu$ node output/object.js
{ name: 'Sam', age: 30, company: 'Oracle' }
● anils-MacBook-Air:typescript-tut anilsidhu$ node output/object.js
{ name: 'Anil sidhu', age: 30, company: 'HCL' }
○ anils-MacBook-Air:typescript-tut anilsidhu$
```

first-program* Launchpad 0 0 1 2

Ln 22, Col 14 Spaces: 4 UTF-8 LF {} TypeScript Go Live



code
sh

Subscribe

typescript-tut

TS object.ts U X

JS object.js U X

TS object.ts > ...

output > JS object.js > ...

```
11 // userData.age=25;
12
13 // console.log(userData);
14
15 var userData:{
16   name:string,
17   age:number,
18   company:string|undefined
19 }={
20   name:'Anil sidhu',
21   age:30,
22   company:undefined
23 }
24 userData.company="HCL"
25
```

```
1 "use strict";
2 // var userData:{
3 //   name:string,
4 //   age:number,
5 //   company:string
6 // }={
7 //   name:'Anil sidhu',
8 //   age:30,
9 //   company:'Oracle'
10 // }
11 // userData.name="Sam";
12 // userData.age=25;
13 // console.log(userData);
14 var userData = {
15   name: 'Anil sidhu',
16   age: 30,
17   company: undefined

```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS GITLENS SPELL CHECKER

To update your account to use zsh, please run `chsh -s /bin/zsh`.
For more details, please visit <https://support.apple.com/kb/HT208050>.

- anils-MacBook-Air:typescript-tut anilsidhu\$ node output/object.js
{ name: 'Sam', age: 30, company: 'Oracle' }
- anils-MacBook-Air:typescript-tut anilsidhu\$ node output/object.js
{ name: 'Anil sidhu', age: 30, company: 'Oracle' }
- anils-MacBook-Air:typescript-tut anilsidhu\$ node output/object.js
{ name: 'Sam', age: 30, company: 'Oracle' }
- anils-MacBook-Air:typescript-tut anilsidhu\$

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Launchpad

0 0 0

Ln 24, Col 23

Spaces: 4

UTF-8

LF

{ } TypeScript

Go Live

node

sh

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TS

Any Data type

- Which allows a variable to hold values of any type.
- If you are not sure about data type of variable you can use any

first-program Launchpad 0 0 0

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Any and Unknown

- What is any data type.
- When to use any.
- Example
- What is unknown data type
- Example of unknown.
- Interview Questions.



first-program Launchpad 0 0 0 0 Live

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typescript-tut

TS object.ts U X

TS object.ts > [?] userData

```
26 var userData:{
35 }={
36   name:'Anil sidhu',
37   age:30,
38   company:'Oracle',
39   address:{
40     houseNo:'house no 60',
41     sector:'70-a',
42     city:'gurgaon'
43   }
44 }
```

JS object.js U X

output > JS object.js > ...

```
1 "use strict";
2 // var userData:{
3 //   name:string,
4 //   age:number,
5 //   company:string
6 // }={
7 //   name:'Anil sidhu',
8 //   age:30,
9 //   company:'Oracle'
10 // }
11 // userData.name="Sam";
12 // userData.age=25;
13 // console.log(userData);
14 // var userData:{
15 //   [key:string]:string|number| undefined
16 // }={
17 //   name:'Anil sidhu',
```

PROBLEMS 4 OUTPUT DEBUG CONSOLE TERMINAL PORTS GITLENS SPELL CHECKER 4

anils-MacBook-Air:typescript-tut anilsidhu\$ node output/object.js
{ name: 'Sam', age: 30, company: 'Oracle' }
anils-MacBook-Air:typescript-tut anilsidhu\$ node output/object.js
{ name: 'Anil sidhu', age: 30, company: 'Oracle' }
anils-MacBook-Air:typescript-tut anilsidhu\$ node output/object.js
{ name: 'Sam', age: 30, company: 'Oracle' }
anils-MacBook-Air:typescript-tut anilsidhu\$ node output/object.js
{ name: 'Anil sidhu', age: 30, company: 'HCL' }
anils-MacBook-Air:typescript-tut anilsidhu\$

Google Chrome

Ln 33, Col 20 Spaces: 4 UTF-8 LF {} TypeScript Go Live

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Unknown Data type

- The unknown type in TypeScript is similar to any, but it's safer because it forces type checking before performing operations on the value.



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Launchpad

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typescript-tut

TS any_unknown.ts U X

TS any_unknown.ts > ...
1 var value:any ="Anil";
2
3 value=100
4
5 value=['anil'];
6
7 value=true; I
8 value={}

JS any_unknown.js U X

output > JS any_unknown.js > ...
1 "use strict";
2 var value = "Anil";
3 value = 100;
4 value = ['anil'];
5 value = true;
6 value = {};
7

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS GITLENS SPELL CHECKER

[10:59:23 PM] File change detected. Starting incremental compilation...

[10:59:23 PM] Found 0 errors. Watching for file changes.

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Ln 8, Col 9 Spaces: 4 UTF-8 LF {} TypeScript Go Live



nod

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Any Data Type Use

- When migrating JavaScript code to TypeScript.
- When dealing with dynamic values from APIs.
- When working with third-party libraries that lack type definitions.



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typescript-tut

TS function.ts U X

function fruits(): string

```
1 function fruits() {
2   return "Apple"
3 }
```

JS function.js U X

output > JS function.js > ...

```
1 "use strict";
2 function fruits() {
3   return "Apple";
4 }
5
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS GITLENS SPELL CHECKER

```
[2:00:11 AM] File change detected. Starting incremental compilation...
[2:00:11 AM] Found 0 errors. Watching for file changes.
```

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Ln 2, Col 19 Spaces: 4 UTF-8 LF TypeScript Port: 5500



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Return Type in Function



typescript-tut

TS any_unknown.ts U X

TS any_unknown.ts > ...

```
9
10 var value1:unknown ="anil";
11 value1=100;
12
13 value1=['anil'];
14 value1={};
15 value1="Anil sidhu"
16
17 if(typeof value1=='string'){
18
19 console.log(value1.toUpperCase());
20 }
```

JS any_unknown.js U X

output > JS any_unknown.js > ...

```
1 "use strict";
2 var value = "Anil";
3 value = 100;
4 value = ['anil'];
5 value = true;
6 value = {};
7 var value1 = "anil";
8 value1 = 100;
9 value1 = ['anil'];
10 value1 = {};
11 value1 = "Anil sidhu";
12 if (typeof value1 == 'string') {
13     console.log(value1.toUpperCase());
14 }
15
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS GITLENS SPELL CHECKER

[11:02:57 PM] File change detected. Starting incremental compilation...

[11:02:57 PM] Found 0 errors. Watching for file changes.

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Ln 14, Col 2 Spaces: 4 UTF-8 LF () JavaScript Go Live



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Never in TypeScript



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Launchpad

Port: 8500

← → typescript-tut

TS function.ts U X

TS function.ts > complex

```
5 function simple():boolean{
6     return true
7 }
8
9 function complex():number|string|boolean{
10     let data=10;
11     let name="anil sidhu"
12     let type="age";
13
14     if(type=='age'){
15         return data;
16     }else{
17         return name
18     }
19 }
```

JS function.js U X

output > JS function.js > ...

```
1 "use strict";
2 function fruits() {
3     return 10;
4 }
5 function simple() {
6     return true;
7 }
8 function complex() {
9     let data = 10;
10    let name = "anil sidhu";
11    let type = "age";
12    if (type == 'age') {
13        return data;
14    }
15    else {
16        return name;
17    }
18 }
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS GITLENS SPELL CHECKER

[2:03:51 AM] File change detected. Starting incremental compilation...

[2:03:51 AM] Found 0 errors. Watching for file changes.

first-program* Launchpad 0 0 0

Ln 19, Col 2 Spaces: 4 UTF-8 LF TypeScript Port: 5500



typescript-tut

TS function.ts U

TS function.ts > simple

```
1 function fruits():number{
2     return 10
3 }
4
5 function simple():boolean{
6     return true
7 }
8
9 |
```

JS function.js U

output > JS function.js > ...

```
1 "use strict";
2 function fruits() {
3     return 10;
4 }
5 function simple() {
6     return true;
7 }
8
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS GITLENS SPELL CHECKER

[2:01:55 AM] File change detected. Starting incremental compilation...

[2:01:55 AM] Found 0 errors. Watching for file changes.

first-program* Launchpad 0 0

Ln 9, Col 1 Spaces: 4 UTF-8 LF TypeScript Port: 5500



EXPLORER

OPEN EDITORS

GROUP 1

TS never.ts U

GROUP 2

JS never.js output U

TYPESCRIPT-TUT

JS app.js

TS app.ts

TS array.ts

TS bigint.ts

TS function.ts U

inputs.html

TS inputs.ts

TS never.ts U

JS null.js

TS null.ts

JS number.js

TS number.ts

TS object.ts

package-lock.json

package.json

OUTLINE

TIMELINE

TS never.ts U X

TS never.ts > simple1

1 function loopFunction():never{

2

3 while (true) {

4 console.log("loop")

5 }

6 }

7 }

8

9 function simple1():never {

10 throw new Error("output and data no

11 }

12 }

13 }

JS never.js U X

output > JS never.js > ...

1 "use strict";

2 function loopFunction() {

3 while (true) {

4 console.log("loop");

5 }

6 }

7 function simple1() {

8 throw new Error("output and data no

9 }

10 }

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS GITLENS

[12:29:16 PM] File change detected. Starting incremental compilation...

[12:29:17 PM] Found 0 errors. Watching for file changes.

first-program* Launchpad 0 0

Ln 12, Col 3 Spaces: 4 UTF-8 LF TypeScript Port : 5500



typescript-tut

EXPLORER

OPEN EDITORS

GROUP 1

- TS never.ts 1, U

GROUP 2

- JS never.js output U

TYPESCRIPT-TUT

- JS app.js
- TS app.ts
- TS array.ts
- TS bigint.ts
- TS function.ts U
- inputs.html
- TS inputs.ts
- TS never.ts 1, U
- JS null.js
- TS null.ts
- JS number.js
- TS number.ts
- TS object.ts
- package-lock.json
- package.json
- OUTLINE
- TIMELINE

TS never.ts 1, U X

```
1 function loopFunction():never{
2
3     while (true) {
4         console.log("loop")
5     }
6
7 }
8
9 function simple1():never {
10     console.log("simple1")
11 }
```

JS never.js U X

```
output > JS never.js > ...
1 "use strict";
2 function loopFunction() {
3     while (true) {
4         console.log("loop");
5     }
6 }
7 function simple1() {
8     console.log("simple1");
9 }
10
```

PROBLEMS 1 OUTPUT DEBUG CONSOLE TERMINAL PORTS GITLENS ...

never.ts:9:21 - error TS2534: A function returning 'never' cannot have a reachable

```
function simple1():never {
```

[12:28:25 PM] Found 1 error. Watching for file changes.

first-program* Launchpad 1 0 Ln 9, Col 26 (5 selected) Spaces: 4 UTF-8 LF () TypeScript Port: 5500



typescript-tut

EXPLORER

OPEN EDITORS

TYPESCRIPT-TUT

- inputs.html
- TS inputs.ts
- TS never.ts
- JS null.js
- TS null.ts
- JS number.js
- TS number.ts
- TS object.ts
- { } package-lock.json
- { } package.json
- JS string.js
- TS string.ts
- TS symbol.ts
- TS test.ts
- tsconfig.json
- TS tuple.ts
- JS undefine.js
- TS undefine.ts
- user.html

OUTLINE

TIMELINE

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS GITLENS SPELL CHECKER

```
/dev/fd/13:18: command not found: compdef
anilsidhu@anils-MacBook-Air typescript-tut % touch never.ts
anilsidhu@anils-MacBook-Air typescript-tut % tsc --watch
```

first-program* Launchpad 0 0 0

Port : 5500

typescript-tut

TS union.ts U X

TS union.ts > fruitsData

```
2
3 studentData = 9999;
4
5 studentData = []
6
7 function fruitsData(): String | string[] | number {
8     var item = 1
9     if (item > 1) {
10         return ['apple', 'banana']
11     } else {
12         return "Apple"
13     }
14 }
15
16
17 console.log(fruitsData())
```

JS union.js U X

output > JS union.js > ...

```
1 "use strict";
2 var studentData = "Anil";
3 studentData = 9999;
4 studentData = [];
5 function fruitsData() {
6     var item = 1;
7     if (item > 1) {
8         return ['apple', 'banana'];
9     }
10    else {
11        return "Apple";
12    }
13 }
14 console.log(fruitsData());
15
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS GITLENS SPELL CHECKER

[6:37:40 PM] File change detected. Starting incremental compilation...

[6:37:40 PM] Found 0 errors. Watching for file changes.

first-program* Launchpad 0 0

Ln 7, Col 45 (21 selected) Spaces: 4 UTF-8 LF () TypeScript Port: 5500

typescript-tut

TS union.ts 1, U

TS union.ts > fruitsData

```
1 var studentData:string| number | []="Anil";
2
3 studentData=9999;
4
5 studentData=[]
6
7 function fruitsData():string {
8   var item=1
9   if(item>1){
10     return ['apple','banana']
11   }else
12     return "A"
13   }
14
15 console.log(fruitsData());
```

PROBLEMS 1

union.ts:10:9 - error TS2322: Type 'string[]' is not assignable to type 'string'.

10 return ['apple','banana']

JS union.js U X

output > JS union.js > ...

```
1 "use strict";
2 var studentData = "Anil";
3 studentData = 9999;
4 studentData = [];
5 function fruitsData() {
6   var item = 1;
7   if (item > 1) {
8     return ['apple', 'banana'];
9   }
10   return "Anil";
11 }
12 console.log(fruitsData());
```

[6:37:07 PM] File change detected. Starting incremental compilation...

union.ts:10:9 - error TS2322: Type 'string[]' is not assignable to type 'string'.

10 return ['apple','banana']

[6:37:07 PM] Found 1 error. Matching for file changes.

first-program*

Launchpad

1 0

Ln 11, Col 9 Spaces: 4 UTF-8 LF () TypeScript Port: 5500



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Union Types

- What is Union Types.
- Use union type with variable data type.
- Use union type with function return type.
- Use union type with function params type.
- Type check with Union type





first-program

Launched

0 0 0

Port: 5500

typescript-tut

TS interface.ts U

```
1 interface Info {
2   name:string,
3   age:number,
4   college:string
5 }
6
7 var studentObj:Info = {
8   name:'anil',
9   age:30,
10  college:'IET alwar'
11 }
12
```

JS interface.js U X

```
1 "use strict";
2 var studentObj = {
3   name: 'anil',
4   age: 30,
5   college: 'IET alwar'
6 };
7
```

PROBLEMS 2 OUTPUT DEBUG CONSOLE TERMINAL PORTS GITLENS SPELL CHECKER 2

```
[3:29:52 AM] File change detected. Starting incremental compilation...
[3:29:52 AM] Found 0 errors. Watching for file changes.
```

first-program* Launchpad 0 0 2

Ln 12, Col 1 Spaces: 4 UTF-8 LF TypeScript Port: 5500



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Interface

- An **interface** is a way to define the structure of an object, function etc.





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Address bar: <https://slides.com/anilsidhu/code-f82053/edit>



Interface

- What is Interface.
- Define Interface
- How to use Interface
- Extend Interface



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Intersection Types in TypeScript

- What is Intersection Types.
- How to use it with type
- How to use it with interface
- Interview Questions



typescript-tut

TS interface.ts U X

TS interface.ts > TeacherType > subject

```
6
7 interface TeacherType extends Info {
8   subject:string
9 }
10
11 var studentObj:Info={
12   name:'anil',
13   age:30,
14   college:'IET alwar'
15 }
16 var teacherObj:TeacherType={
17   name:'Sam',
18   age:40,
19   college:'IET alwar',
20   subject:'Math'
21 }
22
```

JS interface.js U X

output > JS interface.js > ...

```
1 "use strict";
2 var studentObj = {
3   name: 'anil',
4   age: 30,
5   college: 'IET alwar'
6 };
7 var teacherObj = {
8   name: 'Sam',
9   age: 40,
10  college: 'IET alwar',
11  subject: 'Math'
12 };
13 var management = {
14   name: 'Sam',
15   age: 40,
16   college: 'IET alwar'
17 };

```

PROBLEMS 6

OUTPUT

DEBUG CONSOLE

TERMINAL

PORTS

GITLENS

SPELL CHECKER 6

```
[3:32:08 AM] File change detected. Starting incremental compilation...
[3:32:08 AM] Found 0 errors. Watching for file changes.
```

first-program*

Launchpad

0 0 6

Ln 8, Col 12 (7 selected)

Spaces: 4

UTF-8

LF

() TypeScript

Port: 5500



typescript-tut

TS interface.ts U X

TS interface.ts > [0] management

```
7 var studentObj:Info={
8   name:'anil',
9   age:30,
10  college:'IET alwar'
11 }
12 var teacherObj:Info={
13   name:'Sam',
14   age:40,
15   college:'IET alwar'
16 }
17
18 var management:Info={
19   name:'Sam',
20   age:40,
21   college:'IET alwar'
22 }
```

JS interface.js U X

output > JS interface.js > ...

```
1 "use strict";
2 var studentObj = {
3   name: 'anil',
4   age: 30,
5   college: 'IET alwar'
6 };
7 var teacherObj = {
8   name: 'Sam',
9   age: 40,
10  college: 'IET alwar'
11 };
12 var management = {
13   name: 'Sam',
14   age: 40,
15   college: 'IET alwar'
16 };
17
```

PROBLEMS 0

OUTPUT

DEBUG CONSOLE

TERMINAL

PORTS

GIT LENS

SPELL CHECKER 0

[3:30:52 AM] File change detected. Starting incremental compilation...

[3:30:52 AM] Found 0 errors. Watching for file changes.

first-program*

Launchpad

0 0 0 6

Ln 22, Col 2

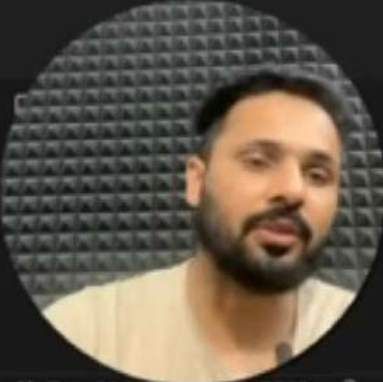
Spaces: 4

UTF-8

LF

() TypeScript

Port: 5500



typescript-tut

TS intersection.ts U X JS intersection.js U X

```
TS intersection.ts > PersonDataC
1 // type personTA={name:string}
2 // type personTB={age:number}
3 // type personTC=personTA & personTB
4
5 interface personTA {name:string}
6 interface personTB {age:number}
7 type personTC = personTA & personTB
8
9
10 var PersonDataA:personTA={name:'Anil sidhu'}
11 var PersonDataB:personTB={age:30}
12
13
14 var PersonDataC : personTC=
15 {name:'Sidhu',age:30}
```

```
output > JS intersection.js > ...
1 "use strict";
2 // type personTA={name:string}
3 // type personTB={age:number}
4 // type personTC=personTA & personTB
5 var PersonDataA = { name: 'Anil sidhu' };
6 var PersonDataB = { age: 30 };
7 var PersonDataC = { name: 'Sidhu', age: 30 };
8
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS GITLENS SPELL CHECKER

```
[2:33:48 AM] File change detected. Starting incremental compilation...
[2:33:48 AM] Found 0 errors. Watching for file changes.
```

first-program* Launchpad 0 0

Ln 15, Col 22 Spaces: 4 UTF-8 LF () TypeScript Port : 5500

typescript-tut

TS intersection.ts U

JS intersection.js U

TS intersection.ts > ...

output > JS intersection.js > ...

```
1 type personTA={name:string}
2 type personTB={age:number}
3 type personTC=personTA & personTB
4
5
6
7 var PersonDataA:personTA={name:'Anil sidhu'}
8 var PersonDataB:personTB={age:30}
9
10
11 var PersonDataC : personTC=
12   {name:'Sidhu',age:30}
```

```
1 "use strict";
2 var PersonDataA = { name: 'Anil sidhu' };
3 var PersonDataB = { age: 30 };
4 var PersonDataC = { name: 'Sidhu', age: 30 };
5
```

PROBLEMS

OUTPUT

DEBUG CONSOLE

TERMINAL

PORTS

GITLENS

SPELL CHECKER

[2:30:44 AM] File change detected. Starting incremental compilation...

[2:30:44 AM] Found 0 errors. Watching for file changes.

first-program*

Launchpad

0 0

Ln 4, Col 1

Spaces: 4

UTF-8

LF

() TypeScript

Port : 5500



Edit: Code

TypeScript Tutorial in Hindi 2

<https://slides.com/anilsidhu/code-f82053/edit>





- intersection types allow you to Combine multiple types into one





The image shows a screenshot of the Visual Studio Code (VS Code) editor interface. The top bar displays the file explorer, search, and other standard VS Code icons. The left sidebar contains the Explorer view, showing a project named 'typescript-tut' with several files listed, including 'type.ts' which is currently selected. The main editor area displays the content of 'type.ts', which contains the following TypeScript code:

```
1 type DataType={name:string,email:string}
2 var empData1:DataType={
3     name:'anil',
4     email:'anil@test.com'
5 }
6
7 var studentData1DataType={
8     name:'sam',
9     email:'sam@college.com'
10 }
```

The code defines a TypeScript interface `DataType` with properties `name` and `email`, both of type `string`. It then declares two variables: `empData1` and `studentData1DataType`, both of type `DataType`. `empData1` is initialized with the values `name: 'anil'` and `email: 'anil@test.com'`. `studentData1DataType` is initialized with the values `name: 'sam'` and `email: 'sam@college.com'`. A yellow lightbulb icon is visible next to line 9, indicating a suggestion or tip.

Below the editor area, the TERMINAL view is active, showing the command prompt `anilsidhu@anils-MacBook-Air typescript-tut %`. A circular video overlay of a man with a beard and short dark hair is positioned in the bottom right corner of the terminal area. The status bar at the bottom of the window shows the current file is 'type.ts', the cursor is at line 9, column 28, and the file is encoded in UTF-8 with LF line endings.

Edit: Code

TypeScript Tutorial in Hindi 2

https://slides.com/anilsidhu/code-f82053/edit



Type in TypeScript

- Define Custom data type for Object, function etc







Edit: Code x TypeScript Tutorial in Hindi 2 x +

https://slides.com/anilsidhu/code-f82053/edit



Type in TypeScript

- What is Type.
- Define type
- How to use type
- Difference between interface and type



Edit: Code

TypeScript Tutorial In Hindi 2

https://slides.com/anilsidhu/code-f82053/edit



Enum Data Type

- What is Enum Type
- How to define enums
- How to use enums
- Interview Questions



typescript-tut

EXPLORER

OPEN EDITORS 1 unsaved

- TS type.ts U

TYPESCRIPT-TUT

- JS null.js
- TS null.ts
- JS number.js
- TS number.ts
- TS object.ts
- () package-lock.json
- () package.json
- JS string.js
- TS string.ts
- TS symbol.ts
- TS test.ts
- tsconfig.json
- TS tuple.ts
- TS type.ts U
- JS undefine.js
- TS undefine.ts
- TS union.ts
- o user.html

OUTLINE

TIMELINE

TS type.ts U

TS type.ts > *o a

```
1 type DataType={name:string,email:string}
2
3 interface a {name:string};
4 interface b extends a {email:string};
5
6
7
8 var empData1:DataType={
9     name:'anil',
10    email:'anil@test.com'
11 }
12
13 var studentData1DataType={
14     name:'sam',
15     email:'sam@college.com'
16 }
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS GITLENS SPELL CHECKER

anilsidhu@anils-MacBook-Air typescript-tut %

Ln 3, Col 13 Spaces: 4 UTF-8 LF () TypeScript 5500



EXPLORED

OPEN EDITORS

TS type.ts U

TYPESCRIPT-TUT

JS null.js

TS null.ts

JS number.js

TS number.ts

TS object.ts

{ } package-lock.json

{ } package.json

JS string.js

TS string.ts

TS symbol.ts

TS test.ts

tsconfig.json

TS tuple.ts

TS type.ts U

JS undefine.js

TS undefine.ts

TS union.ts

user.html

OUTLINE

TIMELINE

TS type.ts U X

TS type.ts > ...

1 type DataType={name:string,email:string}

2

3 type a ={name:string};

4 type b= {email:string};

5

6 type c =a| b;

7

8 var empData1:DataType={

9 name:'anil',

10 email:'anil@test.com' I

11 }

12 |

13 var studentData1DataType={

14 name:'sam',

15 email:'sam@college.com'

16 }

PROBLEMS

OUTPUT

DEBUG CONSOLE

TERMINAL

PORTS

GITLENS

SPELL CHECKER

zsh

anilsidhu@anils-MacBook-Air typescript-tut %

Ln 12, Col 1

Spaces: 4

UTF-8

LF

{ } TypeScript

5500

typescript-tut

EXPLORER

OPEN EDITORS

TS enum.ts U

typescript-tut

- TS any_unknown.ts
- JS app.js
- TS app.ts
- TS array.ts
- TS bigint.ts
- TS enum.ts U
- TS function.ts
- TS functionParams.ts
- inputs.html
- TS inputs.ts
- TS interface.ts
- TS intersection.ts
- TS never.ts
- JS null.js
- TS null.ts
- JS number.js
- TS number.ts
- TS object.ts
- OUTLINE
- TIMELINE

TS enum.ts U X

```
1 enum whoType{
3     teacher="teacher",
4     management="management",
5     labStaff="labStaff",
6 }
7
8
9 var who:whoType= whoType.teacher;
10
11 who =whoType.teacher
12
13 console.log(whoType.teacher);
14
15
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS GITLENS SPELL CHECKER

anilsidhu@anils-MacBook-Air typescript-tut %

Ln 13, Col 28 Spaces: 4 UTF-8 LF () TypeScript Port : 5500

The image shows a screenshot of the Visual Studio Code (VS Code) editor interface. The top bar displays the file explorer, search, and view icons. The Explorer panel on the left shows the project structure, including a folder named 'TYPESCRIPT-TUT' containing several TypeScript files. The file 'enum.ts' is selected and open in the editor. The editor window shows the following TypeScript code:

```
1 enum whoType{
2     student="student",
3     teacher="teacher",
4     management="management",
5     labStaff="labStaff",
6 }
7
8
9 var who:whoType= whoType.teacher;
10
11 who =whoType.teacher
12
```

The bottom panel of the editor shows the 'TERMINAL' tab, which contains the command prompt output: 'anilsidhu@anils-MacBook-Air typescript-tut %'. The status bar at the bottom indicates the current file is 'Ln 11, Col 21', the encoding is 'UTF-8', the line ending is 'LF', and the language is 'TypeScript'. A video call overlay is visible in the bottom right corner, showing a man with a beard and short dark hair, wearing a dark shirt, against a dark background with a grid pattern.

Edit: Code

TypeScript Tutorial in Hindi 2

https://slides.com/anilsidhu/code-f82053/edit





Enum Data Type

- A data type that allows you to define a set of named constants



Edit: Code x (1) TypeScript Tutorial in Hindi x +

https://slides.com/anilsidhu/code-f82053/edit

New Chrome available

TS

DOM Handling and TypeCasting

- How to access DOM element.
- Use exclamatory sign
- Typecasting
- Interview Questions

Subscribe

typescript-tut

0%

typescript-tut

EXPLORER

OPEN EDITORS

TS enum.ts U

TYPESCRIPT-TUT

output

JS union.js

TS any_unknown.ts

JS app.js

TS app.ts

TS array.ts

TS bigint.ts

TS enum.ts U

TS function.ts

TS functionParams.ts

inputs.html

TS inputs.ts

TS interface.ts

TS intersection.ts

TS never.ts

JS null.js

TS null.ts

JS number.js

OUTLINE

TIMELINE

TS enum.ts U X

TS enum.ts > Roles > manager

27 enum Roles{

28 admin,

29 manager,

30 qa,

31 developer,

32 user

33 }

34

35 var userRole:Roles=Roles.qa;

36

37 console.log(userRole);

38

PROBLEMS

OUTPUT

DEBUG CONSOLE

TERMINAL

PORTS

GITLENS

SPELL CHECKER

anils-MacBook-Air:output anilsidhu\$ node enum.js

management

cherry

0

anils-MacBook-Air:output anilsidhu\$ node enum.js

management

cherry

2

anils-MacBook-Air:output an

Google Chrome

Ln 29, Col 8

Spaces: 4

UTF-8

LF

() TypeScript

EXPLORED

OPEN EDITORS

TS enum.ts U

TYPESCRIPT-TUT

output

JS union.js

TS any_unknown.ts

JS app.js

TS app.ts

TS array.ts

TS bigint.ts

TS enum.ts U

TS function.ts

TS functionParams.ts

inputs.html

TS inputs.ts

TS interface.ts

TS intersection.ts

TS never.ts

JS null.js

TS null.ts

JS number.js

OUTLINE

TIMELINE

TS enum.ts U X

TS enum.ts > ...

14

15 enum Fruit {

16 a="Apple",

17 b="banana",

18 c="cherry"

19 }

20

21 var myFruit:Fruit=Fruit.a

22

23 myFruit=Fruit.c

24

25 console.log(myFruit);

26

PROBLEMS

OUTPUT

DEBUG CONSOLE

TERMINAL

PORTS

GIT LENS

SPELL CHECKER

anils-MacBook-Air:typescript-tut anilsidhu\$ cd output/

anils-MacBook-Air:output anilsidhu\$ node enum.js

teacher

anils-MacBook-Air:output anilsidhu\$ node enum.js

management

anils-MacBook-Air:output anilsidhu\$ node enum.js

Ln 25, Col 20

Spaces: 4

UTF-8

LF

() TypeScript

typescript-tut

EXPLORER

OPEN EDITORS

- dom.html U
- TS dom.ts U
- JS dom.js output U

TYPESCRIPT-TUT

- output
- any_unknown.ts
- JS app.js
- TS app.ts
- TS array.ts
- TS bigint.ts
- dom.html U
- TS dom.ts U
- enum.ts
- function.ts
- functionParams.ts
- inputs.html
- TS inputs.ts
- TS interface.ts
- TS intersection.ts
- literal.ts

OUTLINE

TIMELINE

dom.html U TS dom.ts U X JS dom.js U

TS dom.ts > ...

```
1 var headingEl=document.querySelector('h1');
2 console.log(headingEl?.textContent)
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS GITLENS SPELL CH

[1:46:21 PM] File change detected. Starting incremental compilation...

[1:46:21 PM] Found 0 errors. Watching for file changes.

first-program* Launchpad 0 0

Ln 2, Col 23 Spaces: 4 UTF-8 LF {} TypeScript Port : 5500



Subscribe

typescript-tut

dom.html U X TS dom.ts U X

dom.html > html > body > script

```
1 <!DOCTYPE html>
2 <html lang="en">
3 <head>
4   <title>DOM handling</title>
5 </head>
6 <body>
7   <h1>Heading</h1>
8   <a href="/home">go to home</a>
9   <script src="./output/dom.js"></script>
10 </body>
11 </html>
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS GITLENS SPELL CH

```
[1:44:29 PM] Starting compilation in watch mode...
[1:44:30 PM] Found 0 errors. Watching for file changes.
```

Ln 9, Col 33 Spaces: 4 UTF-8 LF () HTML Port : 5500



Subscribe

typescript-tut

EXPLORER

OPEN EDITORS

dom.html

TYPESCR...

- TS null.ts
- JS number.js
- TS number.ts
- TS object.ts
- { } package-lock.json
- { } package.json
- JS string.js
- TS string.ts
- TS symbol.ts
- TS test.ts
- tsconfig.json
- TS tuple.ts
- TS type.ts
- JS undefine.js
- TS undefine.ts
- TS union.ts
- <> user.html

dom.html

```
1 <!DOCTYPE html>
2 <html lang="en">
3 <head>
4   <title>DOM handling</title>
5 </head>
6 <body>
7   <h1>Heading</h1>
8   <a href="/home">go to home</a>
9 </body>
10 </html>
```

Ln 10, Col 8 Spaces: 4 UTF-8 LF HTML Port : 5500



Subscribe

typescript-tut

EXPLORER

OPEN EDITORS

- dom.html U
- TS dom.ts U
- JS dom.js output U

TYPESCRIPT-TUT

- output
- any_unknown.ts
- JS app.js
- TS app.ts
- TS array.ts
- TS bigint.ts
- dom.html U
- TS dom.ts U
- TS enum.ts
- TS function.ts
- TS functionParams.ts
- inputs.html
- TS inputs.ts
- TS interface.ts
- TS intersection.ts
- TS literal.ts

OUTLINE

TIMELINE

dom.html U TS dom.ts U X JS dom.js U

TS dom.ts > ...

```
1 var headingEl=document.querySelector('h1')!;  
2 var anchorEl=document.querySelector('a')!;  
3 var anchorElClass=document.querySelector('.anchorStyle')! as HTMLAnchorElement;  
4 var headingElClass=document.querySelector('.headingStyle')! as HTMLHeadingElement;  
5 console.log(headingEl.classList)  
6 console.log(anchorEl.href)  
7 console.log(anchorElClass.href)  
8 console.log(headingElClass.classList)  
9  
10  
11
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS GITLENS SPELL CH

[1:51:15 PM] File change detected. Starting incremental compilation...

[1:51:15 PM] Found 0 errors. Watching for file changes.

Ln 10, Col 1 Spaces: 4 UTF-8 LF {} TypeScript Port : 5500



Subscribe

typecript-tut

dom.html U X TS dom.ts U X JS dom.js U

dom.html > html > body > h1.headingStyle

```
1 <!DOCTYPE html>
2 <html lang="en">
3 <head>
4   <title>DOM handling</title>
5 </head>
6 <body>
7   <h1 class="headingStyle">Heading h1</h1>
8   <a class="anchorStyle" href="/home">go to home</a>
9   <script src="./output/dom.js" ></script>
10 </body>
11 </html>
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS GITLENS SPELL CH

[1:50:23 PM] File change detected. Starting incremental compilation...
[1:50:23 PM] Found 0 errors. Watching for file changes.
[]

EXPLORER

OPEN EDITORS

dom.html U
TS dom.ts U
JS dom.js output U

TYPESCRIPT-TUT

output

any_unknown.ts
JS app.js
TS app.ts
TS array.ts
TS bigint.ts
dom.html U
TS dom.ts U
TS enum.ts
TS function.ts
TS functionParams.ts
inputs.html
TS inputs.ts
TS interface.ts
TS intersection.ts
OUTLINE
TIMELINE

first-program* Launchpad 0 0 Ln 7, Col 28 (12 selected) Spaces: 4 UTF-8 LF HTML Port : 5500



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typescript-tut

EXPLORER

OPEN EDITORS

- dom.html U
- TS dom.ts U
- JS dom.js output U

TYPESCRIPT-TUT

- output
- any_unknown.ts
- JS app.js
- TS app.ts
- TS array.ts
- TS bigint.ts
- dom.html U
- TS dom.ts U
- TS enum.ts
- TS function.ts
- TS functionParams.ts
- inputs.html
- TS inputs.ts
- TS interface.ts
- TS intersection.ts
- TS literal.ts

OUTLINE

TIMELINE

dom.html U TS dom.ts U X JS dom.js U

TS dom.ts > headingEl

```
1 var headingEl=document.querySelector('h1')
2 console.log(headingEl.classList)
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS GITLENS SPELL CHECK

[1:47:26 PM] File change detected. Starting incremental compilation...

[1:47:26 PM] Found 0 errors. Watching for file changes.

Ln 1, Col 44 (1 selected) Spaces: 4 UTF-8 LF {} TypeScript Port : 5500



Subscribe

typescript-tut

TS product.ts U x

TS product.ts > Product > addToCart

```
1 class Product{
3   price:number;
4   pId:number
5   inCart=false;
6   isOrdered=false;
7   constructor(name:string,price:number,pid:number){
8     this.name=name;
9     this.price=price;
10    this.pId=pid
11  }
12  addToCart():void{
13    this.inCart=true
14  }
15 }
16
17 var product= new Product('Samsung',100000,101)
```

JS product.js U x

output > JS product.js > ...

```
1 "use strict";
2 class Product {
3   name;
4   price;
5   pId;
6   inCart = false;
7   isOrdered = false;
8   constructor(name, price, pid)
9     this.name = name;
10    this.price = price;
11    this.pId = pid;
12  }
13  addToCart() {
14    this.inCart = true;
15  }
16 }
17 var product = new Product('Samsung
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS GITLENS SPELL CHECKER

[1:39:40 PM] File change detected. Starting incremental compilation...

[1:39:40 PM] Found 0 errors. Watching for file changes.

first-program* Launchpad 0 0

Ln 14, Col 6 Spaces: 4 UTF-8 LF () TypeScript 3500



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typescript-tut

TS product.ts U

TS product.ts > Product

```
1 class Product {
2   name:string;
3   price:number;
4   pId:number
5   constructor(name:string,price:number,pid:number){
6     this.name=name;
7     this.price=price;
8     this.pId=pId
9   }
10  add
11
12
13  var product= new Product('Samsung',100000,101)
```

JS product.js U x

output > JS product.js > ...

```
1 "use strict";
2 class Product {
3   name;
4   price;
5   pId;
6   constructor(name, price, pid)
7     this.name = name;
8     this.price = price;
9     this.pId = pid;
10  }
11
12  var product = new Product('Samsung
13
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS GITLENS SPELL CHECKER

node + - [] [x] ... ^ x

[1:38:54 PM] File change detected. Starting incremental compilation...

[1:38:54 PM] Found 0 errors. Watching for file changes.

first-program* Launchpad 0 0

Ln 10, Col 8 Spaces: 4 UTF-8 LF () TypeScript 3500



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Edit: Code

anil sidhu

https://slides.com/anilsidhu/code-f82053/edit

New Chrome available





Classes in TypeScript

- What is Class
- Example of class
- Define Data type to class property and functions
- Interview Questions



Subscribe

typescript-tut

TS product.ts U X

TS product.ts > ...
1 class Product{
22 }
23 }
24
25 var product= new Product('Samsung',100000,101);
26 product.addToCart();
27 console.log(product.buyProduct())
28
29
30 var product= new Product('I phobne',200000,301);
31 product.addToCart();
32 console.log(product.buyProduct())

JS product.js U X

output > JS product.js > ...
1 "use strict";
2 class Product {
3 name;
4 price;
5 pId;
6 inCart = false;
7 isOrdered = false;
8 constructor(name, price, pid)
9 {
10 this.name = name;
11 this.price = price;
12 this.pId = pid;
13 }
14 addToCart() {
15 this.inCart = true;
16 }
17 buyProduct() {
18 if (this.inCart) {

PROBLEMS 2 OUTPUT DEBUG CONSOLE TERMINAL PORTS GITLENS SPELL CHECKER 2

anilsidhu@anils-MacBook-Air typescript-tut % cd output
anilsidhu@anils-MacBook-Air output % node product.js
product Samsung is ordered in 100000
product I phobne is ordered in 200000
anilsidhu@anils-MacBook-Air output %

Ln 29, Col 1 Spaces: 4 UTF-8 LF {} TypeScript

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typescript-tut

TS product.ts U X

TS product.ts > ...
1 class Product{
15 buyProduct():string{
17 return `product \${this.name} is ordered in \${this.price}`
18 }else{
19 return `no product in cart`
20 }
21 }
22 }
23 }
24 }
25 var product= new Product('Samsung',100000,101);
26 console.log(product.buyProduct())

JS product.js U X

output > JS product.js > ...
1 "use strict";
2 class Product {
3 name;
4 price;
5 pId;
6 inCart = false;
7 isOrdered = false;
8 constructor(name, price, pid)
9 {
10 this.name = name;
11 this.price = price;
12 this.pId = pid;
13 }
14 addToCart() {
15 this.inCart = true;
16 }
17 buyProduct() {
18 if (this.inCart) {

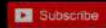
PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS GITLENS SPELL CHECKER

/dev/fd/13:18: command not found: compdef
● anilsidhu@anils-MacBook-Air typescript-tut % cd output
● anilsidhu@anils-MacBook-Air output % node product.js
no product in cart
○ anilsidhu@anils-MacBook-Air output %

first-program* Launchpad 0 0

Ln 26, Col 25 Spaces: 4 UTF-8 LF {} TypeScript 500

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Inheritance in typeScript

- What is Inheritance?
- Example.
- Interview Questions.











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typescript-tut

EXPLORER

OPEN EDITORS 1 unsaved

- TS product.ts

TYPESCRIPT-TUT

- JS null.js
- TS null.ts
- JS number.js
- TS number.ts
- TS object.ts
- { } package-lock.json
- { } package.json
- TS product.ts
- JS string.js
- TS string.ts
- TS symbol.ts
- TS test.ts
- tsconfig.json
- TS tuple.ts
- TS type.ts
- JS undefine.js
- TS undefine.ts
- TS union.ts

OUTLINE

TIMELINE

TS product.ts

```
1 class Product {
2   private name:string;
3   price:number;
4   pId:number
5   inCart=false;
6   isOrdered=false;
7   constructor(name:string,price:number,pid:number){
8     this.name=name;
9     this.price=price;
10    this.pId=pid
11  }
12  addToCart():void{
13    this.inCart=true
14  }
15  buyProduct():string{
16    if(this.inCart){
17      return `product ${this.name} is ordered in ${this.price}`
18    }else{
19      return `no product in cart`
20    }
21  }
22 }
23
24
25 var product= new Product('Samsung',100000,101);
26 product.addToCart();
```

Ln 2, Col 11 Spaces: 4 UTF-8 LF {} TypeScript Port : 5500



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Access Modifiers



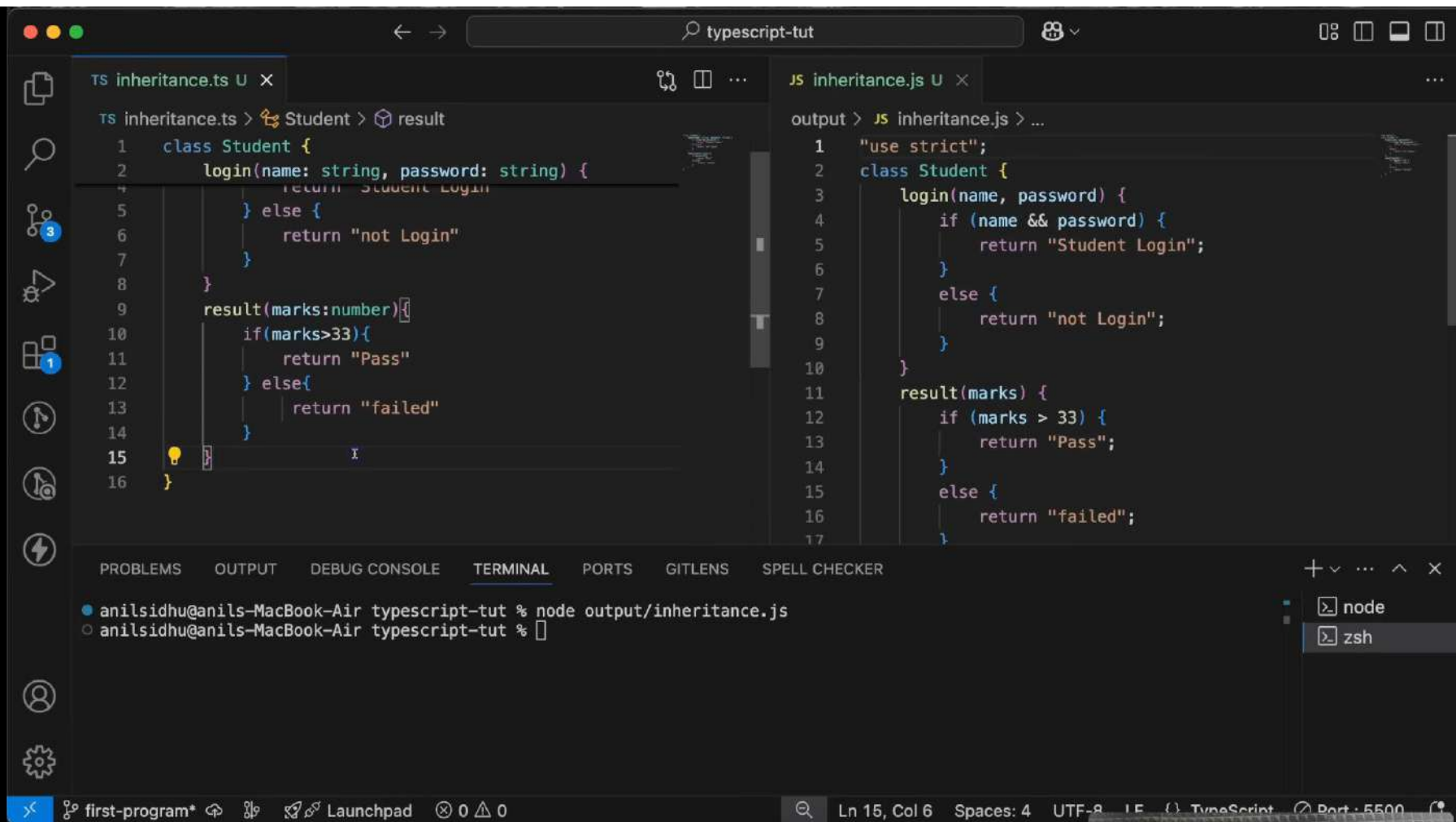
- What is Access Modifiers.
- Public.
- Private.
- Protected.
- Interview Questions.

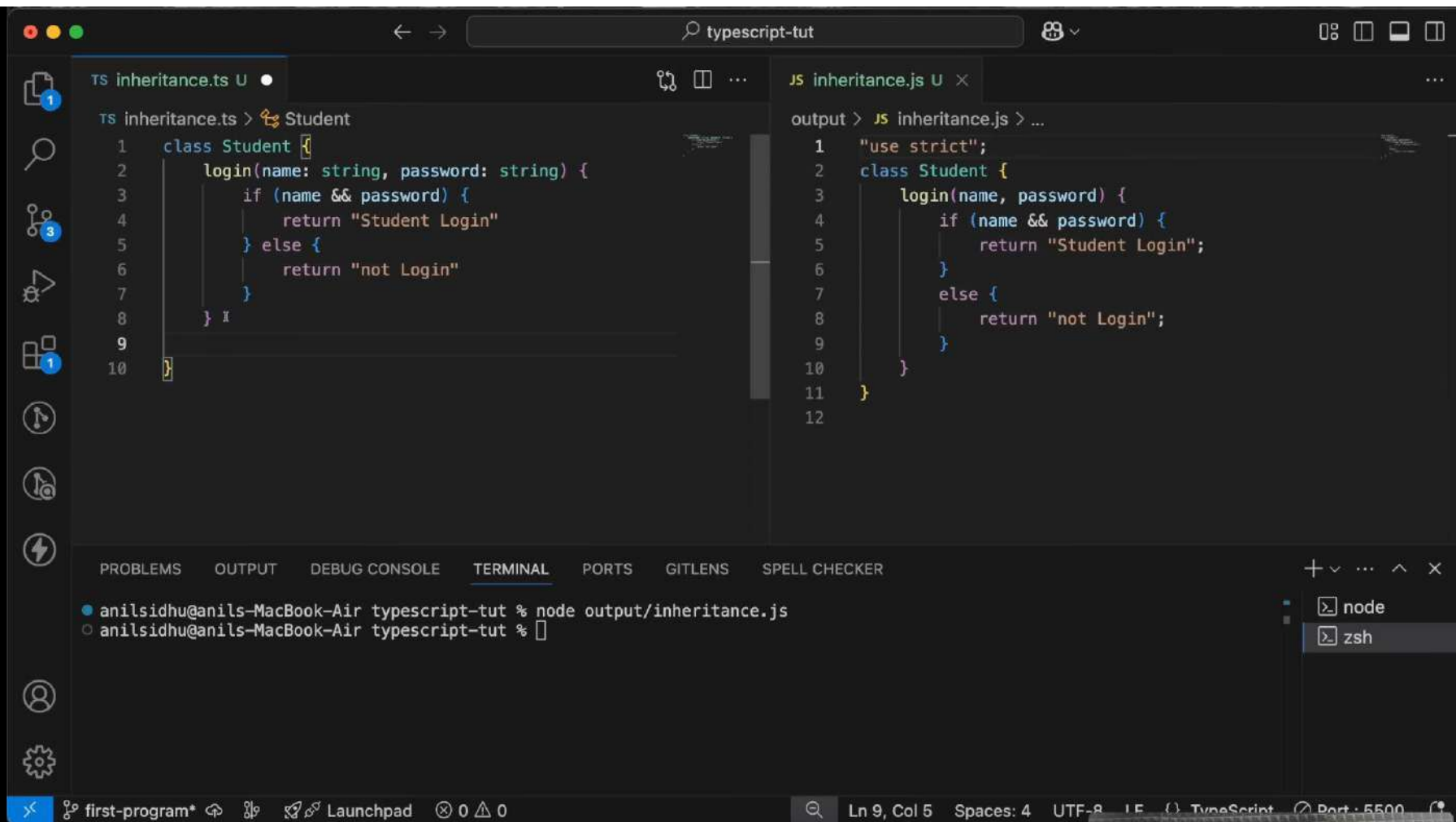
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TS

Inheritance

```
classDiagram
    class Bird {
        +color: String
        +fly()
        +tweet()
    }
    class Parrot {
        +talk()
    }
    Bird <|-- Parrot
```

The diagram illustrates inheritance. The **Bird** class is the superclass, containing attributes like `+color: String` and methods `+fly()` and `+tweet()`. The **Parrot** class is the subclass, inheriting from **Bird** and adding the `+talk()` method. An arrow points from Parrot to Bird, indicating the inheritance relationship.

superclass

subclass

Subscribe

typescript-tut

TS inheritance.ts U x

TS inheritance.ts > Teacher

```
21
22 var s1= new Student();
23 console.log(s1.result(60));
24
25
26 class Teacher extends Auth {
27
28     subject(subject:number){
29         return "he tech"+subject
30     }
31 }
32
33 var t1 = new Teacher();
34 console.log(t1.login("sam","12345"));
35
36
```

JS inheritance.js U x

output > JS inheritance.js > ...

```
1 "use strict";
2 class Auth {
3     login(name, password) {
4         if (name && password) {
5             return "Student Login";
6         }
7         else {
8             return "not Login";
9         }
10    }
11 }
12 class Student extends Auth {
13     result(marks) {
14         if (marks > 33) {
15             return "Pass";
16         }
17         else {
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS GITLENS SPELL CHECKER

anilsidhu@anils-MacBook-Air typescript-tut % node output/inheritance.js

anilsidhu@anils-MacBook-Air typescript-tut % node output/inheritance.js

Pass

anilsidhu@anils-MacBook-Air typescript-tut % node output/inheritance.js

Pass

Student Login

anilsidhu@anils-MacBook-Air typescript-tut % node output/inheritance.js

Pass

Student Login

anilsidhu@anils-MacBook-Air typescript-tut %

node

zsh

first-program* Launchpad 0 0 0 Ln 26, Col 28 Spaces: 4 UTF-8 LF TypeScript Port: 5500

typescript-tut

TS inheritance.ts U x

TS inheritance.ts > Teacher > subject

```
20
21
22 class Teacher {
23     login(name: string, password: string) {
24         if (name && password) {
25             return "Student Login"
26         } else {
27             return "not Login"
28         }
29     }
30     subject(subject: number) {
31         return "he tech"+subject
32     }
33 }
34
35
```

JS inheritance.js U x

output > JS inheritance.js > ...

```
1 "use strict";
2 class Student {
3     login(name, password) {
4         if (name && password) {
5             return "Student Login";
6         }
7         else {
8             return "not Login";
9         }
10    }
11    result(marks) {
12        if (marks > 33) {
13            return "Pass";
14        }
15        else {
16            return "failed";
17        }
18    }
19 }
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS GITLENS SPELL CHECKER

anilsidhu@anils-MacBook-Air typescript-tut % node output/inheritance.js

anilsidhu@anils-MacBook-Air typescript-tut % node output/inheritance.js

Pass

anilsidhu@anils-MacBook-Air typescript-tut %

node

zsh

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typescript-tut

TS inheritance.ts U X

JS inheritance.js U X

TS inheritance.ts > ...

output > JS inheritance.js > ...

```
1 class Student {
9   result(marks:number){
10     if(marks>33){
11       return "Pass"
12     } else{
13       return "failed"
14     }
15   }
16 }
17
18 var s1= new Student();
19 console.log(s1.result(60));
20
```

```
1 "use strict";
2 class Student {
3   login(name, password) {
4     if (name && password) {
5       return "Student Login";
6     }
7     else {
8       return "not Login";
9     }
10  }
11  result(marks) {
12    if (marks > 33) {
13      return "Pass";
14    }
15    else {
16      return "failed";
17    }
18  }
19 }
20
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS GITLENS SPELL CHECKER

anilsidhu@anils-MacBook-Air typescript-tut % node output/inheritance.js

anilsidhu@anils-MacBook-Air typescript-tut % node output/inheritance.js

Pass

anilsidhu@anils-MacBook-Air typescript-tut %

node

zsh

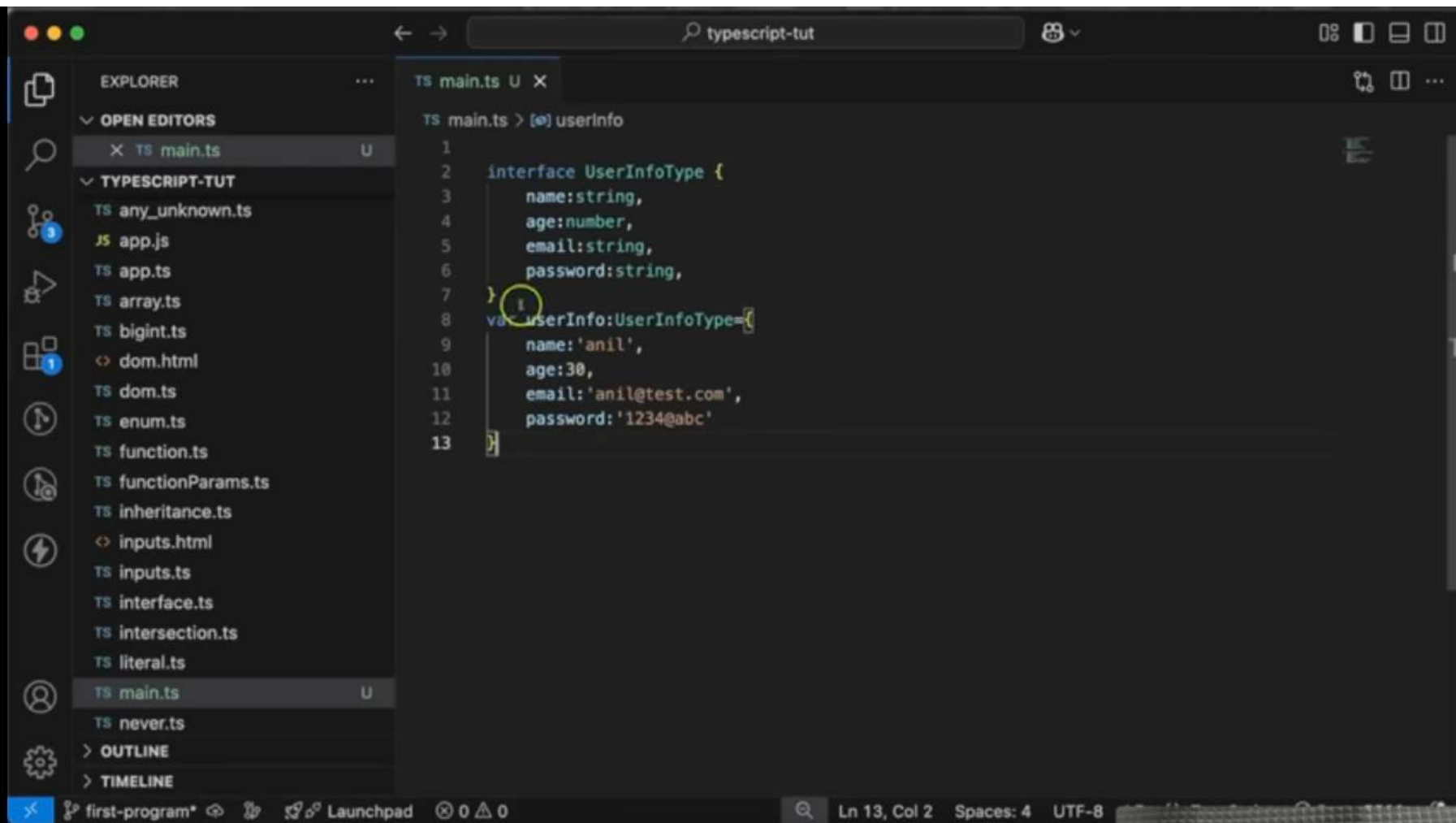
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Launchpad

0 0

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What is Module?



Self-contained unit of code that encapsulates related functionalities, such as classes, functions, and variables

ScreenPal - 3.16.1

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Module in typeScript



- What is modules in TS?
- Example.
- Interview Questions.

ScreenPal - 3.16.1

< >

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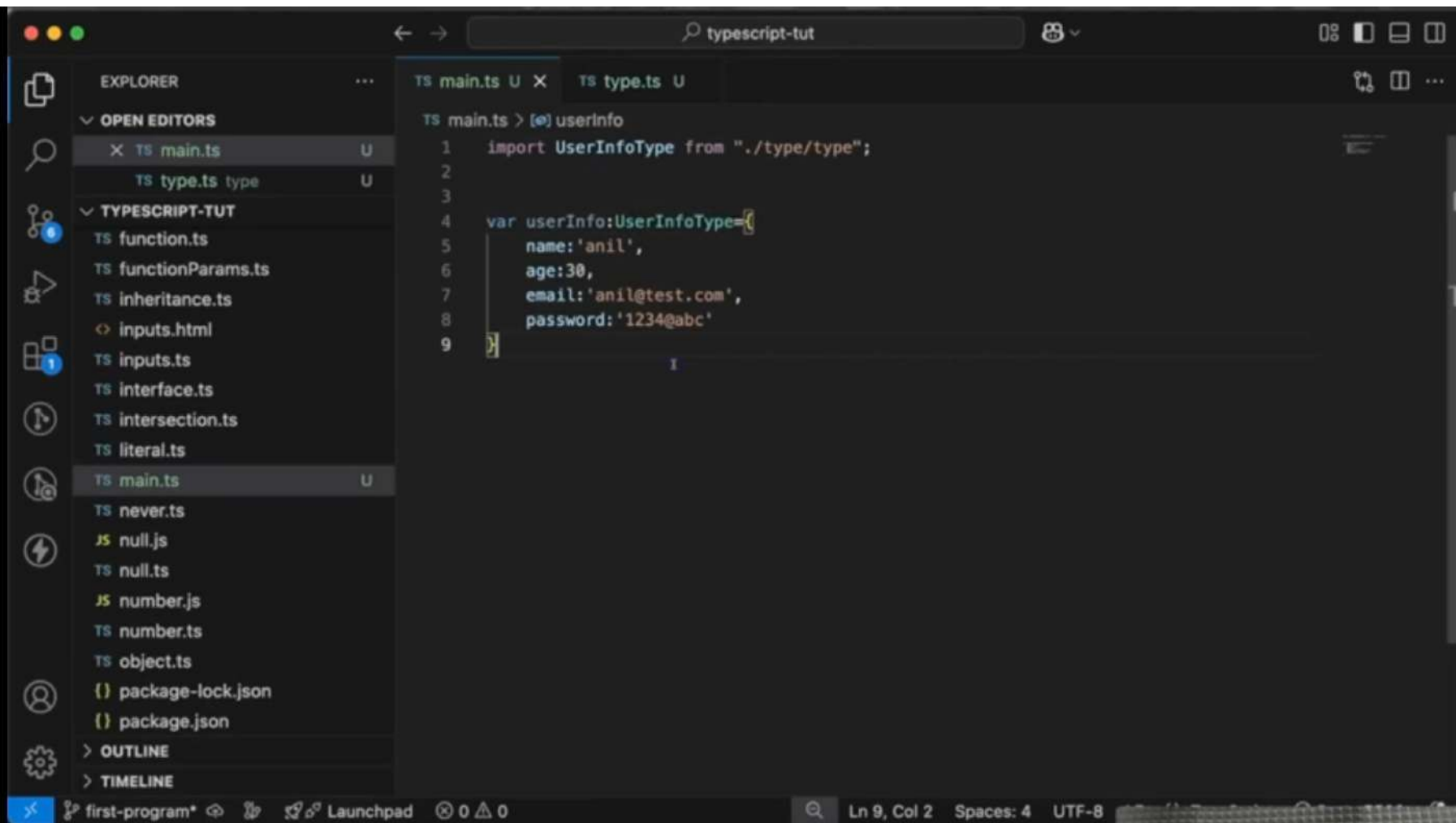
Getter and Setter in Typescript

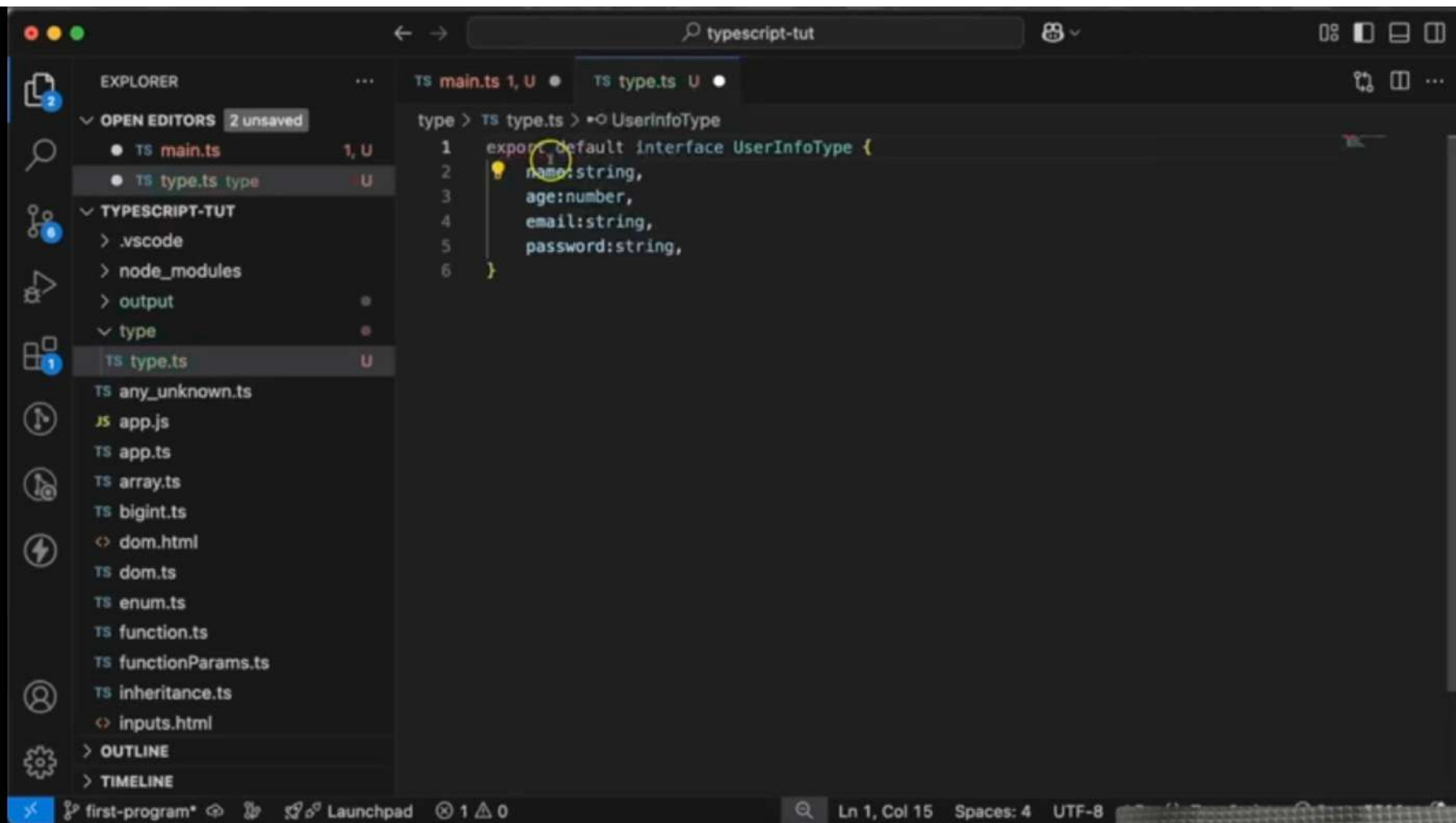


- What is Getter and Example.
- What is Setter and Example.
- Interview Questions.



Background image: A laptop keyboard on a dark desk.





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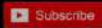




Interface with Class







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typescript-tut

TS get_set.ts U X

TS get_set.ts > ...
1 class EmpInfo{
2 get name():string{
3
4
5
6
7 }
8
9 set email(val:string){
10 this._email="emp"+val
11 }
12 }
13
14 var emp1= new EmpInfo();
15
16 emp1.email="peter@test.com";
17 console.log(emp1._email);
18

JS get_set.js U X

output > JS get_set.js > ...
1 "use strict";
2 class EmpInfo {
3 _name = "Anil";
4 _email = "anil@test.com";
5 get name() {
6 return "MR. " + this._name;
7 }
8 set _email(val) {
9 this._email = "emp" + val;
10 }
11 }
12 var emp1 = new EmpInfo();
13 emp1.email = "peter@test.com";
14 console.log(emp1._email);
15

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS GITLENS SPELL CHECKER

anilsidhu@anils-MacBook-Air output % node get_set.js
anilsidhu@anils-MacBook-Air output % node get_set.js
Anil
anilsidhu@anils-MacBook-Air output % node get_set.js
Anil
anilsidhu@anils-MacBook-Air output % node get_set.js
Anil
anilsidhu@anils-MacBook-Air output % node get_set.js
MR. Anil
anilsidhu@anils-MacBook-Air output % node get_set.js

node

Ln 16, Col 29 Spaces: 4 UTF-8 LF () TypeScript

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typescript-tut

TS get_set.ts U X

TS get_set.ts > EmpInfo > (get) name

```
1 class EmpInfo{
2   _name:string="Anil";
3
4   get name():string{
5     return "MR. " + this._name;
6   }
7 }
8
9 var emp1= new EmpInfo();
10 console.log(emp1.name);
11
```

JS get_set.js U X

output > JS get_set.js > ...

```
1 "use strict";
2 class EmpInfo {
3   _name = "Anil";
4   get name() {
5     return "MR. " + this._name;
6   }
7 }
8 var emp1 = new EmpInfo();
9 console.log(emp1.name);
10
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS GITLENS SPELL CHECKER

anilsidhu@anils-MacBook-Air output % node get_set.js

anilsidhu@anils-MacBook-Air output % node get_set.js

Anil

anilsidhu@anils-MacBook-Air output % node get_set.js

Anil

anilsidhu@anils-MacBook-Air output % node get_set.js

Anil

anilsidhu@anils-MacBook-Air output % node get_set.js

MR. Anil

anilsidhu@anils-MacBook-Air output %

node



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0 0 0

Ln 5, Col 21 (5 selected)

Spaces: 4

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LF

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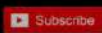


Static Keywords

- use to define static property and methods.
- Memory Efficiency
- Utility Methods
- Global Constants







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Static Keyword



- Why static keyword?
- Advantage of static keyword?
- How to use static property
- How to use static functions
- Example.
- Interview Questions.



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typescript-tut

TS collegeData.ts U X

TS collegeData.ts > CollegeDataType

```
1 interface CollegeDataType{
2     name:string;
3     // displayTeachersName():void;
4 }
5
6 class CollegeData implements CollegeDataType{
7     name:string;
8     constructor(cName:string){
9         this.name=cName
10    }
11
12 }
13
14 var college1= new CollegeData('IIT Delhi')
```

JS collegeData.js U X

output > JS collegeData.js > ...

```
1 "use strict";
2 class CollegeData {
3     name;
4     constructor(cName) {
5         this.name = cName;
6     }
7 }
8 var college1 = new CollegeData('IIT Delhi');
9
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS GITLENS SPELL CHECKER

```
/dev/fd/13:18: command not found: compdef
● anilsidhu@anils-MacBook-Air typescript-tut % cd output
● anilsidhu@anils-MacBook-Air output % node collegeData.js
○ anilsidhu@anils-MacBook-Air output %
```

+ v ... ^ x

node

zsh output

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Launchpad

0 0

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Typeguard in typescript

- What is TypeGuard.
- Types of TypeGuard.
- Examples.
- Interview Questions.



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is math a class in javascript

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🌐 En 🔊 Listen

Math is not a class in either JavaScript or TypeScript. Instead, it is a built-in object in JavaScript that provides properties and methods for mathematical constants and functions. TypeScript inherits this behavior from JavaScript, treating Math as an object as well. 🗣️

Since Math is an object and not a class, you do not create instances of it using the new keyword. All its properties and methods are static, meaning you can access them directly using Math.property or Math.method().

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🔗 Stack Overflow

2 answers · 5 years ago ⋮

Is Math an object or a class? - javascript



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typescript-tut

TS static.ts U X

TS static.ts > ...
1 class Company{
2 | static name:string="Google"
3 }
4
5
6 var c1= new Company();
7 // console.log(c1.name);
8 console.log(Company.name);
9
10

JS static.js U X

output > JS static.js > ...
1 "use strict";
2 class Company {
3 | static name = "Google";
4 }
5 var c1 = new Company();
6 // console.log(c1.name);
7 console.log(Company.name);
8

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS GITLENS SPELL CHECKER

/dev/fd/13:18: command not found: compdef
● anilsidhu@anils-MacBook-Air typescript-tut % cd output
● anilsidhu@anils-MacBook-Air output % node static.js
Google
● anilsidhu@anils-MacBook-Air output % node static.js
Google
○ anilsidhu@anils-MacBook-Air output %

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TypeScript Tutorial in Hindi 20

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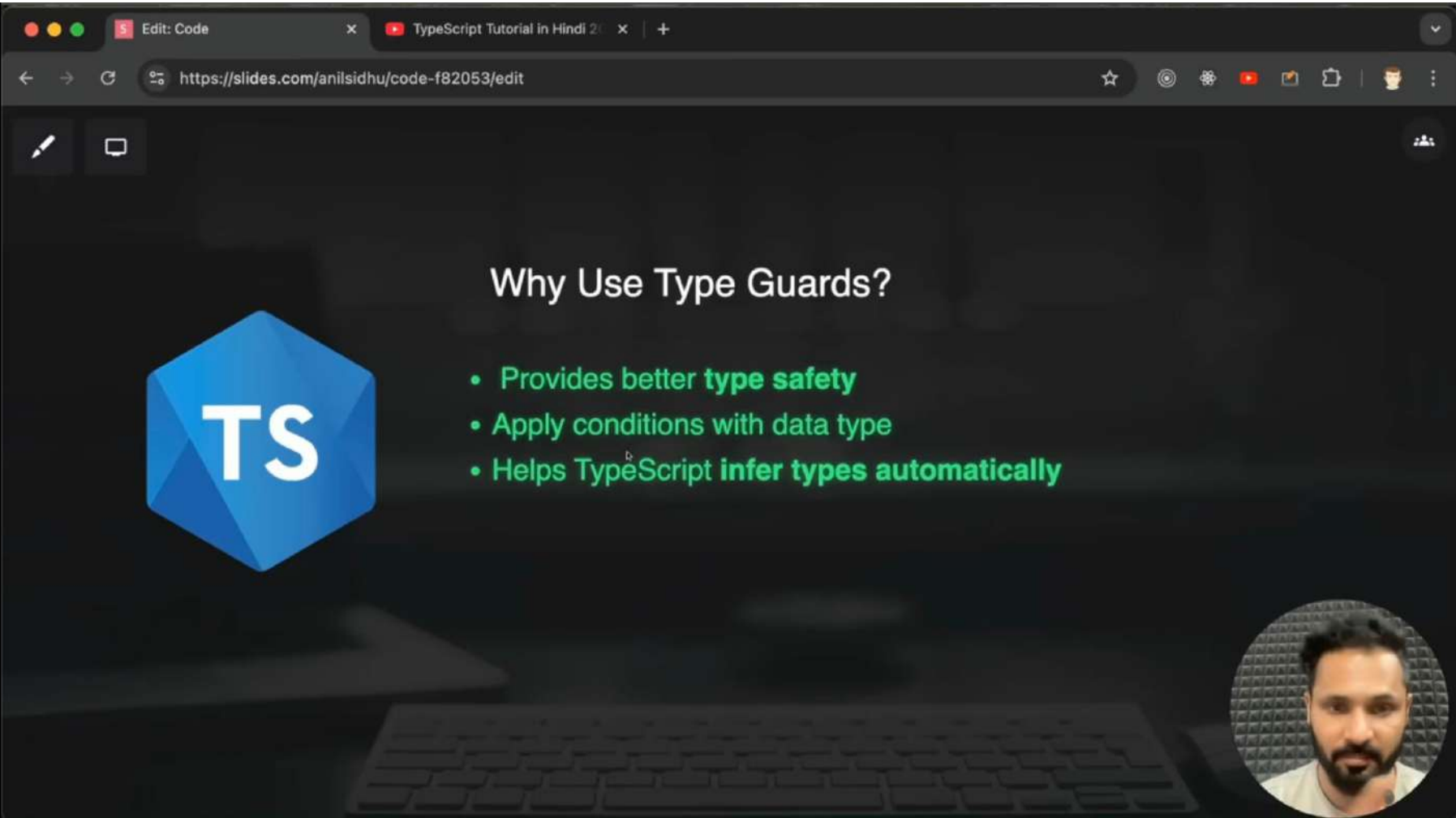


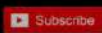


Why Use Type Guards?

- Provides better **type safety**
- Apply conditions with data type
- Helps TypeScript **infer types automatically**







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Typeguard in typescript

- `typeof`
- `instanceof`
- Custom Type



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Typeguard in typescript



- A **Type Guard** in TypeScript is a technique used to narrow down the type of a variable within a conditional block



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typescript-tut

TS typeGuards.ts U X

JS typeGuards.js U X

TS typeGuards.ts > o1

output > JS typeGuards.js > ...

```
37 var p1 = new Product21()
38 class Order21 {
39 }
40
41
42 var o1 = new Order21();
43
44
45 function checkDetails(data :Order21 |Product21 ){
46     if(data instanceof Order21){
47         console.log('this is a order');
48     }else{
49
50         console.log('this is a product');
51     }
52 }
53
```

```
1 "use strict";
2 let userData20 = "Anil sidhu";
3 // userData20=true
4 // if(typeof userData20 == "boolean"){
5 //     console.log('this is a bool data type');
6 // }
7 // else if(typeof userData20 == "string"){
8 //     console.log('this is a string data type');
9 // }else{
10 //     console.log("this is a number");
11 // }
12 // }
13 // function checkDataType(data:string | number){
14 //     if(typeof data == 'number'){
15 //         console.log("this is a number");
16 //     }else{
17 //         console.log("this is a string");
18 //     }
19 }
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS GITLENS SPELL CHECKER

Product21 {}

● anilsidhu@anils-MacBook-Air output % node typeGuards.js

this is a string

this is a product

● anilsidhu@anils-MacBook-Air output % node typeGuards.js

this is a product

● anilsidhu@anils-MacBook-Air output % node typeGuards.js

this is a order

○ anilsidhu@anils-MacBook-Air output %

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Ln 42, Col 7 Spaces: 4 UTF-8 LF TypeScript



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typescript-tut

TS typeGuards.ts U X

JS typeGuards.js U X

TS typeGuards.ts > checkDataType

output > JS typeGuards.js > ...

```
14
15 }
16
17 function checkDataType(data:string | number){
18
19     if(typeof data == 'number'){
20         console.log("this is a number");
21     }else{
22         console.log("this is a string");
23     }
24 }
25
26
27
28 checkDataType(29)
```

```
1 "use strict";
2 let userData20 = "Anil sidhu";
3 // userData20=true
4 if (typeof userData20 == "boolean") {
5     console.log('this is a bool data type');
6 }
7 else if (typeof userData20 == "string") {
8     userData20;
9     console.log('this is a string data type');
10 }
11 else {
12     console.log("this is a number");
13 }
14 function checkDataType(data) {
15     if (typeof data == 'number') {
16         console.log("this is a number");
17     }
18     else {
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS GITLENS SPELL CHECKER

anilsidhu@anils-MacBook-Air output % node typeGuards.js
this is a string data type
anilsidhu@anils-MacBook-Air output % node typeGuards.js
this is a string data type
29
anilsidhu@anils-MacBook-Air output % node typeGuards.js
this is a string data type
this is a number
anilsidhu@anils-MacBook-Air output %

first-program* Launchpad 0 0 0

Ln 28, Col 15 Spaces: 4 UTF-8 LF () TypeScript



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typescript-tut

TS typeGuards.ts U X

JS typeGuards.js U X

TS typeGuards.ts > ...

output > JS typeGuards.js > ...

```
1 let userData20:number|string|boolean="Anil sidhu"
2
3 userData20=true
4 if(typeof userData20 == "boolean"){
5     console.log('this is a bool data type');
6 }
7
8 else if(typeof userData20 == "string"){
9     console.log('this is a string data type');
10 }
11 }else{
12     console.log("this is a number");
13 }
14 }
```

```
1 "use strict";
2 let userData20 = "Anil sidhu";
3 userData20 = true;
4 if (typeof userData20 == "boolean") {
5     console.log('this is a bool data type');
6 }
7 else if (typeof userData20 == "string") {
8     console.log('this is a string data type');
9 }
10 else {
11     console.log("this is a number");
12 }
13
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS GITLENS SPELL CHECKER

/dev/fd/13:18: command not found: compdef

● anilsidhu@anils-MacBook-Air typescript-tut % cd output

○ anilsidhu@anils-MacBook-Air output %

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Ln 12, Col 34 Spaces: 4 UTF-8 LF TypeScript



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typescript-tut

TS typeGuards.ts U X

TS typeGuards.ts > [email] userData21 > email

```
74 var userData21:userData | userInfo={
75   email:'anil@test.com'
76 }
77
78 function checkUserInfo(data:userData|userInfo){
79   if((data as userData).name!==undefined){
80     console.log('this is a user data');
81   }else{
82     console.log('this is a user info');
83   }
84 }
85
86 checkUserInfo(userData21)
```

JS typeGuards.js U X

output > JS typeGuards.js > ...

```
1 "use strict";
2 let userData20 = "Anil sidhu";
3 // userData20=true
4 // if(typeof userData20 == "boolean"){
5 //   console.log('this is a bool data type');
6 // }
7 // else if(typeof userData20 == "string"){
8 //   console.log('this is a string data type');
9 // }else{
10 //   console.log('this is a number');
11 // }
12
13 // function checkDataType(data:string | number){
14 //   if(typeof data == 'number'){
15 //     console.log("this is a number");
16 //   }else{
17 //     console.log("this is a string");
18 //   }
19 }
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS GITLENS SPELL CHECKER

Product21 {}

- anilsidhu@anils-MacBook-Air output % node typeGuards.js
this is a string
this is a product
- anilsidhu@anils-MacBook-Air output % node typeGuards.js
this is a product
- anilsidhu@anils-MacBook-Air output % node typeGuards.js
this is a order
- anilsidhu@anils-MacBook-Air output %

first-program* Launchpad 0 0

Ln 76, Col 12 Spaces: 4 UTF-8 LF () TypeScript



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typescript-tut

TS typeGuards.ts U X

TS typeGuards.ts > ...

```
50
57
58 interface userData{
59     name:string,
60     city:string
61 }
62
63 interface userInfo{
64     id:number,
65     email:string
66 }
67 var userData22:userData | userInfo
68
```

JS typeGuards.js U X

output > JS typeGuards.js > ...

```
1  "use strict";
2  let userData20 = "Anil sidhu";
3  // userData20=true
4  // if(typeof userData20 == "boolean"){
5  //     console.log('this is a bool data type');
6  // }
7  // else if(typeof userData20 == "string"){
8  //     console.log('this is a string data type');
9  // }else{
10 //     console.log("this is a number");
11 // }
12 // }
13 // function checkDataType(data:string | number){
14 //     if(typeof data == 'number'){
15 //         console.log("this is a number");
16 //     }else{
17 //         console.log("this is a string");
18 //     }
19 }
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS GITLENS SPELL CHECKER

Product21 {}

- anilsidhu@anils-MacBook-Air output % node typeGuards.js
this is a string
this is a product
- anilsidhu@anils-MacBook-Air output % node typeGuards.js
this is a product
- anilsidhu@anils-MacBook-Air output % node typeGuards.js
this is a order
- anilsidhu@anils-MacBook-Air output %

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typescript-tut

TS typeGuards.ts U

JS typeGuards.js U

TS typeGuards.ts > ...

output > JS typeGuards.js > ...

```
45 function checkDetails(data :Order21 |Product21 ){
46     console.log('this is a order');
47 }
48
49 }else{
50
51     console.log('this is a product');
52 }
53 }
54
55 checkDetails(o1);
56
57
58 |
```

```
1 "use strict";
2 let userData20 = "Anil sidhu";
3 // userData20=true
4 // if(typeof userData20 == "boolean"){
5 //     console.log('this is a bool data type');
6 // }
7 // else if(typeof userData20 == "string"){
8 //     userData20
9 //     console.log('this is a string data type');
10 // }else{
11 //     console.log("this is a number");
12 // }
13 // function checkDataType(data:string | number){
14 //     if(typeof data == 'number'){
15 //         console.log("this is a number");
16 //     }else{
17 //         console.log("this is a string");
18 //     }
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS GITLENS SPELL CHECKER

Product21 {}

● anilsidhu@anils-MacBook-Air output % node typeGuards.js

this is a string

this is a product

● anilsidhu@anils-MacBook-Air output % node typeGuards.js

this is a product

● anilsidhu@anils-MacBook-Air output % node typeGuards.js

this is a order

○ anilsidhu@anils-MacBook-Air output %

first-program* Launchpad 0 0

Ln 58, Col 1 Spaces: 4 UTF-8 LF () TypeScript



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typescript-tut

TS generic.ts 1, U X

JS generic.js U X

TS generic.ts > ...

output > JS generic.js > ...

```
1 function fruits(name:string|number):string|number{
2     return name
3 }
4
5 fruits("apple")
6 fruits(100)
7 fruits(true)
8
9
```

```
1 "use strict";
2 function fruits(name) {
3     return name;
4 }
5 fruits("apple");
6 fruits(100);
7 fruits(true);
8
```

PROBLEMS 1 OUTPUT DEBUG CONSOLE TERMINAL PORTS GITLENS SPELL CHECKER

generic.ts:7:8 - error TS2345: Argument of type 'boolean' is not assignable to parameter of type 'string | number'.

7 fruits(true)

[12:55:01 AM] Found 1 error. Watching for file changes.

first-program* Launchpad 1 0

Ln 6, Col 12 Spaces: 4 UTF-8 LF () TypeScript Port: 5500



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Generic in TypeScript

- Generics in TypeScript allow you to create reusable components that work with a variety of data types while maintaining type safety



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Generic in TypeScript



- What is Generic in TypeScript
- Example
- Interview Questions



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typescript-tut

TS generic.ts U X

JS generic.js U X

TS generic.ts > Users17

output > JS generic.js > ...

```
9 // let onlyFruit=fruitsAny("apple")
10 // let onlyNum=fruitsAny(100)
11 // let onlyBool =fruitsAny(true)
12
13 let onlyFruit=fruits10("apple")
14 let onlyNum=fruits10(100)
15 let onlyBool =fruits10(true)
16
17 function Users17<T>(data:T[]):T {
18   return data[0]
19 }
20
21 let userCollection = Users17(['anil','sam'])
22
```

```
1 "use strict";
2 function fruits10(name) {
3   return name;
4 }
5 function fruitsAny(name) {
6   return name;
7 }
8 // let onlyFruit=fruitsAny("apple")
9 // let onlyNum=fruitsAny(100)
10 // let onlyBool =fruitsAny(true)
11 let onlyFruit = fruits10("apple");
12 let onlyNum = fruits10(100);
13 let onlyBool = fruits10(true);
14 function Users17(data) {
15   return data[0];
16 }
17 let userCollection = Users17(['anil', 'sam']);
18
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS GITLENS SPELL CHECKER

[1:00:05 AM] File change detected. Starting incremental compilation...

[1:00:05 AM] Found 0 errors. Watching for file changes.

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Ln 17, Col 28 Spaces: 4 UTF-8 LF TypeScript Port: 5500

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typescript-tut

TS generic.ts U X

JS generic.js U X

```
TS generic.ts > ...
9 // let onlyFruit=fruitsAny("apple")
10 // let onlyNum=fruitsAny(100)
11 // let onlyBool =fruitsAny(true)
12
13 let onlyFruit=fruits10("apple")
14 let onlyNum=fruits10(100)
15 let onlyBool =fruits10(true)
16
17 function Users17<T>(data:T):T {
18     return data
19 }
20
21 let userCollection = Users17(['anil','sam'])
22 |
```

```
JS generic.js > ...
1 "use strict";
2 function fruits10(name) {
3     return name;
4 }
5 function fruitsAny(name) {
6     return name;
7 }
8 // let onlyFruit=fruitsAny("apple")
9 // let onlyNum=fruitsAny(100)
10 // let onlyBool =fruitsAny(true)
11 let onlyFruit = fruits10("apple");
12 let onlyNum = fruits10(100);
13 let onlyBool = fruits10(true);
14 function Users17(data) {
15     return data;
16 }
17 let userCollection = Users17(['anil', 'sam']);
18
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS GITLENS SPELL CHECKER

[12:59:31 AM] File change detected. Starting incremental compilation...

[12:59:31 AM] Found 0 errors. Watching for file changes.

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Ln 22, Col 1 Spaces: 4 UTF-8 LF TypeScript Port: 5500



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typescript-tut

TS generic.ts U X

JS generic.js U X

TS generic.ts > ...

output > JS generic.js > ...

```
1 function fruits10<T>(name:T):T{
2     return name
3 }
4
5 let onlyFruit=fruits10("apple")
6 let onlyNum=fruits10(100)
7 let onlyBool =fruits10(true)
8
9
```

```
1 "use strict";
2 function fruits10(name) {
3     return name;
4 }
5 let onlyFruit = fruits10("apple");
6 let onlyNum = fruits10(100);
7 let onlyBool = fruits10(true);
8
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS GITLENS SPELL CHECKER

[12:56:38 AM] File change detected. Starting incremental compilation...

[12:56:38 AM] Found 0 errors. Watching for file changes.

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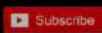


keyOf

- The `keyof` keyword in TypeScript is used to get the keys of a type as a union of string.
- It is primarily used for type safety when working with objects.







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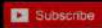




keyOf

- The `keyof` keyword in TypeScript is used to get the keys of a type as a union of string.
- It is primarily used for type safety when working with objects.





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keyOf in TypeScript

- What is keyof?
- Example
- How to use object keys with keyof
- Interview Questions



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typescript-tut

EXPLORER

OPEN EDITORS

GROUP 1

TS keyof.ts U

GROUP 2

JS keyof.js o... U

TYPESCRIPT-TUT

TS generic.ts

TS get_set.ts

TS inheritance.ts

inputs.html

TS inputs.ts

TS interface.ts

TS intersection.ts

TS keyof.ts U

TS literal.ts

TS main.ts

TS never.ts

JS null.js

TS null.ts

JS number.js

TS number.ts

OUTLINE

TIMELINE

TS keyof.ts U

TS keyof.ts > [0] UserX

6 let PersonData:PersonT={

10 }

11 type PersonX=keyof PersonT;

12 // let PersonDataX:PersonX;

13 let PersonDataX:keyof PersonT;;

14 PersonDataX="name";

15 PersonDataX="age";

16 PersonDataX="isEmp";

17

18 let UserX:keyof type PersonData="name"

19

20

21

JS keyof.js U

output > JS keyof.js > ...

1 "use strict";

2 let PersonData = {

3 name: 'Anil sidhu',

4 age: 30,

5 isEmp: true

6 };

7 // let PersonDataX:Perso

8 let PersonDataX;

9 ;

10 PersonDataX = "name";

11 PersonDataX = "age";

12 PersonDataX = "isEmp";

13 let UserX = "name";

14

PROBLEMS

OUTPUT

DEBUG CONSOLE

TERMINAL

PORTS

GITLENS

SPELL CHECKER

[11:45:29 AM] File change detected. Starting incremental compilation...

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Ln 18, Col 23 (6 selected)

Spaces: 2

UTF-8

LF

() TypeScript

Live



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typescript-tut

EXPLORER

OPEN EDITOR... 1 unsaved

GROUP 1

- TS keyof.ts 1, U

GROUP 2

- JS keyof.js o... U

TYPESCRIPT-TUT

- TS generic.ts
- TS get_set.ts
- TS inheritance.ts
- inputs.html
- TS inputs.ts
- TS interface.ts
- TS intersection.ts
- TS keyof.ts 1, U
- TS literal.ts
- TS main.ts
- TS never.ts
- JS null.js
- TS null.ts
- JS number.js
- TS number.ts
- OUTLINE
- TIMELINE

TS keyof.ts 1, U

TS keyof.ts > ...

```
6 let PersonData:PersonT={
8     age:30,
9     isEmp:true
10 }
11 type PersonX=keyof PersonT;
12 let PersonDataX:PersonX;
13
14 PersonDataX="name";
15 PersonDataX="age";
16 PersonDataX="isEmp";
17 PersonDataX="city";
18
19
```

JS keyof.js U

output > JS keyof.js > ...

```
1 "use strict";
2 let PersonData = {
3     name: 'Anil sidhu',
4     age: 30,
5     isEmp: true
6 };
7 let PersonDataX;
8 PersonDataX = "name";
9 PersonDataX = "age";
10 PersonDataX = "isEmp";
11
```

PROBLEMS 1 OUTPUT DEBUG CONSOLE TERMINAL PORTS GITLENS SPELL CHECKER

```
[11:44:01 AM] File change detected. Starting incremental compilation...
[11:44:01 AM] Found 0 errors. Watching for file changes.

```

Ln 17, Col 18 Spaces: 2 UTF-8 LF TypeScript Live



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EXPLORER

OPEN EDITORS

GROUP 1

TS keyof.ts U

GROUP 2

JS keyof.js o... U

TYPESCRIPT-TUT

TS generic.ts

TS get_set.ts

TS inheritance.ts

inputs.html

TS inputs.ts

TS interface.ts

TS intersection.ts

TS keyof.ts U

TS literal.ts

TS main.ts

TS never.ts

JS null.js

TS null.ts

JS number.js

TS number.ts

OUTLINE

TIMELINE

TS keyof.ts U X

TS keyof.ts > [0] PersonData

1 type PersonT={

2 name:string,

3 age:number,

4 isEmp:boolean

5 }

6 let PersonData:PersonT={

7 name:'Anil sidhu',

8 age:30,

9 isEmp:true

10 }

JS keyof.js U X

output > JS keyof.js > ...

1 "use strict";

2 let PersonData = {

3 name: 'Anil sidhu',

4 age: 30,

5 isEmp: true

6 };

7

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS GITLENS SPELL CHECKER

[11:42:05 AM] File change detected. Starting incremental compilation...

[11:42:05 AM] Found 0 errors. Watching for file changes.

Ln 10, Col 2 Spaces: 2 UTF-8 LF () TypeScript Live

Code.Steps

Subscribe

typescript-tut

TS index_signature.ts U X

JS index_signature.js U X

TS index_signature.ts > [0] userData7

output > JS index_signature.js > ...

```
1 type userData7Type={
2   mobile:number,
3   id:number,
4   marks:number,
5   age:number
6 }
7
8 var userData7:userData7Type={
9   mobile:9999,
10  id:10,
11  marks:90,
12  age:40
13 }
```

```
1 "use strict";
2 var userData7 = {
3   mobile: 9999,
4   id: 10,
5   marks: 90,
6   age: 40
7 };
8
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS GITLENS SPELL CHECKER

[5:03:59 PM] File change detected. Starting incremental compilation...

[5:03:59 PM] Found 0 errors. Watching for file changes.

node + ... ^ x



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Ln 11, Col 14 Spaces: 4 UTF-8 LF () TypeScript Go Live

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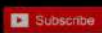




Index signature in TypeScript

- An **index signature** in TypeScript allows you to define objects with dynamic keys while specifying the type of their values.





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Index signature in TypeScript

- What is Index Signature.
- How to use it with Flexible Object Shapes.
- How to use readonly key.
- Interview Questions.



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typescript-tut

TS index_signature.ts U X

JS index_signature.js U X

TS index_signature.ts > [?] userData7

output > JS index_signature.js > ...

```
1 type userData7Type={
2   [key:string]:number|string
3 }
4
5 var userData7:userData7Type={
6   mobile:9999,
7   id:10,
8   marks:90,|
9   age:40,
10  semester:3,
11  name:'anil'
12 }
```

```
1 "use strict";
2 var userData7 = {
3   mobile: 9999,
4   id: 10,
5   marks: 90,
6   age: 40,
7   semester: 3,
8   name: 'anil'
9 };
10
```

PROBLEMS OUTPUT DEBUG CONSOLE **TERMINAL** PORTS GITLENS SPELL CHECKER

11 name:'anil'

index_signature.ts:2:5
2 [key:string]:number

The expected type comes from this index signature.

[5:05:48 PM] Found 1 error. Watching for file changes.

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Ln 8, Col 14 Spaces: 4 UTF-8 LF TypeScript Go Live

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typescript-tut

TS index_signature.ts 2, U

JS index_signature.js U

TS index_signature.ts > ...

output > JS index_signature.js > ...

1 type userData7Type={

2 [key:string]:number

3 }

4

5 var userData7:userData7Type={

6 mobile:9999,

7 }

8

9 }

1 "use strict";

2 var userData7 = {

3 mobile: 9999,

4 };

5

PROBLEMS 2 OUTPUT DEBUG CONSOLE TERMINAL PORTS GITLENS SPELL CHECKER

node + ... ^ x

7 ,

index_signature.ts:9:1 - error TS1128: Declaration or statement expected.

9 }

[5:05:25 PM] Found 2 errors. Watching for file changes.

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Ln 7, Col 4 Spaces: 4 UTF-8 LF TypeScript Go Live

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typescript-tut

TS index_signature.ts 1, U

JS index_signature.js U

TS index_signature.ts > [U] userData7

output > JS index_signature.js > ...

```
1 type userData7Type={
3   id:number,
4   marks:number,
5   age:number,
6   semester:number
7 }
8
9 var userData7:userData7Type={
10   mobile:9999,
11
12   marks:90,
13   age:40,
14   semester:3
15 }
```

```
1 "use strict";
2 var userData7 = {
3   mobile: 9999,
4   id: 10,
5   marks: 90,
6   age: 40,
7   semester: 3
8 };
9
```

PROBLEMS 1 OUTPUT DEBUG CONSOLE TERMINAL PORTS GITLENS SPELL CHECKER

[5:04:33 PM] File change detected. Starting incremental compilation...

[5:04:34 PM] Found 0 errors. Watching for file changes.

node + ... ^ x



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Ln 11, Col 2 Spaces: 4 UTF-8 LF () TypeScript Go Live

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5 Edit: Code x TypeScript Tutorial in Hindi 2 x +

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✎ 🖨️ 👤

Utility Types in TypeScript



- What is Utility Types ?
 - Partial
 - Required
 - Readonly
 - Pick
- Example
- Interview Questions

< >

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typescript-tut

TS index_signature.ts 1, U X

TS index_signature.ts > [❌] userData7Type

```
14 type userData7Type = {
16   id:number,
17   mobile:number,
18   readonly [key: string]: number | string
19 }
20
21 var userData7: userData7Type = {
22   name: 'anil',
23   mobile: 9999,
24   id: 10,
25   marks: 90,
26   age: 40,
27   semester: 3,
28 }
29 userData7.city='noida'
```

JS index_signature.js U X

output > JS index_signature.js > ...

```
1 "use strict";
2 // type userData7Type={
3 //   [key:string]:number|string
4 // }
5 var userData7 = {
6   name: 'anil',
7   mobile: 9999,
8   id: 10,
9   marks: 90,
10  age: 40,
11  semester: 3,
12 };
13 userData7.city = 'noida';
14
```

PROBLEMS 1 OUTPUT DEBUG CONSOLE TERMINAL PORTS GITLENS SPELL CHECKER

[5:08:08 PM] File change detected. Starting incremental compilation...

index_signature.ts:29:1 - error TS2542: Index signature in type 'userData7Type' only permits reading.

29 userData7.city='noida'

[5:08:08 PM] Found 1 error. Watching for file changes.

⌵

first-program* Launchpad 1 ⚠ 0

Ln 18, Col 13 (8 selected) Spaces: 4 UTF-8 LF () TypeScript Go Live

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⚙️

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typescript-tut

TS index_signature.ts 1, U X

JS index_signature.js U X

TS index_signature.ts > [❌] userData7Type

output > JS index_signature.js > ...

```
10 // semester:3,
11 // name:'anil'
12 // }
13
14 type userData7Type = {
15   name:string,
16   id:number,
17   mobile:number,
18   [key: string]: number | string
19 }
20
21 var userData7: userData7Type = {
22   name: 'anil',
23   mobile: 9999,
24   // id: 10,
```

```
1 "use strict";
2 // type userData7Type={
3 //   [key:string]:number|string
4 // }
5 var userData7 = {
6   name: 'anil',
7   mobile: 9999,
8   // id: 10,
9   marks: 90,
10  age: 40,
11  semester: 3,
12 };
13
```

PROBLEMS 1 OUTPUT DEBUG CONSOLE TERMINAL PORTS GITLENS SPELL CHECKER

node + ... ^ x



21 var userData7: userData7Type = {

index_signature.ts:16:5

16 id:number,

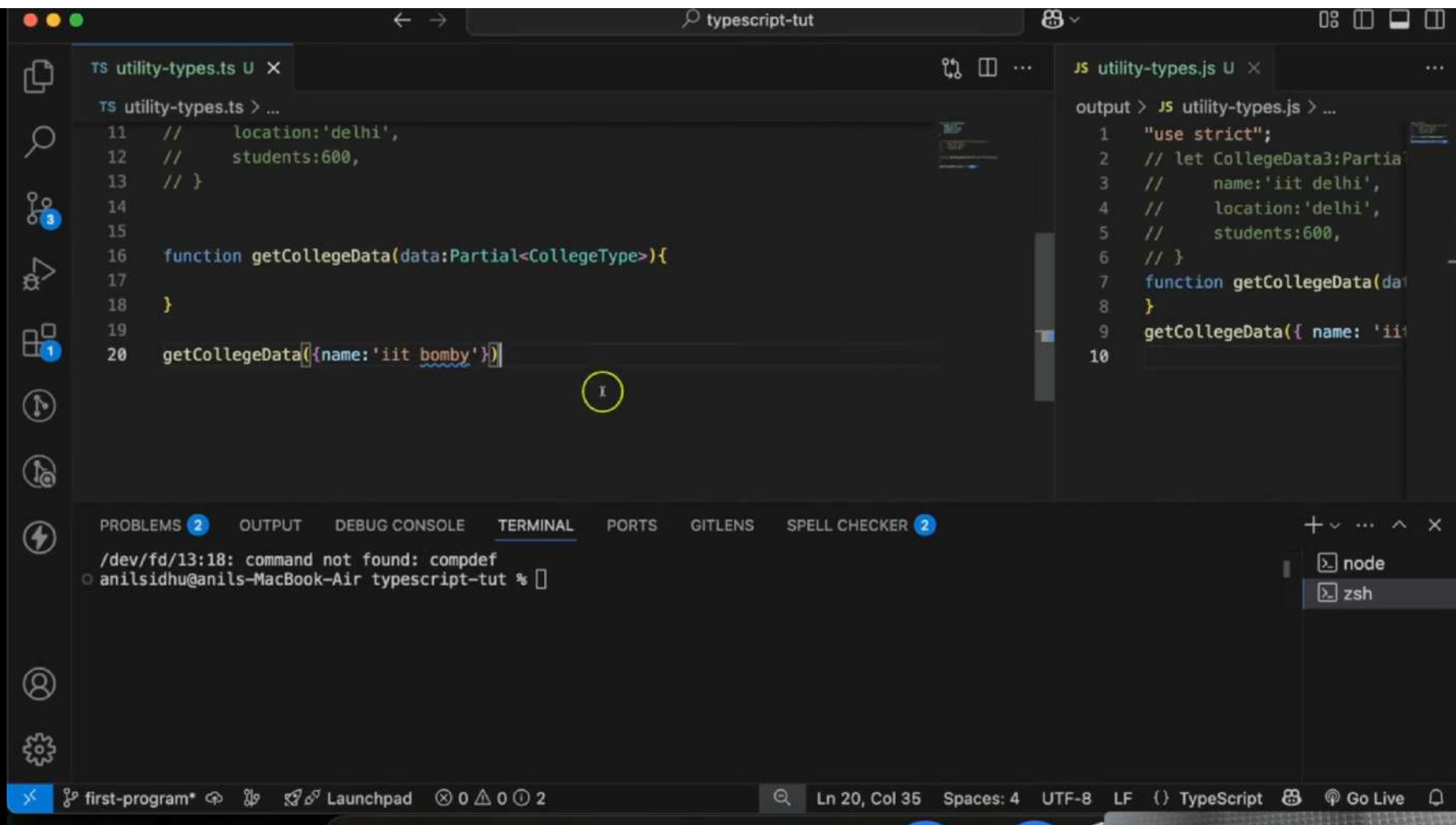
'id' is declared here.

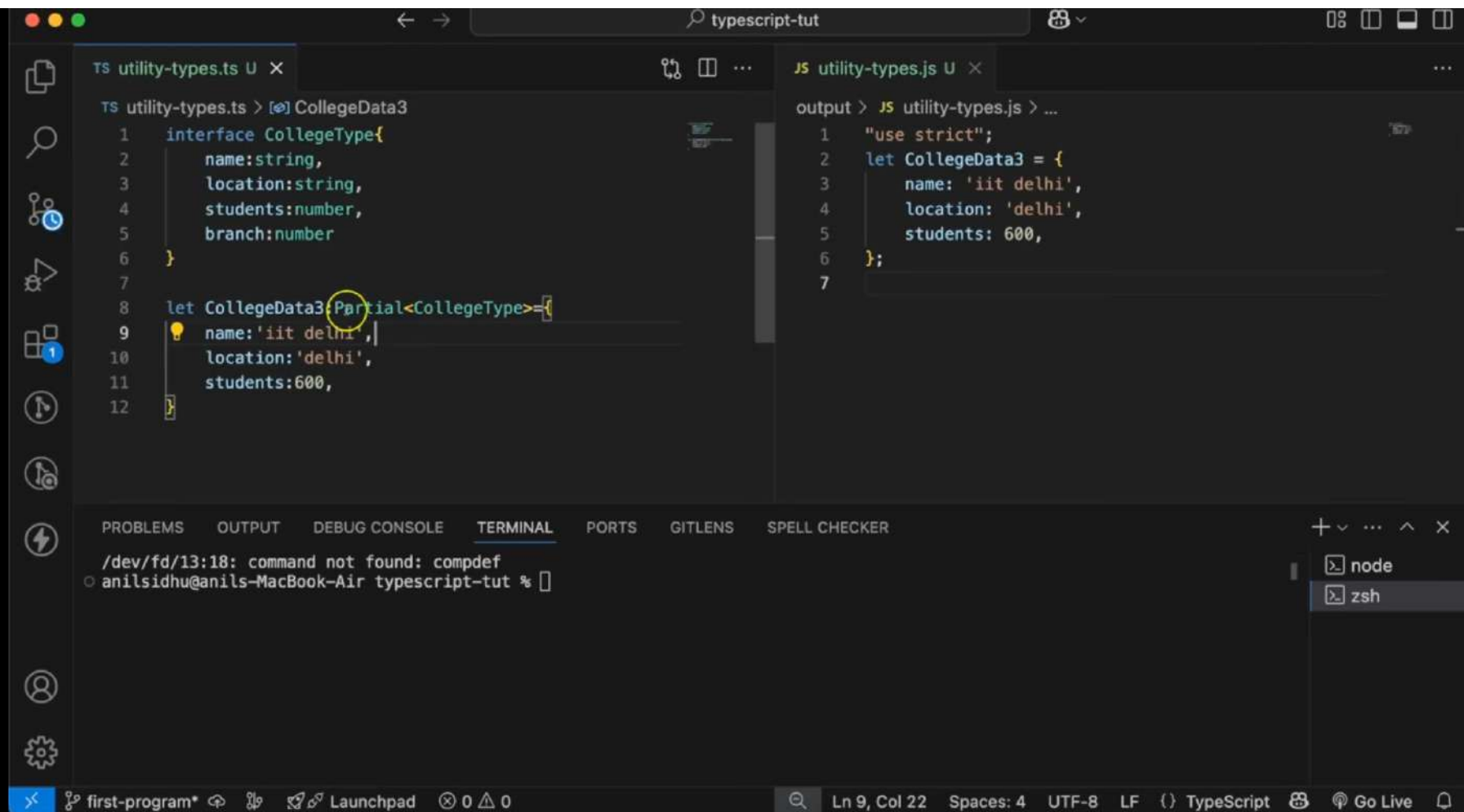
[5:07:05 PM] Found 1 error. Watching for file changes.

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🖥️

👥

Utility Types in TypeScript



- In TypeScript, **Utility Types** are built-in types that help transform or manipulate other types in a convenient way.

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⚙️

📄

typescript-tut

TS namespace.ts U X

TS namespace.ts > Auth

```
1 class Auth {
2   login() {
3     console.log("user login function");
4   }
5 }
6
7 function getList() {
8   console.log("list of users");
9 }
10
```

JS namespace.js U X

output > JS namespace.js > ...

```
1 "use strict";
2 class Auth {
3   login() {
4     console.log("user login function");
5   }
6 }
7 function getList() {
8   console.log("list of users");
9 }
10
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS GITLENS SPELL CHECKER

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TS

Namespace in Typescript

- What is namespace
- Make two name space
- How to run class, functions with namespace
- how to import namespace
- Interview Question



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Utility Types in TypeScript

- Omit
- Exclude
- Extract
- NonNullable
- Record
- Example
- Interview Questions



typescript-tut

TS namespace.ts U

```
1 namespace UserNameSpace {
2   export class Auth {
3     login() {
4       console.log("user login function");
5     }
6   }
7 }
8 export function getList() {
9   console.log("list of users");
10 }
11
12
13
14
15 namespace AdminNameSpace {
```

JS namespace.js U

```
1 "use strict";
2 var UserNameSpace;
3 (function (UserNameSpace) {
4   class Auth {
5     login() {
6       console.log("user login function");
7     }
8   }
9   UserNameSpace.Auth = Auth;
10  function getList() {
11    console.log("list of users");
12  }
13 })(UserNameSpace || (UserNameSpace = {}));
14 var AdminNameSpace;
15 (function (AdminNameSpace) {
```

PROBLEMS 1 OUTPUT DEBUG CONSOLE TERMINAL PORTS GITLENS SPELL CHECKER 1

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Ln 8, Col 11 Spaces: 4 UTF-8 LF () TypeScript



typescript-tut

TS namespace.ts U X

TS namespace.ts > {} UserNameSpace

```
1 namespace UserNameSpace{
8   function getList(){
11
12
13
14
15 namespace AdminNameSpace{
16   class Auth{
17     login(){
18       console.log("user login function");
19     }
20   }
21 }
22 function getList(){
23   console.log("List of users");
```

JS namespace.js U X

output > JS namespace.js > ...

```
1 "use strict";
2 var UserNameSpace;
3 (function (UserNameSpace) {
4   class Auth {
5     login() {
6       console.log("user login function");
7     }
8   }
9   function getList() {
10     console.log("List of users");
11   }
12 })(UserNameSpace || (UserNameSpace = {}));
13 var AdminNameSpace;
14 (function (AdminNameSpace) {
15   class Auth {
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS GITLENS SPELL CHECKER

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Ln 13, Col 2 Spaces: 4 UTF-8 LF () TypeScript



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typescript-tut

TS namespace.ts U X

```
TS namespace.ts > {} UserNameSpace
1 namespace UserNameSpace{
2   class Auth{
3     login(){
4       console.log("user login function");
5     }
6   }
7   function getList(){
8     console.log("list of users");
9   }
10 }
11
12
13
14
```

JS namespace.js U X

```
output > JS namespace.js > ...
1 "use strict";
2 var UserNameSpace;
3 (function (UserNameSpace) {
4   class Auth {
5     login() {
6       console.log("user login function");
7     }
8   }
9   function getList() {
10     console.log("list of users");
11   }
12 })(UserNameSpace || (UserNameSpace = {}));
13
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS GITLENS SPELL CHECKER

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Decorators in Typescript

- What are decorators?
- How to use it ?
- Example with class and property
- Interview Question



typescript-tut

TS namespace.ts U X

TS namespace.ts > ...
15 namespace AdminNameSpace{
27 }
28
29
30 var user8=new UserNameSpace.Auth()
31 user8.login()

JS namespace.js U X

output > JS namespace.js > ...
1 "use strict";
2 var UserNameSpace;
3 (function (UserNameSpace) {
4 class Auth {
5 login() {
6 console.log("user login function");
7 }
8 }
9 UserNameSpace.Auth = Auth;
10 function getList() {
11 console.log("list of users");
12 }
13 UserNameSpace.getList = getList;
14 })(UserNameSpace || (UserNameSpace = {}));
15 var AdminNameSpace;

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typescript-tut

TS namespace.ts U X

```
1 namespace UserNameSpace {
2   export class Auth {
3     login() {
4       console.log("user login function");
5     }
6   }
7 }
8 export function getList() {
9   console.log("list of users");
10 }
11
12
13
14
15 namespace AdminNameSpace {
```

JS namespace.js U X

```
1 "use strict";
2 var UserNameSpace;
3 (function (UserNameSpace) {
4   class Auth {
5     login() {
6       console.log("user login function");
7     }
8   }
9   UserNameSpace.Auth = Auth;
10   function getList() {
11     console.log("list of users");
12   }
13 })(UserNameSpace || (UserNameSpace = {}));
14 var AdminNameSpace;
15 (function (AdminNameSpace) {
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS GITLENS SPELL CHECKER

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typescript-tut

TS decorators.ts U X

TS decorators.ts > ...
1 function classLogger(constructor:Function){
2 console.log(constructor.name);
3
4 }
5 |
6 class CustomMaths{
7 value1:number;
8 value2:number;
9 constructor(x:number,y:number){
10 this.value1=x;
11 this.value2=y;
12 }
13 }
14
15 var cm1 = new CustomMaths(10,20);

JS decorators.js U X

output > JS decorators.js > ...
1 "use strict";
2 function classLogger(constructor) {
3 console.log(constructor.name);
4 }
5 class CustomMaths {
6 value1;
7 value2;
8 constructor(x, y) {
9 this.value1 = x;
10 this.value2 = y;
11 }
12 }
13 var cm1 = new CustomMaths(10, 20);
14

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS GITLENS SPELL CHECKER

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typescript-tut

TS decorators.ts U X

```
1 class CustomMaths{
2   value1:number;
3   value2:number;
4   constructor(x:number,y:number){
5     this.value1=x;
6     this.value2=y;
7   }
8 }
9
10 var cm1 = new CustomMaths(10,20);
```

JS decorators.js U X

```
1 "use strict";
2 class CustomMaths {
3   value1;
4   value2;
5   constructor(x, y) {
6     this.value1 = x;
7     this.value2 = y;
8   }
9 }
10 var cm1 = new CustomMaths(10, 20);
11
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS GITLENS SPELL CHECKER

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Decorators in Typescript



- In TypeScript, **decorators** are a special kind of declaration that can be attached to classes, methods, **properties**, or parameters to modify their behavior.

```
1 @Logger
2 class Person {
3   constructor(public name: string) {}
4 }
```



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Override function with Decorator in Typescript

Speaker: A man with a beard and glasses, wearing a light green shirt, is speaking into a microphone. He is positioned in the bottom right corner of the slide.

Background: A dark, blurred image of a laptop keyboard and a desk.

Bottom right corner: A red "Subscribe" button with a YouTube icon.

typescript-tut

TS decorators.ts U X

TS decorators.ts > CustomMaths

```
5 function getKeyDetails(target:any,key:any){
6     console.log(key);
7 }
8
9
10
11 @classLogger
12 class CustomMaths{
13
14     @getKeyDetails
15     value1:number;
16     value2:number;
17     constructor(x:number,y:number){
18         this.value1=x;
19         this.value2=y;
20     }
21 }
```

JS decorators.js U X

output > JS decorators.js > ...

```
1 "use strict";
2 var __esDecorate = (this && this._
3     function accept(f) { if (f !==
4         var kind = contextIn.kind, key =
5         var target = !descriptorIn && c
6         var descriptor = descriptorIn |
7         var __, done = false;
8         for (var i = decorators.length -
9             var context = {};
10             for (var p in contextIn) con
11             for (var p in contextIn.acce
12             context.addInitializer = fun
13             var result = (0, decorators
14             if (kind === "accessor") {
15                 if (result === void 0) {
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS GITLENS SPELL CHECKER

CustomMaths123

- anilsidhu@anils-MacBook-Air output % node decorators.js
CustomMaths
- anilsidhu@anils-MacBook-Air output % node decorators.js
{
 kind: 'field',
 name: 'value1',
 static: false,
 private: false,
 access: { has: [Function: has], get: [Function: get], set: [Function: set] },
 metadata: undefined,
 addInitializer: [Function (anonymous)]

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Ln 13, Col 3 Spaces: 2 UTF-8 LF () TypeScript



Subscribe

typescript-tut

TS decorators.ts U X

TS decorators.ts > CustomMaths

```
1 function classLogger(constructor:Function){
2     console.log(constructor.name);
3 }
4
5
6 @classLogger
7 class CustomMaths{
8
9     value1:number;
10    value2:number;
11    constructor(x:number,y:number){
12        this.value1=x;
13        this.value2=y;
14    }
15 }
```

JS decorators.js U X

output > JS decorators.js > ...

```
1 "use strict";
2 var __esDecorate = (this && this.__esDecorate) || function (decorators, contextIn, initializer, ...rest) {
3     function accept(f) { if (f !== undefined) {
4         var kind = contextIn.kind, key = contextIn.key, target = !descriptorIn && contextIn.target, descriptor = descriptorIn | target, done = false;
5         for (var i = decorators.length - 1; i >= 0; i--) {
6             var context = {};
7             for (var p in contextIn) context[p] = contextIn[p];
8             for (var p in contextIn.accessors) context[p] = contextIn.accessors[p];
9             context.addInitializer = function () { if (done) return; done = true; initializer.call(this); };
10            var result = (0, decorators[i])(kind === "accessor" ? { kind: "accessor", name: key, descriptor, target } : { kind: kind, name: key, descriptor, target }, context);
11            if (result !== void 0) {
12                if (kind === "accessor" || kind === "getter" || kind === "setter") {
13                    descriptor.accessor = result;
14                } else {
15                    // plain data property
16                    descriptor.value = result;
17                }
18                contextIn.descriptor = descriptor;
19            }
20        }
21    }
22 }
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS GITLENS SPELL CHECKER

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Launchpad

0 0

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Subscribe

typescript-tut

TS decorator-with-function.ts M X

TS decorator-with-function.ts > CustomMaths1

```
1 function updatedSum(target:any,key:string,descriptor:PropertyDescriptor)
2     descriptor.value=function sum(x:number,y:number){
3         let output=x+y
4         return "the output of "+x+" and "+y+" is : " + output
5     }
6 }
7
8 class CustomMaths1{
9     You, 1 second ago * Uncommitted changes
10     sum(x:number,y:number){
11         return x+y
12     }
13 }
14 var cm2= new CustomMaths1();
15 console.log(cm2.sum(10,20));
```

JS decorator-with-function.js M X

```
2 function updatedSum(target, key,
3     descriptor.value = function
4
5         return "the output of "
6     };
7 }
8 You, 1 second ago | 1 author (You)
9 class CustomMaths1 {
10     sum(x, y) {
11         return x + y;
12     }
13 }
14 var cm2 = new CustomMaths1();
15 console.log(cm2.sum(10, 20));
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS GITLENS SPELL CHECKER

```
● anilsidhu@anils-MacBook-Air output % node decorator-with-function.js
30
● anilsidhu@anils-MacBook-Air output % node decorator-with-function.js
30
○ anilsidhu@anils-MacBook-Air output %
```

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typescript-tut

TS decorator-with-function.ts M X

TS decorator-with-function.ts > updatedSum

You, 1 second ago | 1 author (You)

```
1 function updatedSum(target:any,key:string,descriptor:PropertyDescriptor)
2   |
3   |
4   |
5 You, 1 second ago | 1 author (You)
6 class CustomMaths1{
7   sum(x:number,y:number){
8     return x+y
9   }
10 }
11 var cm2= new CustomMaths1();
12 console.log(cm2.sum(10,20));
13
```

JS decorator-with-function.js M X

output > JS decorator-with-function.js > ...

You, 4 seconds ago | 1 author (You)

```
3 }
4 class CustomMaths1 {
5   sum(x, y) {
6     return x + y;
7   }
8 }
9 var cm2 = new CustomMaths1();
10 console.log(cm2.sum(10, 20));
11
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS GITLENS SPELL CHECKER

anilsidhu@anils-MacBook-Air output % node decorator-with-function.js

30

anilsidhu@anils-MacBook-Air output %

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typescript-tut

EXPLORER

OPEN EDITORS

GROUP 1

- TS decorator... M

GROUP 2

- JS decorator... M

TYPESCRIPT-TUT

- type
- TS any_unknown.ts
- JS app.js
- TS app.ts
- TS array.ts
- TS bigint.ts
- TS collegeData.ts
- TS decorator-with... M
- TS decorators.ts
- dom.html
- TS dom.ts
- TS enum.ts
- TS function.ts
- TS functionParams.ts
- TS generic.ts
- OUTLINE
- TIMELINE

TS decorator-with-function.ts M X

TS decorator-with-function.ts > CustomMaths1 > sum

You, 5 seconds ago | 1 author (You)

```
1 class CustomMaths1 {
2
3     sum(x:number,y:number) {
4         return x+y;
5     }
6 }
7 var cm2= new CustomMaths1();
8 console.log(cm2.sum(10,20));
9
```

You, 6 seconds ago • Uncommitted changes

JS decorator-with-function.js M X

output > JS decorator-with-function.js > ...

```
2 class CustomMaths1 {
3     sum(x, y) {
4         return x + y;
5     }
6 }
7 var cm2 = new CustomMaths
8 console.log(cm2.sum(10, 2
9
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS GITLENS SPELL CHECKER

anilsidhu@anils-MacBook-Air output % node decorator-with-function.js

30

anilsidhu@anils-MacBook-Air output %

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Code Steps



Edit: Code

TypeScript Tutorial in Hindi 2

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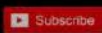




Typed Promise in TypeScript

- What is Promise.
- How to define type for promise.
- Define Custom type in promise.
- Interview Question





typescript-tut

TS decorator-with-function.ts M X

TS decorator-with-function.ts > CustomMaths1 > sum

```
1 function updatedSum(target:any,key:string,descriptor:PropertyDescriptor)
2     descriptor.value=function sum(x:number,y:number){
3         let output=x+y
4         return "the output of "+x+" and "+y+" is : " + output
5     }
6 }
7
8 class CustomMaths1{
9
10     @updatedSum
11     sum(x:number,y:number){
12         return x+y
13     }
14 }
15 var cm2= new CustomMaths1();
```

JS decorator-with-function.js M X

```
2 var __decorate = (this && this._
4     if (typeof Reflect === "obje
5     else for (var i = decorators
6     return c > 3 && r && Object.
7 };
8 function updatedSum(target, key,
9     descriptor.value = function
10         let output = x + y;
11         return "the output of "
12     };
13 }
14 class CustomMaths1 {
15     sum(x, y) {
16         return x + y;
17     }
18 }
```

output > JS decorator-with-function.js > ...

```
2 var __decorate = (this && this._
4     if (typeof Reflect === "obje
5     else for (var i = decorators
6     return c > 3 && r && Object.
7 };
8 function updatedSum(target, key,
9     descriptor.value = function
10         let output = x + y;
11         return "the output of "
12     };
13 }
14 class CustomMaths1 {
15     sum(x, y) {
16         return x + y;
17     }
18 }
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS GITLENS SPELL CHECKER

```
● anilsidhu@anils-MacBook-Air output % node decorator-with-function.js
30
● anilsidhu@anils-MacBook-Air output % node decorator-with-function.js
30
● anilsidhu@anils-MacBook-Air output % node decorator-with-function.js
the output of 10 and 20 is : 30
○ anilsidhu@anils-MacBook-Air output %
```

first-program* Launchpad 0 0 Not Committed Yet Ln 10, Col 16 Spaces: 4 UTF-8 LF {} TypeScript 5500



EXPLORED

OPEN EDITORS

TS promise.ts U

TYPESCRIPT-TUT

output

JS useNameSpace.js

TS utility-types.d.ts

JS utility-types.js

type

TS type.ts

TS any_unknown.ts

JS app.js

TS app.ts

TS array.ts

TS bigint.ts

TS collegeData.ts

TS decorator-with... M

TS decorators.ts

dom.html

TS dom.ts

TS enum.ts

TS function.ts

OUTLINE

TIMELINE

TS promise.ts U X

TS promise.ts > then() callback

```
1 function complexLogic():Promise<string> {
2   return new Promise((resolved)=>{
3     setTimeout(() => {
4       resolved("result is here")
5     }, 2000);
6   })
7 }
8
9
10 complexLogic().then((data:string)=>{
11   console.log(data);
12 })
13
```

PROBLEMS

OUTPUT

DEBUG CONSOLE

TERMINAL

PORTS

GIT LENS

SPELL CHECKER

node

zsh output

anilsidhu@anils-MacBook-Air output % node promise.js
test1
test2
anilsidhu@anils-MacBook-Air output % node promise.js
test2
test1
anilsidhu@anils-MacBook-Air output % node promise.js
result is here
test2
anilsidhu@anils-MacBook-Air output %

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Launchpad

0 0

Ln 10, Col 33 (6 selected)

Spaces: 4

UTF-8

LF

{ } TypeScript

Go Live

EXPLORED

OPEN EDITORS

TS promise.ts U

TYPESCRIPT-TUT

output

useNameSpace.js

utility-types.d.ts

utility-types.js

type

type.ts

any_unknown.ts

app.js

app.ts

array.ts

bigint.ts

collegeData.ts

decorator-with... M

decorators.ts

dom.html

dom.ts

enum.ts

function.ts

OUTLINE

TIMELINE

TS promise.ts U X

TS promise.ts > ...

```
1 function complexLogic() {
2   return new Promise((resolved)=>{
3     setTimeout(() => {
4       resolved("result is here")
5     }, 2000);
6   })
7 }
8
9
10 complexLogic().then((data)=>{
11   console.log(data);
12 })
13
14 test2()
15 }
16 function test2() {
17   console.log("test2");
18 }
```

PROBLEMS

OUTPUT

DEBUG CONSOLE

TERMINAL

PORTS

GIT LENS

SPELL CHECKER

node

zsh output

anilsidhu@anils-MacBook-Air output % node promise.js

test1

test2

anilsidhu@anils-MacBook-Air output % node promise.js

test2

test1

anilsidhu@anils-MacBook-Air output % node promise.js

result is here

test2

anilsidhu@anils-MacBook-Air output %

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Launchpad

0 0

Ln 8, Col 1

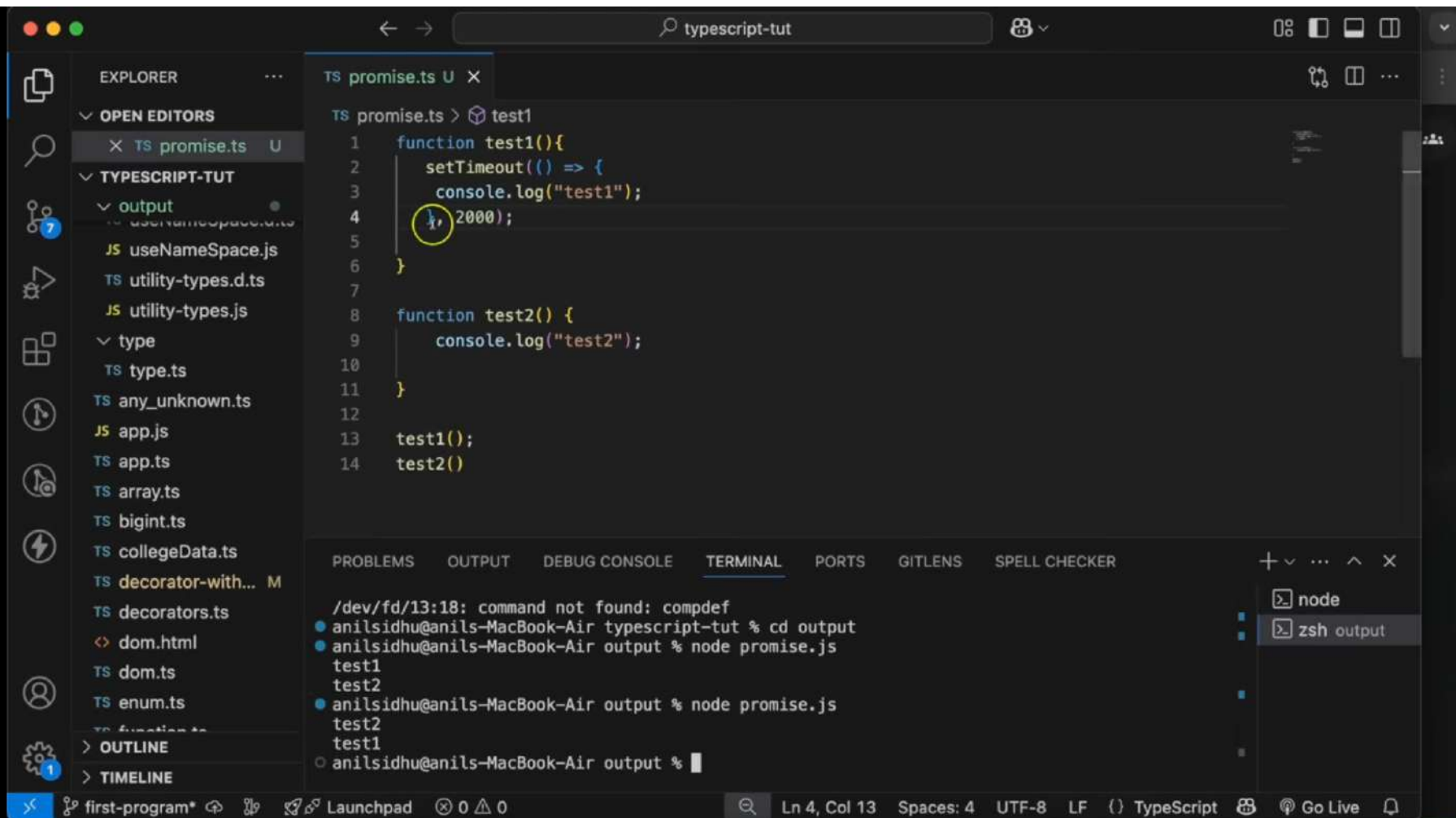
Spaces: 4

UTF-8

LF

{ } TypeScript

Go Live



Try it

Run this code here, in a console or from any site:

```
fetch('https://jsonplaceholder.typicode.com/todos/1')  
  .then(response => response.json())  
  .then(json => console.log(json))
```

Run script

```
{}
```

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Google Search Results for "dummy api call"

Results for **Block F, Carterpur Village, Sector 23A, Gurugram** · Choose area

JSONPlaceholder
<https://jsonplaceholder.typicode.com>

JSONPlaceholder - Free Fake REST API

JSONPlaceholder is a free online REST API that you can use whenever you need some fake data. It can be in a README on GitHub, for a demo on CodeSandbox, in ...

[My JSON Server](#) · [Guide](#) · [Typicode's blog](#)

DummyJSON
<https://dummyjson.com>

DummyJSON - Free Fake REST API for Placeholder JSON Data

DummyJSON provides a free fake REST API with placeholder JSON data for development, testing, and prototyping. Access realistic data quickly for your ...

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Beeceptor
<https://beeceptor.com/docs/sample-api-for-testing>

Sample API For Testing

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Slide content:



API call in TypeScript

- Write code for API call.
- Define type for response.
- Apply data type to API response.
- Interview Question

Navigation: back arrow

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typescript-tut

EXPLORER

OPEN EDITORS

TS apicall.ts U

TYPESCRIPT-TUT

output

JS union.js

TS useNamespace.d.ts

JS useNamespace.js

TS utility-types.d.ts

JS utility-types.js

type

TS type.ts

TS any_unknown.ts

TS apicall.ts U

JS app.js

TS app.ts

TS array.ts

TS bigint.ts

TS collegeData.ts

TS decorator-with-func...

TS decorators.ts

OUTLINE

TIMELINE

TS apicall.ts U X

TS apicall.ts > ...

1

2

3 async function apiCallHandling(){

4 const result = await fetch('https://jsonplaceholder.typicode.com/todos/3');

5 const data = result.json();

6 console.log(data);

7 }

8

9

10 apiCallHandling();|

PROBLEMS

OUTPUT

DEBUG CONSOLE

TERMINAL

PORTS

GITLENS

node

[10:01:10 PM] File change detected. Starting incremental compilation...

[10:01:10 PM] Found 0 errors. Watching for file changes.

first-program*

Launchpad

0 0 0

Ln 10, Col 19

Spaces: 4

UTF-8

LF

{ } TypeScript

Go Live

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Address bar: <https://jsonplaceholder.typicode.com/todos/1>

Buttons: ☆, 🎯, ⚙️, 📺, 📁, 📄, 📥, 👤, ⋮

Pretty print ☐

```
{
  "userId": 1,
  "id": 1,
  "title": "delectus aut autem",
  "completed": false
}
```

Try it

Run this code here, in a console or from any site:

```
fetch('https://jsonplaceholder.typicode.com/todos/1')  
  .then(response => response.json())  
  .then(json => console.log(json))
```

Run script

```
{}
```

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typescript-tut

TS apicall.ts U X

TS apicall.ts > ...

```
7 }
8 async function apiCallHandling():Promise<APIType>{
9     const result = await fetch('https://jsonplaceholder.typicode.com/todos/4');
10    const data = await result.json();
11    // console.log(data);
12    return data
13 }
14 }
15
16 apiCallHandling().then((data)=>{
17     console.log(data);
18 })
19 })
20
```

apicall.ts U

JS apicall.js U X ...

output > JS apicall.js > ...

```
1 "use strict";
2 async function apiCall
3     const result = awa
4     const data = await
5     // console.log(dat
6     return data;
7 }
8 apiCallHandling().then
9     console.log(data);
10 });
11
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS GITLENS SPELL CHECKER

node

zsh output

```
Promise { <pending> }
● anilsidhu@anils-MacBook-Air output % node apicall.js
{ userId: 1, id: 3, title: 'fugiat veniam minus', completed: false }
● anilsidhu@anils-MacBook-Air output % node apicall.js
{ userId: 1, id: 4, title: 'et porro tempora', completed: true }
● anilsidhu@anils-MacBook-Air output % node apicall.js
Promise { <pending> }
● anilsidhu@anils-MacBook-Air output % node apicall.js
{ userId: 1, id: 4, title: 'et porro tempora', completed: true }
○ anilsidhu@anils-MacBook-Air output %
```

first-program* Launchpad 0 0 0 Ln 20, Col 1 Spaces: 4 UTF-8 LF {} TypeScript Go Live

typescript-tut

TS apicall.ts U

JS apicall.js U

TS apicall.ts > apiCallHandling

output > JS apicall.js > ...

```
2 type APIType={
3   userId: string
4   id: number,
5   title: string
6   completed: boolean
7 }
8 async function apiCallHandling():Promise<APIType>{
9   const result = await fetch('https://jsonplaceholder.typicode.com/todos/4');
10  const data = await result.json();
11  console.log(data);
12  return data
13 }
14
15
16 apiCallHandling();
```

```
1 "use strict";
2 async function apiCall
3   const result = awa
4   const data = await
5   console.log(data);
6 }
7 apiCallHandling();
8
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS GITLENS SPELL CHECKER

/dev/fd/13:18: command not found: compdef

● anilsidhu@anils-MacBook-Air typescript-tut % cd output

● anilsidhu@anils-MacBook-Air output % node apicall.js

Promise { <pending> }

● anilsidhu@anils-MacBook-Air output % node apicall.js

{ userId: 1, id: 3, title: 'fugiat veniam minus', completed: false }

● anilsidhu@anils-MacBook-Air output % node apicall.js

{ userId: 1, id: 4, title: 'et porro tempora', completed: true }

○ anilsidhu@anils-MacBook-Air output %

node

zsh output

first-program*

Launchpad

0 0

Ln 11, Col 23 Spaces: 4 UTF-8 LF {} TypeScript

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typescript-tut

TS apicall.ts U X

TS apicall.ts > apiCallHandling

```
1
2
3 async function apiCallHandling(){
4     const result = await fetch('https://jsonplaceh
5     const data = await result.json();
6     console.log(data);
7
8
9
10 apiCallHandling();
```

TS apicall.ts U

JS apicall.js U X

output > JS apicall.js > ...

```
1 "use strict";
2 async function apiCallHandling() {
3     const result = await fetch('https://jsonplaceh
4     const data = await result.json();
5     console.log(data);
6 }
7 apiCallHandling();
8
```

PROBLEMS

OUTPUT

DEBUG CONSOLE

TERMINAL

PORTS

GITLENS

SPELL CHECKER

/dev/fd/13:18: command not found: compdef

● anilsidhu@anils-MacBook-Air typescript-tut % cd output

● anilsidhu@anils-MacBook-Air output % node apicall.js

Promise { <pending> }

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Do not use this

- Number, String, Boolean and Object

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TS

Key Differences

Feature	<code>string</code>	<code>String</code>
Type	Primitive	Object wrapper
Memory/Performance	Lightweight and faster	Heavier due to object overhead
Usage	Recommended for typing	Avoid using in type annotations
Example	<code>let x: string = 'abc';</code>	<code>let x: String = new String('abc');</code>

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